CHAPTER ONE

THE PROBLEM OF FREE WILL

In "The Dilemma of Determinism," William James writes that he cannot prove that the will is free, but he hopes to persuade his audience to assume that it is. He states that if we are free, our first act of that freedom "ought in all inward propriety to be to affirm that we are free" (1977, p. 588). As this work is in defense of free will, it seems that I am following James' advice. I assume that free will exists, and in this work hope to show how that existence is possible.

Nevertheless, it is also helpful to keep in the mind the admonition Daniel Dennett offers in his article "Some Observations on the Psychology of Thinking About Free Will" (2008),[1] where he notes that there is much poorly considered work done about free will by otherwise intelligent people. People sense "that it really matters, and they just don't want to contemplate the implications straightforwardly, in case the truth is too horrible to live with... People don't do their best work when the stakes are astronomically high" (p. 249).

One reason for this is the common view that we must be in control of our actions to be held responsible for them. If there is no control, there is no free will, and therefore no responsibility. Then no one can justly be punished for any crimes. Obviously, as we do punish those who have committed crimes, the common belief is that we are responsible for our own actions (at least most of the time), and therefore have free

[1] Cited by Timpe (2010).

will.

A simple example is sometimes used to explain how our will is free. As Henrik Walter (2001) states, when people want to raise their arm, they raise it. This is an action they feel to be freely made. But, one must ask, where did that desire itself come from? Was it as a result of a choice arising entirely from themselves, or from their environment, or from some combination of the two, or another, unrelated cause. One could further ask where the desire to understand this all has come from.

These questions are difficult to answer, and yet it is easy to wonder whether the decisions we make are unavoidable or by our choosing. If we do not understand why an action is taking place, it can be very confusing, and we may begin to ask how we can be responsible for it. In most cases, it is more comforting to think that we are freely in control of our actions. It is easier to accept the consequences for one's actions if they are the result of one's own decisions. When we do not want to do this, it is usually because we have made a bad decision. We do not often use a defense of determinism to explain our acts of kindness in the same way we might want to use it when we are accused of a crime. We are inclined to want the freedom of will and to think that we have it (though if we are inclined to want it, is it really free?).

Practically speaking, one might ask whether finding an answer to this question is even necessary. Regardless of why we do it, we most often seek to understand human behavior on a less particular level, and usually without much consideration for the nature of the will. Students of American politics study elections and look to explain why one candidate was elected over the other without considering whether the minds of voters were free to make the choice they did. One person will vote for the president because he likes the way the candidate looks, another because of party preference, and another because of economic fears. Things can be analyzed further by looking at how these attitudes formed, which seems to be through some combination of our environmental and genetic backgrounds, as one might conclude from reading Matt Ridley's book, *Nature via Nurture*[2]. The consideration of the freedom of those voters' choices is typically limited to them being made as free as possible from coercive external influences. Thus, in America, ballots are cast secretly, money spent on campaigning is regulated, and a substantial number of people are eligible to hold office. Perhaps understanding this kind of freedom is all that is necessary to ease our concerns about responsibility. Dennett (1984) does mention that it is these things that seem to be the "real threats to human freedom" (p. 169).

This might be the case if one could be restrained from wondering what the justification for a government is. In other words, why is a republican government better than a communist or monarchical government? We praise the freedom that the American government brings because such freedoms are thought to be rights that all men possess. But what makes these freedoms rights? One possible answer is that they are rights because they are correspondent to human nature, and humans are by nature free. The basis for the formation of republics such as the American regime is sometimes justified with the ideas that have emerged from theorists such as John Locke, who states that governments are legitimate only if they are the expression of the will of the people. People are

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[2] Indeed, the idea of a nature versus nurture dichotomy may be a way of looking at things that completely distorts the reality of the world, as Susan Oyama suggests in *The Ontogeny of Information* (2000).

only willing to give up so much freedom as will allow them to be secure, and no more than that, but this freedom is given up by a free choice. If there are no truly free choices, then is not the basis for the formation of legitimate governments undermined?

This is a reason, it would seem, why the question of the freedom of man's will is still debated. This freedom of choice is also tied to particular institutions of government. In most cases, it is governments that are in control of the legitimate means of punishing injustice. If a man is to be punished for his actions, the justification for that punishment is that there has been a violation of the agreement that was entered into by being a member of society, and that this violating act was done so by a responsible agent. In judging someone, we take circumstances into account, and when those circumstances curtail one's freedom, then that person is judged less harshly. Even when one takes another's life, which is in some ways the ultimate destruction of another's freedom, there are cases where the action can be found innocent of guilt. Sometimes this is for reasons of self-defense, and sometimes it is because one was truly not in control of his own actions, perhaps due to some temporary insanity or other mental defect. In other cases, one might not be judged innocent, but charged with a lesser crime, such as when one takes an action that results in another's death only unintentionally.

One way that responsibility may be lessened is due to brain defect. Modern science has come far in understanding how the brain works, and there is a debate over how much of our behavior is due to processes in the brain that happen without any conscious control. It could be argued that inasmuch as this does happen, such actions would seem to be out of one's control, and therefore provide reason for us to be held less accountable for our actions.

However, even if this argument is entirely accepted, and the conscious will thought to be an illusion, or epiphenomenal[3], it is debatable precisely what impact such a determinism can have on the law. Walter (2001)points out that punishment under the law can be in order to influence future behavior. This punishment is intended to correct one's behavior, regardless of whether the individual was in control or not. Ridley (2003) points out that knowing that a criminal is not in control of his actions is very unpersuasive in absolving him of the need for correction. If a criminal has admitted he has acted in a way out of his control, that he was conditioned to do so by his brain, his genes, or some other physiological processes, then it seems fairly certain that he will do so again without effort at correction. Still, even if we are not going to allow the inability to control one's actions to be the basis for acquittal, it would seem that the law should be adjusted to take this more into account if this is the case.

Given the above, it does seem that the freedom of the will is something that should be considered and is relevant. In terms of practice, perhaps one can get around the necessity of knowing the will's precise nature, but the better understanding one has, the more soundly political and legal institutions can be founded or adapted. A proper knowledge of why and how we do things can lead to better governments and better laws in those governments. For example, one can see that in a certain sense, both communism and democracy are concerned with the need for freedom. One might say that a Communist belief sees freedom to be

[3] If consciousness is epiphenomenal, it exists, but has no effect on decisions that are made.

found in the equality of everyone. Everyone being equal means that no one will have much of an advantage over anyone else, and in that equality there exists the most freedom that it is possible for an individual to have. The modern democratic conception of freedom would state that freedom is best maintained by the government securing certain minimal rights, namely those of life, liberty, and property (as one finds in the thought of John Locke). Other than ensuring the security of these rights, the government should leave its citizens to do as their own desires dictate. The debate in democracies exists over how much needs to be done to make this possible.

That both can potentially be seen as ways of achieving freedom points to the difficulty in defining what freedom is. Both systems might have freedom as a goal, but the outcome will look quite different. Even more difficult than describing the freedom of a citizen is describing the freedom of that citizen's own will. How one defines that freedom can determine whether one decides that the will is free. One explanation of what free will is, and which concludes that free will does exist, may not be an explanation that another would find adequate. This is compounded by the fact that countless philosophers, theologians, and scientists have attempted to explain what the nature of this freedom is, and very few of them have agreed with each other.

Of course, in *An Inquiry Concerning Human Understanding*, David Hume (1748) states that men really do agree about what human liberty is, and the problem arises because of "some ambiguous expressions, which keep the antagonists still at a distance, and hinder them from grappling with each other" (p. 62). If we had a common way of speaking about these issues, then perhaps an answer would be easier to find.

Nonetheless, despite the fact that Hume made an attempt to clear up the confusion, the debate continues.

Within the contemporary debate on free will, there are many categories that can define one's position. One can be a determinist, who believes that the will play little to no role in deciding our actions. Within the category of determinism, one can be a hard determinist, who believes that the will plays no role whatsoever in how people make their decisions, or a soft determinist, who believes that in some ways, the will is determined and in some ways it is free. One can also be a compatibilist, which is essentially another way of categorizing a soft determinist[4].

Also, there are many ways in which one can state the will is determined. It might be determined by God, or biology, or fate, or environment, or physics, or some combination of these and/or other influences[5]. With so many possibilities, it seems like there can be little room for any opinion of one's own that has not really been determined by factors outside oneself.

At the other extreme, one can be an indeterminist, who believes that the will is not determined by anything other than itself. Sometimes this position is also referred to as libertarianism or contra-causal free will.

It is here where one can see how the aforementioned confusion arises. A compatibilist can say that we have free will, and mean that our will is not entirely determined, but an indeterminist will disagree with him, and define free will as arising solely from the will itself.

4 As subsequent chapters will show, most opinions about free will tend to be some kind of compatibilism, including the view that will be presented here. This still leaves plenty of room for disagreement.

5 See Doyle (n.d.) for a more exhaustive list.

They will both say we have free will, but what one sees as free, the other sees as illogical. After all, how can something be both determined and undetermined, or how can something be completely undetermined? We do usually say that every effect must have a cause after all.

In his essay, "How to Think About the Problem of Free Will," Peter van Inwagen (n.d.) suggests that to keep the discussion of free will more coherent, philosophers should confine themselves to three categories: determinism, indeterminism, and compatibilism, "libertarianism, hard determinism, and soft determinism, are conjunctions of more fundamental theses" (p. 6). Therefore, they should not be used unless talking about the ideas of other philosophers who have used them. In the interest of achieving clarity, this work will follow van Inwagen's advice and try to categorize all opinions reviewed here with these three terms.

If we wanted to make things even simpler, John Searle (2008) argues that compatibilism is an incomprehensible category and that only determinism and free will are valid positions:

> According to the definition of these terms that I am using, determinism and free will are not compatible. The thesis of determinism asserts that all actions are preceded by sufficient causal conditions that determine them. The thesis of free will asserts that some actions are not preceded by sufficient causal conditions. Free will so defined is the negation of determinism (p 47).

This is a disagreement over terminology. Searle seems to be describing what looks to be a compatibilist position, that some actions are not preceded by sufficient causal conditions (and presumably that some other actions are preceded by sufficient causal conditions), but he defines this as being free will. It is not an indeterminist free will (at least not as described here), because at some times, there may be sufficient causal conditions that determine our will, and yet we may still have free will. This variance in the meaning of the terminology serves to further complicate the issue.

As Searle does conclude that we have free will as long as merely some actions are not preceded by sufficient causal conditions, the definition he presents is an appealing one.

Setting confusion over terminology aside, Searle suggests that without this controversy, and other philosophical confusions, the problem of free will may "essentially be a problem about how the brain works" (p. 41). Now, with new technological advances, we are gaining more insight into the workings of the brain. As we learn more, perhaps a greater insight into the nature of human agency will also come. If we understand what causes the brain to make decisions, and see every step of how these decisions are made, then perhaps we will be able to determine the extent to which these decisions are free.

But perhaps not. Perhaps by identifying our decisions as being made by brain processes, we become guilty of committing what Bennett and Hacker (2003) call the mereological fallacy. This fallacy refers to the tendency of neuroscientists and some philosophers to attribute behavior to the brain. They mistake what is really a part of the process of behavior for the whole. Just as we usually do not say it is eyes that see, but rather that it is a person who sees, so too we should say that it is not brains that think, or decide, but rather it is the person who has that brain who thinks or decides. As they write:

> We know what it is for human beings to experience things, to see things, to know or believe things, to make decisions, to interpret equivocal data, to guess and to form hypotheses. We understand what it is for people to reason inductively, to estimate probabilities, to present arguments, to classify and categorize the things they encounter in their experience. We pose questions and

> search for answers, using a symbolism – namely our language – in terms of which we represent things. But do we know what it is for *a brain* to see or hear, for *a brain* to have experiences, to know or believe something? Do we know what it is for a brain (let alone a neuron) *to reason*.... to *estimate probabilities*, to *present arguments*, to *interpret data* and to *form hypotheses* on the basis of its interpretations? (p. 70).

Bennett and Hacker believe this fallacy arises due a form of "mutant Cartesianism." Modern neuroscientists apply to the brain what "dualists ascribe to the immaterial mind" (p. 72). These neuroscientists rejected the substance dualism that came from Descartes, but in its absence, they placed the functions that previously rested in the mind in the brain. Bennett and Hacker state that the explanation for these functions cannot be placed in the brain because "the brain cannot be conscious; only a living creature whose brain it is can be conscious – or unconscious" (p. 72).

It might be argued that when neuroscientists are using these terms, they only mean them metaphorically. Brains do not actually possess these psychological qualities, but they represent "systematic causal connections" (p. 79). Brains do not think in the way a person thinks, but when a person thinks, their brain functions in a certain way. The way their brain functions may be said to be thinking in this metaphorical sense. If this is all that is meant when neuroscientists say that a brain thinks, then Bennett and Hacker have no objection. However, they find that often these metaphors are unclear, and that those who use them sometimes seem to forget that they are merely metaphors.

This is not to say that the study of the brain and how it works while people are thinking is not valuable. Knowing the neural correlates of our thinking does possess a great deal of value. Being aware of how the brain works during the process of thinking and of being conscious provides information that can help resolve the debate over free will. It can give us an understanding of the low level processes that are taking place while we do act. This was information that was not available before neuroscientific advances. It is a mistake to confuse these processes with thinking and consciousness themselves, but they are certainly relevant to the process. One of the things this work hopes to do is explain the relationship between these low level neural processes and the higher level functions they are correlates of, while showing they do not entirely determine those functions. Still, it is important not to fall into the kind of thinking that Bennett and Hacker warn against. The hope here is that this fallacy can be avoided[6].

Definition of Free Will

It would be foolish to spend all this time writing about free will without coming up with the criteria necessary to define it. As noted, part of this debate centers on holding people responsible for their actions, so it would seem appropriate to define free will in a way that preserves that responsibility. To do this, we would need to see a way in which people are somehow in control of their own actions.

Similarly, in *Elbow Room* (1984), Daniel Dennett writes: "The variety of free will we deem worth wanting are those – if there are any – that will secure for us our dignity and responsibility" (p. 153). In determining a definition for free will, we should look for one that creates a line "between exculpating pathology (the sort that is claimed in the notorious 'insanity defense') and varieties of falling-short that

[6] Hopefully the brief theory sketched in this paragraph does not already mean failure in this goal.

still leave agents genuinely culpable" (p. 157). He concludes that this is the kind of free will we have, "as a natural product of our biological endowment, extended and enhanced by our initiation into society" (p. 169).

One possible set of criteria is found in Henrik Walter's book *The Neurophilosophy of Free Will* (2001). He defines free will as existing if: "three pivotal conditions are satisfied in a critical number of his actions and decisions. The person: i. *could* have acted *otherwise* (he acts freely), ii. acts for understandable reasons (intelligible form of volition[7]), and iii. is the originator of his actions" (p. 6). Walter points out that though philosophers disagree on how to define free will, these elements can be found, "in nearly all theories about free will. Those various theories differ solely in the fact that they either deal only with part of the components, or they declare one of them to be particularly significant, or they support variously strong interpretations" (p. 6).

Given this set of criteria, Walter does not believe that free will exists, but does believe that a kind of freedom exists in mankind. What we do have is what he calls natural autonomy:

> We possess natural autonomy when under very similar circumstances we could also do other than we actually do (because of the chaotic nature of our brain). This choice is understandable (intelligible - it is determined by past events, by immediate adaptation processes in the brain, and partially by our linguistically formed environment), and it is authentic (when reflection loops with emotional adjustment we can identify that action). This kind of autonomy suits a compatibilistic concept of responsibility and supplements it in some areas (p. 299).

Walter arrives at this conclusion on the basis of what he has learned of

[7] By volition or will, Walter means "being orientated or having a propensity for something objective or a goal... which is desired... chosen... or attempted (2001, p. 28).

neuroscience. Part of what resulted in him reaching this conclusion is the premise that causality can be somewhat circular.

Of Walter's criteria, the most problematic is the first. It is difficult to know if we could have acted otherwise. It seems beyond the ability of any temporal being to be able to know if we could. No two circumstances are going to be alike. No matter how similar two circumstances might be, the next time such a circumstance occurs, one will have the benefit of knowing at least some of what has resulted from ones previous decision. It seems the only way to know otherwise would involve seeing the same action occur at the same time more than once, which short of time travel is impossible.

Dennett (1984) argues that most of the time we do not really care if someone could have done otherwise when determining responsibility. In fact, sometimes it seems inappropriate to ever do otherwise. He cites the example of Martin Luther rebelling against the Catholic church and declaring: "Here I stand. I can do no other" (p. 133). Luther certainly wanted to be held responsible for his actions, and held that doing anything other than what he thought was right would be unjustifiable. What we really want, according to Dennett, is an "open future," where "the only possible thing that can happen is the thing I decide in the end to do" (p. 139).

Because this criterion is questionable, I do not think it is sufficient to rule out the possibility of free will as it seems to do for Walter. Still, because it is so frequently used as a criterion for free will, I will continue to mention it throughout this work.

John Searle also lays out criteria for what free will would be like if we have it,

> if you had an account of brain processes that explained how the brain produced the unified field of consciousness, together with the experience of acting, and in addition how the brain produced conscious thought processes, in which the constraints of rationality are already built in as constitutive elements, you would, so to speak, get the self for free (2008, p. 72).

If we have free will, there is a explanatory gap between the state of our neurons before a decision is made and the new state of our neurons after a decision is made. If this change cannot entirely be explained by the processes going on at the neuronal level, then perhaps the gap is filled in by our conscious deliberation over the alternatives. There must be more than simple upward causation from the neurons at work during decision-making, affecting the state of our consciousness. There must also be downward causation from the systems-level effect of our consciousness to the neurons. Searle believes that quantum level indeterminacy may play some role in making this downward causation possible, and calls on neuroscientists to look for evidence that might support this hypothesis.

I believe that Walter Freeman's work does well in meeting this criteria. Because of the circular nature of causality, conscious thought will affect the behavior of neuronal processes as well as be affected by them. Freeman will be discussed further in Chapter Four.

John Searle's definition of free will may also meet Walter's three criteria. Where it might fall short is in Walter's assertion that his criteria must be met in a critical number of actions. Searle's definition states that for us to have free will only some actions must not be preceded by sufficiently causal conditions. This could put it in conflict with Walter's first criterion, but again, this is the most questionable of the criteria. As Searle, like Walter, looks to neuroscience to try and answer this question, the second and third criteria seem easy to meet, even if their solutions risk committing the mereological fallacy.

Searle's explanation of what free will should look like is the one that will be the primary focus here. If what is subsequently described in Chapter Four is sufficient to meet his criteria (and I think it is), then I will consider free will adequately defined. By this I mean that it allows us the kind of freedom that Dennett states we really want[8].

Definition of Consciousness

As should be evident from the above discussion, to understand what free will is, and whether we have it, it is also necessary to understand consciousness. Hacker and Bennett (2003) talk about consciousness as though it is irreducible to its neural correlates, and can only be attributed to persons rather than to any particular part, or parts, of them. If we do have free will, it would be because our conscious thoughts are directing our activity.

John Searle (2008) defines consciousness as "subjective, qualitative states of sentience or feeling or awareness." These conscious states are caused "entirely by neuronal processes in the brain and are realized in the brain" (p. 5). This idea of consciousness being entirely caused by neuronal processes is criticized by Hacker and Bennett (2003). They say that this "conflates a causal condition with a cause" (p. 444).

Instead, they suggest that consciousness emerges as a result of living creatures having "more and more complex forms of sensitivity to

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8 Whether Dennett's version of free will gives us what he claims it does will be considered in Chapter Five

the environment and more and more complex forms of response." As a result of this, we experience "sensation, perception, emotion, desire, and purpose" (p. 304).

Evan Thompson (2007) also criticizes Searle's position. He believes that we do not have sufficient scientific knowledge to be able to state this: "We simply do not yet know whether the brain alone is the minimal causal basis for consciousness or whether this basis also includes features of the brain-body interface or brain-body-environment interface" (p. 240). He seems to believe that more factors than just the brain need to be taken into account, as bodily processes and environmental resources are necessary to provide context for consciousness. Even if its emergence is solely caused by the brain, the fact that consciousness is not reducible to neuronal processes suggests to Thompson that it has a "transcendental status in addition to an empirical one" (p. 239).

For Thompson, consciousness is a result of life's need to perpetuate itself: "This immanent purposiveness of life is recapitulated in the temporality and intentionality of consciousness. Consciousness is a self-constituting flow inexorably directed toward the future and pulled by the affective valence of the world" (p. 362). Important to this experience is a consciousness of time, which "includes awareness of external things and their temporal characters, and awareness of experience itself as temporal and as unified across time" (p. 318). This results in a consciousness of the present as having "temporal width" which has an awareness of both past and future (albeit the very recent past and the very recent future). Consciousness is experienced in this way because of "the way the brain dynamic parses its own activity" (p. 334). This is because neuroscientific research shows that events taken on a scale of 1, which would be from 250 milliseconds to several seconds, and which is how they are perceived, are made up of neuronal events taken on a 1/10 scale.

Consciousness here shall be considered the experience of sensations, perceptions, emotions, desires, and purposes. An action is free as a result of using this faculty to make a decision, if, as mentioned above, it truly does make a difference and not merely seem to make a difference. Of course, this does not take into account the unconscious processes that are occurring in the brain, which will have an effect on what one is consciously experiencing, and which will influence that which is experienced. What is important is that those processes are affected by consciousness as they affect it. This results in what Thompson calls a dynamic co-emergence, where "a whole not only arises from its parts, but the parts also arise from the whole. Part and whole co-emerge and mutually specify each other" (p. 38).

This is a complicated concept, and hopefully it will become clearer as we go on and the process of consciousness is explained in more detail.

Conclusion

The central thesis to be presented here is that a neurodynamic understanding of the mind supports the possibility of free will in a way that meets the criterion of John Searle. It also meets two of the three criteria of Henrik Walter. This concept of free will could probably be classified as a compatibilist one, at least as the category is defined here. This understanding is sufficient to preserve the responsibility necessary for us to be held accountable for our actions. It also shows the importance of freedom not only for the will, but also for political regimes.

Much has already been written on this topic, so perhaps it seems unnecessary to add even greater volume to the literature. This may well be true. It is hoped that what will differentiate this work from the others previously written on the subject is the way that it uses theories from the fields of philosophy, theology, neuroscience, politics, the law, and psychology to understand the problem of free will. Though even this has been done before, it is hoped that the way these ideas are discussed here and the conclusions reached will add something new and useful to the free will debate.

In the following chapters I hope to expand upon some of the ideas that have been briefly mentioned in this introduction:

In chapters two and three, I will provide a brief history of philosophical and theological ideas about free will that are particularly relevant to this neuroscience based discussion. These concepts can help us to understand the findings of neuroscientific research and place that research in its proper context. In particular, St. Thomas Aquinas' theory of intentionality can make the neurodynamic theory of the mind more easy to understand.

In chapter four, I will explain Walter Freeman's theory of circular causality, and explain why it, and the larger theory of neurodynamics offer a promising solution to the free will problem.

In chapter five, I will consider other possible neuroscientific based answers to the free will question, and whether they have the potential to provide a more favorable answer than a neurodynamic one.

The final three chapters move towards looking at what some of the implications of a neuroscientific understanding of free will might be.

In chapter six, I will consider the religious implications of this neuroscientific understanding of free will. I examine whether neuroscientific research has eliminated the possibility of the soul. Sometimes it is claimed that neuroscience has done this, and that, by doing so, has eliminated the possibility of free will. I also consider whether a scientific understanding of humanity eliminates the need for religion all together, focusing on the arguments of "new atheist" Daniel Dennett in particular.

In chapter seven, I consider the political implications of neuroscientific research. I begin by examining what political conclusions we might draw from studying the brain using a neurodynamic approach, since it does provide us with the potential for free will. Then I contrast such a view with the political conclusions drawn from neuroscience found in the work of George Lakoff, Drew Westen, William Connolly, and Leslie Paul Thiele.

In chapter eight, I focus on the legal implications of a neuroscientific understanding of free will. I argue that, despite opinions to the contrary, there is no need to radically change the law on the basis of neuroscientific discoveries, because they do not diminish the responsibility we have for our actions. I also look at how neuroscientific evidence is currently being used in legal systems.

CHAPTER TWO

IDEAS OF FREE WILL FROM ARISTOTLE TO AQUINAS

It is conventional, at the start of a work concerning free will, to point out that the vast amount of writing on the subject makes it nearly impossible to present a proper survey of prior thought. This work is no different. The intention in these next two chapters is not to offer an exhaustive account of the past free will debate, but to examine previous ideas that inform the neurodynamic theory of free will to be presented in chapter four. This chapter will concentrate on the thoughts of Aristotle, St. Augustine, and St. Thomas Aquinas.

It has been stated that it was not until Christian doctrine became more fully developed that the question of free will and the implications of that question became clear. For example, Arthur Schopenhauer wrote that, "it is through Augustine and his dispute with the Manichaeans and the Pelagians that philosophy awoke to an awareness of our problem" (Schopenhauer, 1839, p. 67). Simon Harrison (2006) points out that the general consensus regarding Augustine's legacy is that he introduced the concept of the will into philosophy. However, this depends on what means by the will.

Charles Kahn (1997) proposes four ways that the will might be defined. The first way was introduced by Augustine and developed in the Christian church, reaching its culmination with Thomas Aquinas. It is a theological view of the will that states human will is preceded by the will of God, and is in the image of that will.

The second definition of will was introduced by Rene Descartes: "This essentially involves the notion of *volition* as an inner, mental event or act of consciousness which is the cause, accompaniment, or necessary condition for any outer action" (p. 235). Because of this, since Descartes, philosophers have had to deal with this dualistic way of explaining our actions.

The third way of defining the will comes from Immanuel Kant. From Kant comes the idea of the will being "self-legislation and hence as the dimension within which we become aware of ourselves as noumenal, non-empirical entities" (p. 236). Of these theories, this leads to the strongest conception of the will, and was influential on Schopenhauer and Nietzsche. Though an important idea, this definition will not be discussed in much detail here, as it is not important to our larger theory.

The fourth way of defining the will is in contrast with determinism. This is a concern for all ways of thinking about the will, and goes back to the debates of Greek philosophy. It is this debate that Aristotle tries to explain, though not having the idea of a will that is found in any of these other three ways of thinking about it (Kahn points out that the lack of a concept of will in Aristotle can be shown by comparing Aristotle's theory of human action with Aquinas').

Anthony Kenny (1979) has argued that an account of the will can be formulated from Aristotle's writings. To have a satisfying account of the will, one must be able to "relate human action to ability, desire, and belief" (p. viii). Since Aristotle has much to say on these three categories, a theory of will can be constructed on their basis. However, being able to construct a theory of the will from Aristotle's writings is not the same thing as Aristotle having a theory of the will himself. Rather, it seems more accurate to say that though Aristotle did not have a concept that directly corresponds to the will, his writing is relevant to understanding the problems that arise as a result of the concept of the will. This is probably why Aristotle is mentioned by Kahn when he is explaining his fourth way of understanding the will.

Aristotle

Though there is not an explicit concept of will in Aristotle, he does consider the question of responsibility for actions that are out of our control. Aristotle writes in *The Nicomachean Ethics* that actions are worthy of praise or blame inasmuch as they are committed without being forced, or in other words, when they are voluntary. He defines voluntary and involuntary actions as being, "what is involuntary is what is forced or is caused by ignorance, what is voluntary seems to be what has its origin in the agent himself when he knows the particulars that the action consists in" (1111a22-24).

Actions involving desire or emotion are not involuntary, because they are not much different than rational calculation. One can act wrongly in both and rightly in both. The one is no less human than the other because both have their origins in the self (1111a25-b4).

Different than mere action is decision. Decisions are actions taken that are left up to us. These are the actions we take as a result of deliberation, involving reason and thought (1112a15-33). There is a part in us that decides, and this is the dominant part within ourselves (1113a5-9). These actions distinctively originate in us, and are different from those caused by nature, necessity, or fortune (1112a30-33). When our actions are our own, we are responsible for the bad actions that we do. If, "we cannot refer actions back to other origins beyond those in ourselves, then it follows that whatever has its origins in us is itself up to us and voluntary" (1113b20). Furthermore, we are responsible for our characters, since it is our actions that produce our character. For example, by frequently drinking alcohol, one may become an alcoholic. At the point before one had ever taken a drink, he had the choice to abstain. The fact that some people are more predisposed to alcoholism than others does not mean that at a certain point in their lives they were not alcoholics, and therefore bear responsibility for their actions. However, inasmuch as defect is caused by nature, they are not worthy of blame (1114a1-30).

Related to this consideration is Aristotle's explanation of habit.

Virtue, which is the primary focus of the *Nicomachean Ethics*, is not natural in man, but acquired by habit. By constantly repeating similar activities we acquire our characters (1103b20-25). If we can be taught to acquire the right character, then presumably our characters are not determined, because otherwise such teaching would be useless. Especially important is a good upbringing, because it is at this time that habits are most easily formed (1103b23-25). In this, there seems to be great limits to the malleability of man, for Aristotle admits that after a certain point, poor habits cannot be undone (1114a20-22). It is also only those who have had a good upbringing (which presumably is a result of good fortune) who are capable of understanding the political lessons he has to teach, and thus able to become better (1095b5-10).

This ability to become virtuous by habituation suggests that we only become virtuous by our actions in the world (Thiele, 2006). A man attains the virtues "first exercising them... For the things we have to learn before we can do them, we learn by doing them, e.g. men become builders by building and lyreplayers by playing the lyre; so too we become just by doing just acts, temperate by doing temperate acts, brave by doing brave acts" (1103a32-1103b2). Still, though these habits will guide behavior, they will not control it, because one of the habits developed by a good man is that of deliberation (Thiele, p. 24). It would seem that the better one has been educated and habituated into proper action, the more in control of his actions he will be, and thus more responsible for them.

In a sense, this might almost seem paradoxical. If we assume that being more responsible for our actions entails more control over them, it seems what is being said is that the more conditioned we are to act properly, the more in control of our behavior we will be. But more conditioning would seem to imply less choice, and thus the more educated one becomes, the less they will be free. This might be the case if a necessary condition for the freedom of the will really does lie in the ability to act otherwise. Based on what we have shown from Aristotle, the person with the greatest ability to act otherwise would be the uneducated, unhabituated, vicious person. Such a person seems like they could be just as likely to do one thing as do another, having a weaker deliberative capacity to constrain the range of actions available to them. This calls to mind the famous beginning of Tolstoy's *Anna Karenina*, where it is said that "All happy families are alike; each unhappy family is unhappy in its own way."

As noted last chapter, there are many times when it is hoped that an individual can only act a certain way. When we seeing someone being taken advantage of, we hope that someone will help them. That is the virtuous choice, and to do otherwise is worthy of blame. If we are habituated to courage and deliberation, we will help the person in need, and do so in a way that will actually be helpful. Does the necessity of this really mean that the virtuous person is less free? Doyle (n.d.) argues that this is not the case. For Aristotle, so long as these habits were freely developed, and could be changed in the future, then we do have freedom (though, if they are good habits, we would not want to change them).

Aristotle on Causation

Part of difficulty that comes in trying understand free will is that of causation. It is the common opinion that every effect has a cause. When we try to explain the actions we take in this way, it can make them seem determined. For example, because my tooth was chipped, I was in pain, and had no choice but to stop working and go to the dentist. The cause of my stopping work was the pain, the pain was caused by the chipping of my tooth, the chipping of my tooth by a pretzel, and so on. If this sequence of events goes back far enough, eventually we get to a First Cause, or Unmoved Mover, which Aristotle talks about in the *Physics*:

> there is something that embraces them all and is apart from each of them, which is the cause explaining why some exist and some do not exist, and why the change is continuous. This is the cause of motion in these <other movers>, and these are the cause of motion in the other things...
>
> And one mover is sufficient; it will be first and everlasting among the unmoved things, and the principle of motion for the other things (258a3-16).

If this first cause is what put everything into motion, then it would seem that everything must be determined by it and that there is no choice, but this is not necessarily the case.

Recognizing the difficulty that comes from trying to comprehend causation, Aristotle attempted to make the issue clearer by offering a more sophisticated explanation for why things happen that goes beyond mere cause and effect. In the *Physics*, events are explained as happening in four ways. There is a material, formal, efficient, and final cause that occur. Also listed as potential causes are luck and chance. These are coincidental causes. Luck results from chance, but chance does not necessarily result from luck. Luck is chance that is related to some kind of action by a person. These causes cannot be prior to a thing "that is a cause in its own right... Chance and luck are therefore posterior to mind and nature" (198a8-11).

However, Doyle points out that in the *Metaphysics*, Aristotle writes that without chance the world would be entirely determined:

> It is obvious that there are principles and causes which are generable and destructible apart from the actual processes of generation and destruction; for if this is not true, everything will be of necessity: that is, if there must necessarily be some cause, other than accidental, of that which is generated and destroyed. Will *this* be, or not? Yes, if *this* happens; otherwise not. (1027a29)

For Aristotle, everything cannot be "always or usually" (1027a20). Though Doyle gives this fact more importance in his writing than the evidence from Aristotle suggests, the fact remains that Aristotle does recognize that there is some variance in the world that cannot be explained by science (at least not before quantum physics and chaos theory), and therefore the world is not entirely determined (though it can still be mostly determined). Given the evidence above from the *Ethics*, it can at least be allowed that this determination is not sufficient to entirely govern our lives.

Aristotle's understanding of causation will be considered further in Chapter Four.

St. Augustine

When Aristotle stated that when we are the originators of our actions they can be considered voluntary, he did not need to consider the Christian doctrine that states that God's will is what guides the world (though Aristotle does state that there was a First Cause, or unmoved mover, that set the world in motion, he says comparatively little about its purpose beyond this). Part of this problem involves the question of whether God is the cause of evil. If God is perfectly good, and he is the cause of everything, then how could evil exist? This concern did become a problem for Christianity, and thus arose St.

Augustine's need to write about the nature of man's freedom. He did this in part to combat the ideas of the Manichees and the Pelagians.

The Manichees solved the problem of God being the source of evil by stating that God was not omnipotent and was contested by an equally powerful force who was the cause of evil. Augustine was an adherent to this philosophy before he converted to Christianity, and some of his contemporary critics accused him of being unduly influenced by the Manichees in his theological thought (McGrath, 2009b). Despite this, since the Manichean position is quite heretical (since it denied God's omnipotence) this was an unacceptable position for Augustine.

The Pelagians solved the problem of evil by proposing that God had no direct influence over our actions, and therefore that we are responsible for them, including the choice to believe. Our nature makes us capable of being perfect. This belief was seen by Augustine as bearing "little relation to either the New Testament teaching or human experience" (McGrath, 2009b, p. 165). Man's imperfection makes it impossible for him to achieve salvation without the direct intervention of God.

This direct intervention would seem to preclude the will being free. If salvation only comes as a result of the direct intervention of God, how can man be responsible for his own actions? There is disagreement over whether Augustine was able to come up with a coherent answer to this problem (Harrison, 2006; Stump, 2001). His most direct writing on the subject is in his book *On the Free Choice of the Will*.

At the outset of this work, Augustine contends that to solve the problems of the will and of evil that: "Unless you believe, you will not understand" (*lib.arb* 1.2). There are two contradictory propositions, that God created all things, and that God is not the cause of sins. To solve this problem, one should begin with a belief that God is good: "The truest beginning of piety is to think as highly of God as possible" (1.2). We ought to give God the benefit of the doubt. Harrison (2006) argues that this "ought" is meant by Augustine to come from reason. If we do not act in this way, then it will be impossible to understand how God cannot be the cause of evil and yet the cause of all creation. These are two ways of knowing that "differ by their mode of justification. When we believe something, we have it on authority (and it may be good authority), but we do not really 'know' or 'understand' it until we ourselves can give a *ratio* [reason]" (p. 89). Our belief should lead us to want to understand, so that we can give a reasonable account of that belief and be able to defend it.

To understand evil, we must know that it comes about by inordinate desire. When our minds are in control of our desires, our minds will be virtuous, and we act virtuously. God is superior to a virtuous mind, and therefore, cannot cause us to act unjustly. Instead, "only its own will and free choice can make the mind a companion of cupidity" (1.11). Furthermore, in book two, Augustine writes:

> If human beings are good things, and they cannot do right unless they so will, then they ought to have a free will, without which they cannot do right. True, they can also use free will to sin, but we should not therefore believe that God gave them free will so that they would be able to sin. The fact that human beings could not live rightly without it was sufficient reason for God to give it. The very fact that anyone who uses free will to sin is divinely punished shows that free will was given to enable human beings to live rightly, for such punishment would be unjust if free will had been given both for living rightly, and for sinning. After all, how could someone justly be punished for using the will for the very purpose for which it was given? When God punishes a sinner, don't you think he is saying, "Why didn't you use your free will for the purpose for which I gave it to you?" – that is, for living rightly? (2.1).

Humans cannot do right without a will, and so therefore they must have one, or else it would be unjust to punish them if they do wrong. Our freedom lies in the fact that it is subject to the truth, and that truth is God. Augustine quotes from John 8, where it is written that "the truth shall make you free" (2.13).

The movement of our will towards evil is a free one, and it must be free, because it is movement away from God. The source of this movement away from God and towards sin cannot be known, since it is nothing, and Augustine claims that "every defect comes from nothing" (2.20). Nothing, by definition, cannot be known. The one thing that we cannot do voluntarily is find our way back to God, "we cannot pick ourselves up voluntarily as we fell voluntarily" (2.20).

Augustine states that at one time, before the fall into sin, man had the ability to perfectly control his actions, but because he erred in his use of that freedom, God punished him, and thus man is held culpable even when he acts out of ignorance:

> And it is no wonder because of our ignorance we lack the free choice of the will to choose to act rightly, or that even when we do see what is right and will to do it, we cannot do it because of the resistance of carnal habit, which develops almost naturally because of the unruliness of our mortal inheritance. It is indeed the most just penalty for sin that we should lose what we were unwilling to use well, since we could have used it well without the slightest difficulty if only we had willed to do so; thus we who knew what was right but did not do it lost the knowledge of what is right, and we who had the power but not the will to act rightly lost the power even when we had the will (3.18).

It is only by the grace of God that man can now do what is right. The will needs to be liberated by that grace so it can do right. Nonetheless, this does not mean the errors that we have made after the fall are caused by God; they are still our actions, and thus we are still accountable for them.

Even God's foreknowledge of the future does not pose a problem for Augustine. God knows what will happen in the future, and what our wills will do. His foreknowledge does not take away the power of our wills, but rather, makes that power certain, because his foreknowledge never is wrong.

This account makes Augustine seem to fall into the compatibilist category. Our wills are free in a certain respect, in that they can do evil without the influence of God, but they are not free in the respect that they can only do good by God's grace. However, in God's grace we do have freedom. It is not the freedom to do otherwise, but the freedom to do good. Before the fall into sin, man had what might be considered a libertarian free will (Baker, 2003).

I think Augustine's position can be best illustrated by looking at two passages from Galatians 5. In verse one, Paul writes that "It is for freedom that Christ has set us free," and in verse 13: "You, my brothers, were called to be free. But do not use your freedom to indulge the sinful nature, rather, serve one another in love" (NIV). Our freedom comes by God's call, but once that call has been received there is freedom to do good. This freedom seems to include the ability to do evil and reject God's grace, or there would be no need for the exhortation here on how to use that freedom. Perhaps this might mean that from a Christian point of view, only believers have complete free will, at least as long as they are faithful.

We have also seen that God's foreknowledge is compatible with our free choice. God knows the actions we will make before we make them (Doyle, n.d.), but he does not make us make them, at least not our evil actions. However, the fact that everyone does not seem to receive grace is problematic. If good can only be done by the grace of God, and not all people have grace, then how can they be responsible for their actions? Because of this, it can seem as though humans are simply "instruments in the hands of God" (Bonner, 2007). Augustine in the end claims that God is inscrutable to us (Stump, 2001).

Eleonore Stump (2001) proposes that Augustine's problems here exist because he believes there are only two possibilities. The will must either assent to or reject grace. Stump suggests a third possibility. One can assent, one can reject, and one can be quiescent, in other words, do nothing at all. This view she attributes to St. Thomas Aquinas. When the will is quiescent it neither refuses nor accepts grace: "It is thus possible to hold that a human person has it in her power to refuse grace or fail to refuse grace without also holding she has it in her power to form the good act of will that is essential to grace" (p. 140). This forms an explanation that allows us to attribute good will to God's influence, "yet still permits human beings to be the source of their own volitions" (p. 140).

So what then is the relevance of Augustine to this work? If it were merely to show how the problem of free will emerged (or at least transformed from what was spoken of in the pre-Christian world) that would be enough. However, the greater relevance here lies in the idea of belief preceding understanding. The belief that we have free will can guide us to an understanding of that free will that we could not obtain by assuming that it does not exist. Just as for Aristotle we attain the virtues by acting virtuously, Augustine shows that we can come to an understanding of free will by the activity of believing in free will, and then seeking to understand it.

St. Thomas Aquinas

St. Thomas Aquinas' position is not so different from St. Augustine's, but there is a different emphasis that can be seen. Aquinas states that man does have a free will, and it is necessary that he does, for, "otherwise counsels, exhortations, commands, prohibitions, rewards and punishments would be in vain" (*ST*, I, Q. 83, A. 1). Man acts by his judgment when he needs to decide if something should be avoided or pursued. He compares alternative possibilities with the use of reason, and since he may choose one or the other, he has free judgment in these matters. The will is not free in all things, however. A man naturally desires happiness, and cannot do otherwise.

Concerning God as the first cause, Aquinas explains that "it does not necessarily belong to liberty that what is free should be the first cause of itself, as neither for one thing to be cause of another need it be the first cause" (*ST*, I, Q. 83, A. 1). The causal source of our actions lies in our will and our intellect, "human beings have an intellectual form and inclinations of the will resulting from understood forms, and external acts result from these inclinations." (*QDM*, Q. 6). Understood forms are universal, and since such a form deals with particulars which do not include all possibilities,

> inclinations of the will remain indeterminately disposed to many things. For example, if an architect should conceive the form of a house in general, under which different shapes of a house are included, his will can be inclined to build a square house or a round house or a house of another shape (*QDM*, Q. 6, A. 1).

These alternative possibilities seem to arise from what Aquinas calls "phantasms," which exist in the mind: "Even in one man, there may be many phantasms of a stone (perceptions of it from various angles, etc.), but there is only one concept 'stone' (the universal)" (Freeman, 2008, p. 217).

To understand what Aquinas means by forms, it should be noted that there is a difference between forms in matter and forms in the mind. Therefore, reality manifests itself "essentially, in the world, and intentionally, in the individual's mind[9]" (Freeman, 2008, p. 214). Forms are in the world, and are intelligible, because humans have the capability of understanding them. The process by which this understanding is done is as a result of the interaction between the intellect and the world. Freeman quotes Aquinas's formulation of this: *adaequatio rei et intellectus* (p. 214). The truth of a thing is "a relation of conformity between 'a knower intellect' and 'a thing known'" (Di Blasi, n.d.). We know something truthfully if that knowledge represents a real quality of it. This process is facilitated by the phantasms, which "mediate the creation of forms in the mind." By this, we assimilate ourselves to the world. These phantasms are sometimes interpreted as "sense experiences" or "qualia" (Freeman, 2008, p. 214).

What happens as a result is that:

> Knower and reality co-act in a continuous dynamism towards a better understanding of what is, determined not only by what is outside, but by the appetites and goals of the individual that stretches forth the world where she or he abides. The separation means that we are not conscious of ourselves and of our images (and language) when they are in use - and then know things and speak about them. But we may also reconsider our verbal and iconic language and speak about them (p. 218).

The knower participates in this process, and thus the goal towards which he is seeking knowledge may create a different kind of knowledge than it would for another. This second person may have a quite different goal, and this will lead to two different ideas in these individuals, with

[9] See *ST*, I, Q. 85, A. 1

both ideas potentially being true. For example, consider this chapter on the history of free will. It is not my goal to present a comprehensive view of that history, or even a comprehensive view of the theories of free will held by the philosophers discussed here. Instead, I am trying to present what is useful in their theories for understanding the neurodynamics of free will. Even assuming that I have presented the views of these philosophers accurately (and I certainly hope that I have), this abbreviated presentation shows a much different picture than that of someone who chose to concentrate on a particular philosopher's ideas on free will for an entire book. Both presentations can present a truthful view of a philosopher's theories, but they will be very different views.

Aquinas explains this in terms of intentions. For Aquinas, an intention is the power of the mind that leads it to tend towards a particular end. Intentions make us tend towards reality, but are always modified by experience (Di Blasi, n.d.). An intention[10] can be characterized as a likeness or representation of something in the mind (Brower & Brower-Toland, 2008). Another way of thinking of intentionality might be to say that it is the power of the mind that is directing it to tend towards a particular truth. My understanding of Aquinas's theory of free will is intentional in that I am tending towards a particular idea of it, but it becomes modified as I relate what I have in mind to what I continue to read on the subject.

Aquinas sees intention as a part of the will. This is because, as intention does tend towards something, it must be moved in some way. Since it is the will that moves all the powers of the soul to their

[10] Thinking of intentions in terms of motives is the result of its meaning shifting in psychological literature (Freeman, 2008).

ends, intention must be part of the will. (*ST*, I-II, Q. 12, A. 1).

There are two ways in which the powers of intellect and will can be moved: by reference to the subject, which relates to performing an action, and by reference to the object, which relates to that which specifies an action. Regarding the object, the first source of movement is the intellect, and this relates to Aristotle's formal cause. In this respect, the will can be necessarily moved by some things, such as by the desire for the good, but not all things.

Regarding the subject's performing an action, the first source of movement is the will, and this relates to Aristotle's final cause. When performing an action, the will is moved by itself. The will moves itself by deliberation, and because deliberation can lead to contrary conclusions, the result of the will's movement is not a necessary one. In this respect, the will is not at all moved necessarily.

However, something had to first set the will in motion, and this must have been done by God. Since he moves things according to their nature, and the nature of the will is "indeterminately disposed to many things" (*QDM*, Q. 6), this does not happen in a way that necessarily determines the results, though being omnipotent these results would be known by God beforehand.

Stump (2003) points out that this account does not present a complete explanation of what it means to have freedom of the will, however. It explains *liberum arbitrium*[11], and this is only contains part of what Aquinas means when he says that the will is free. To properly understand his views on human freedom, it is necessary to understand "the nature of the intellect and will, the interaction

[11] Which translates to free decision. *De Libero Arbitrio* is also the Latin title of the work from Augustine considered here.

between them, and the emergence of freedom from their interaction" (Stump, 2003, p. 1). She emphasizes the fact that for Aquinas, the will is an "appetite for goodness" (p. 2). The intellect decides what is good, and the will acts because of that decision. In this scenario, the intellect becomes the final cause for the will. Aquinas states: "A thing is said to move in two ways: First, as an end; for instance, when we say that the end moves the agent. In this way the intellect moves the will, because the good understood is the object of the will, and moves it as and end" (*ST*, I, Q. 82, A. 4).

On the other hand, the will can redirect the intellect away from what the intellect thinks is good. This interaction between the two that makes it very difficult to distinguish where one begins and the other ends:

> It is apparent, then, that on Aquinas's account of intellect and will, the will is part of a dynamic feedback system composed primarily of the will and the intellect, but also including the passions. The interaction between will and intellect is so close and the acts of the two powers are so intertwined that Aquinas often finds it difficult to draw the line between them (p. 7).

This kind of dynamic system resembles what will be explained in Chapter Four when we discuss neurodynamics.

Stump explains the process that goes into voluntary human action for Aquinas in five steps:

Step	Intellect	Will
1	Determines that an end is good	Wills that end
2	Determines that an end is currently possible	Intends to act to try to achieve the end
3	Determines means most suitable to achieve ends	Accepts available means
4	Of all possible means, determines best one	Accepts best means
5	Issues command to act	Exerts control over

		something subject to will, for example, an arm, to achieve end

Really, by this account, human freedom does not lie in the will for Aquinas so much as freedom is a property of humans that the will contributes to. In this sense, it does not seem as important to determine whether the will is free so much as it does to say that humans are free.

To understand the nature of human freedom, the interaction between man and God still needs to be made clearer. Aquinas agrees with Augustine that God acts on the will to lead man to salvation, but for Aquinas, man seems to play a greater role than for Augustine. Though free will is not sufficient in achieving salvation, it is by a free act of the will that salvation is achieved: "Hence in him who has the use of reason, God's motion to justice does not take place without a movement of the free will; but He so infuses the gift of justifying grace that at the same time He moves the free-will to accept the gift of grace, in such as are capable of being moved thus" (*ST*, I-II, Q. 113, A. 3).

I think this formulation of Aquinas's thought on free will would make him some kind of compatibilist, though a very moderate one. Most of the time, it seems that we are free to do as the interaction between our wills and intellects decide, but this is not always the case, especially when the issue of divine salvation is considered. Stump argues that Aquinas can possibly be though of as some kind of libertarian, because the causes of all our actions will emanate from the will and intellect. Outside events will be causally influential, but not causally determinative, and this is enough to place Aquinas within the libertarian category, because she argues that for compatibilism, outside events can be causally deterministic and we can still be considered to have free will. The difference in our categorization of Aquinas is likely because of our different definitions of those categories.

Conclusion

After considering the views of these three philosophers, a theme should become apparent. The way we understand the world is as a result of the interaction we take with the world. Virtue is the result of acting virtuously for Aristotle; understanding is a result of the need to justify belief for Augustine; meaning emerges as the result of the assimilative process between the self and the world, and freedom emerges as the result of the dynamic interaction of the will and the intellect for Aquinas. This way of understanding the world foreshadows the way the brain has been discovered to work, as there are many parallels between a neurodynamic understanding of brain functions and these philosophical systems. This is why it is not surprising that a neuroscientist (Walter Freeman) wrote one of the article quoted from in this chapter, which compares Aquinas's thought with our modern, neuroscientific understanding of the mind. Understanding these philosophical ideas can help us to better understand the ideas that neuroscience contributes to the question of free will.

CHAPTER THREE

IDEAS OF FREE WILL FROM THE ENLIGHTENMENT TO THE 20TH CENTURY

This chapter continues the survey started in chapter two. It looks at ideas relevant to a neurodynamic theory of free will held by Rene Descartes, David Hume, Jonathan Edwards, William James, Maurice Merleau-Ponty, and Hans Jonas.

Rene Descartes

Starting with Rene Descartes, the question of free will began to slowly move away from its theological origins. Descartes contributed to this with the way he explained the division of mind and body. For Descartes, human bodies are unthinking objects with no mental or psychological activity. They exist in the physical world. Human minds are immaterial, and they do our thinking and are what feel pain and pleasure. The body and mind are connected (through the pineal gland) so that the mind experiences the sensory input of the body and uses it in its mental processes. Bodies and minds are different because they are "distinct kinds of things or substances." Each of those substances can only experience one kind of property, either a mental property (like constructing a theory of free will), or a physical property (like being six feet, four inches tall). Our identities are bound to our minds, which would be the same in one body or another (Corcoran, 2006, pp. 28-29). This view is known as substance dualism. Generally speaking, this, or any other kind of dualism, is a very unpopular view at present. As Dennett (1984) puts it: "At best, we are grateful to him [Descartes] in the same way we might be grateful to someone who gave us the wrong directions, but whose direction led us on a fascinating misadventure from which we happened to learn a great deal" (p. 18). However, as was noted in Bennett and Hacker (2003), the residue of Cartesian dualism remains in many current neuroscientific theories of human behavior.

Among the consequences of this view was the separation of science and religion, the one belonging to the physical world, and the other to the spiritual world, with any interaction between them occurring through the pineal gland: "Science ceded the soul and the conscious mind to religion and kept the material world for itself" (Schwartz & Begley, 2002, p. 33).

Obviously then, for Descartes, free will must exist entirely in the mind. In fact, free will, is the greatest faculty that humans possess. Free will comes from God, and how we know that we have his "image and likeness (Descartes, 1632, p. 38). Free will "consists solely in the fact that when something is proposed to us by our intellect either to affirm or deny, to pursue or shun, we are moved in such a way that we sense we are determined to it by no external force" (pp. 38-39). So, much as it was for Aquinas, there is an interaction between the intellect and the will as decision making occurs, but it seems that for Descartes, the will is the more powerful of the two, "the will extends further than the intellect" (p. 39). This is the source of error, because the will may choose without having enough information, which is brought to it by the intellect. This can be explained by the fact that the will is infinite (since it most resembles God's powers), but the intellect is finite (Berman, 2004).

What this means is that having the ability to do otherwise, while we do possess it, is actually the product of defect:

> In order to be free I need not be capable of being moved in each direction; on the contrary, the more I am inclined toward one

> direction – either because I clearly understand that there is in it an aspect of the good and the true, or because God has thus disposed the inner recesses of my thought - the more freely do I choose that direction. Nor indeed does divine grace or natural knowledge ever diminish one's freedom; rather, they increase and strengthen it. However, the indifference that I experience when there is no reason moving me more in one direction than in another is the lowest grade of freedom; it is, a certain negation in knowledge. Were I always to see clearly what is true and good, I would never deliberate about what is to be judged or chosen. In that event, although I would be entirely free, I could never be indifferent (p. 39).

Furthermore, much like Aquinas and Augustine, Descartes points out that rather than diminishing free will, God's influence increases our freedom.

Descartes is included here because of the influence of his conception of dualism, though such conception is largely discredited, the remnants of its influence remain. Evan Thompson argues that Descartes did make a positive advance in his recognition of a consciousness that that had a "first-person, phenomenological orientation missing from the concept of soul[12]" (2004, p. 228). It may also be the case that this conception was the first formation of the idea of consciousness. Thompson quotes Gareth Matthews, who claims that Descartes created a new concept, "which includes thinking plus the 'inner part' (so to speak) of sensation and perception" (Matthews, 1991, p. 68).

However, because the body is simply a mechanism for Descartes, and completely different from the mind, he fails to explain how they "form a distinct unity in each individual human being" (p. 229). This would seem to be the primary flaw in Descartes theory.

Given that Descartes does believe that we have the ability to do

[12] By this, Thompson can be taken to mean previous concepts of the soul, as Descartes has such a concept as well.

otherwise, it would seem that he could be classified as a libertarian. Despite this, it is important that he points out that this represents a lesser form of free will. The will at its most perfect will know precisely what to do, and there can only be one course of action for it, the right way.

David Hume

As Hume understands the question of will, every decision we make arises necessarily from our experience and our character. If an outside observer had perfect knowledge of our past actions and disposition, he would easily be able to predict the decisions we would make (Hume, 1739). Though we may take actions intended to show our freedom, these actions merely demonstrate the necessity we feel to show we are free. Hume mentions that theology has aided in the deception that leads people to think that they have free will, but that free will is not necessary to religion. In point of fact, it is the necessity of actions that allows praise or blame to be attributed to them: "Actions are by their very nature temporal and perishing; and where they proceed not from some cause in the characters and disposition of the person, who perform'd them, they infix themselves not upon him, and can neither redound to his honour if good, nor his infamy, if evil" (p. 411). If actions do not happen as a result of this internal necessity, we cannot be blamed for them: "But the person is not answerable for them [his actions]; and as they proceeded from nothing in him, that is durable and constant, and leave nothing of that nature behind them, it is impossible he can, upon their account, become the object of punishment or vengeance" (Hume, 1748, p. 74).

The necessity of actions arises "entirely from the uniformity, observable in the operations of nature; where similar objects are constantly conjoined together, and the mind is determined by custom to infer the one from the appearance of the other" (p. 63). This is how necessity ought to be defined, as a constant conjunction of events, and we should not seek to find any deeper level of necessity (p. 71). When events arise that seem to be without cause, or have uncertain cause, this is a result of imperfect information. What is seemingly uncertain is the result of "the secret opposition of contrary causes" (p. 67). This may be very difficult to determine. Though, as noted, a perfect knowledge of our past actions and dispositions can lead an observer to perfectly predict all our actions, such perfect knowledge is impossible. The variable behavior of people should be considered "in the same manner as the winds, rain, clouds, and other variations of the weather are supposed to be governed by steady principles; though not easily discoverable by human sagacity and enquiry" (p. 67).

The fact that actions happen by necessity must be the case, because otherwise the effort of constructing a political science, or system of morality would be a worthless effort. If we don't have some idea of how people will behave, then on what basis will we be able to offer guidance and prohibitions.

Despite this, people do tend to think that there is a difference between acts that occur in nature and acts that occur in their own minds. Acts in nature seem more determined than acts in our own minds, which arise from our thoughts and reasoning. Hume thinks they are the same, for the reason mentioned above, that necessity should be thought as nothing more than a conjunction of events. To look for more in either case is to go beyond what we can discover. Because of this rather light definition of necessity, Hume claims that liberty can be found in necessity: "By *liberty*, then, we can only mean *a power of acting or not acting, according to the determinations of the will*" (p. 72).

Furthermore, contrary to Aristotle, St. Augustine, and St. Aquinas, Hume believes that reason is governed by the passions, and subordinate to them (1739, p. 415). Philosophers have gone wrong in the past because they supposed the direction of the will was determined entirely by only one of the possible passions within a person, the passion to be reasonable. A passion may be stronger than another at one time, and the same passion weaker than another at a different time. It must be recognized that it is different passions that determine actions at different times. What is said to be strength of mind comes from the dominance of calm passions over violent ones (p. 418).

Hume's position here is a compatibilist one. Paul Russell (2007) notes that Hume's position is one of the most influential in favor of a compatibilist position. However, his idea of necessity guiding our actions seems to be a rather weak distinction, especially as he defines necessity. This forms a rather weak foundation for defining liberty:

> In general, on Hume's account, in so far as we have any knowledge of “causal connections” they are always *subjective* (i.e. in the mind of the observer) and never *objective* (i.e. discovered in the objects themselves). The fact that we suppose that the objects themselves, *qua* cause, possess a power or efficacy by which they are connected with their effects is entirely due to the influence of experience or “custom” on the human mind; there are no (known) corresponding features in these objects (see, e.g., T, 1.3.14.14-25/162-7; cp. EU, 8.26-30/74-9). Given this, Hume's account of liberty of spontaneity seems vulnerable to the very same objections that the antilibertarian argument raised against the libertarian position. That is, given the necessity argument, we still have no reason to believe that such agents are (really) connected with their actions, although our subjective

experience may make us *feel* that they are (Russell, 2007).

Russell points out that Hume likely formulates his theory of the will in this way because of his ideas concerning the passions and moral evaluation. It is by our feelings that we come to regard an individual as responsible rather than by our judgments. Reason is insufficient to solve philosophical problems, and feeling is essential in allowing us to do so:

> Necessity is psychologically essential to ascriptions of responsibility, because in the absence of the relevant regularities and inferences, the regular mechanism which produces our moral sentiments would simply fail to function. Similarly, liberty of spontaneity is (psychologically) essential to responsibility for action because it is only in these circumstances (i.e. in which we discover constant conjunctions between motives and actions) that it is possible for us to draw the specific kinds of inferences required to generate the moral sentiments. If conduct is produced by external violence, no moral sentiment is aroused and, thus, we do not (as a matter of fact) hold the person responsible.

It is in stressing the importance of emotion or feelings in our decision making that Hume's thought has the most relevance for this work. Neuroscientific research (Damasio 1999) seems to show that those who have impaired functioning in the ventromedial prefrontal cortex (which is a part of the brain that deals with processing emotions) have trouble making decisions, and when they do choose, they tend to choose poorly. Hume's recognition of this fact is what guided the construction of his theory of free will, and the value of emotions in considering how the will can be guided is important.

Jonathan Edwards

Jonathan Edwards was an American protestant theologian of the Reformed tradition who lived in the 18th century. He wrote a book, *Freedom of the Will*, against the Arminian theology. Somewhat like the

Pelagians, the Arminians believed that man plays a significant role in coming to faith. Edwards conflates these two beliefs, and defines them as believing that the will must be self determining; that the mind must be indifferent or in equilibrium before deciding to act; and that the will must be contingent, or opposed to any kind of necessity (Edwards, 1754). Contrary to this, Edwards defines the will as the act of choosing. The will is moved when, "there is some preponderance in the mind, one way rather than another; and the soul had rather have or do one thing than another, or than not to have or do that thing; and where there is absolutely no preferring or choosing, but a perfect, continuing equilibrium, there is no volition" (p. 12). The will is moved when a motive is stronger than any other, and determines how the will chooses.

Edwards believes that we have freedom, but it does not lie in the will. The will is not an agent, but rather a power of choosing. Freedom and volition are properly attributed to humans, as the possessors of will. Edwards attributes this idea to John Locke in his *Essay on the Human Understanding*. We can be considered to have liberty if we are not constrained from acting as our will determines. It is our selves that have freedom, and these selves exist as a result of memory. We only have the ideas that we are now thinking of now as a result of past ideas and sensations. This view of Edwards would seem to be one that might meet with approval by Bennett and Hacker, as it avoids the mereological fallacy they warn against by attributing the power of freedom to the whole human person rather than one of our faculties.

Just as ideas arise from our past experience; motives, and the subsequent act of will, arise from past motives. Instead, volition "is caused by previous motive and inducement, is not caused by the Will exercising a sovereign power over itself, to determine, cause, and excite Volitions in itself" (p. 83). We cannot be absolutely indifferent and still choose. Decisions come about as a result of several steps, and none of them are indifferent. Edwards uses the example of touching a square on a chess board to explain how the steps of making a decision are not indifferent, even if one tries to choose by chance. If one is going to touch a square of the chess board, first comes a general determination to touch a square, then comes a general determination to touch a square at random, and finally comes the particular determination to touch one of the squares: "Now it is apparent that in none of these several steps does the mind proceed in absolute indifferent, but in each of them is influenced by a preponderant inducement" (p. 62). Even the second step, of touching a square at random involves a preference. One chooses this because "it appears at the time a convenient and requisite expedient in order to fulfill the general purpose" (p. 62). To say that we are indifferent when we choose is to say that we have a preference while at the same time we have no preference (p. 68). This process of decision making will present an important criticism to the free will experiments of Benjamin Libet discussed in chapter five.

Concerning the cause of volitions, Edwards writes that the same volition can be both a cause and an effect, though this must be in different respects. This is because the soul may be both active and passive "in the same thing in different respects; active with relation to one thing, and passive with relation to another" (p. 191). In order to understand this, one must see the sense in which action and passivity are contrary relations,

> contrary relations may belong to the same things, at the same time, with respect to different things. So, to suppose that there are acts of the soul by which a man voluntarily moves, and acts upon objects, and produces effects which yet themselves are effects of something else, and wherein the soul itself is the object of something acting upon, and influencing that, does not at all confound action and passion" (p. 192).

What Edwards might mean here is that action is somewhat circularly causal, or at least that causality can be very confusing. For example, let us say that I decide to turn off the computer and take a nap. Perhaps one can say that this action is passive, in that I am no longer seeing the computer screen, and active, in that I have pushed a button to turn it off. My action of turning off the computer leads to the effect of no image on the screen. This action is probably the effect of being tired. The absence of the image on the screen makes it so I am more likely to go to sleep. The cause was the desire to go to sleep. The effect of turning off the computer affects the cause in that the desire to go to sleep should now be stronger, as there are fewer distractions to keep me awake. In other words, the effect that was part of the cause has now contributed to the later existing cause. The point Edwards wants to make is that it is not essential to internal action for the agent to be self-determined and the will to be the cause of it.

Part of the reason that causality is confusing is the common idea that external actions are the result of the will causing the motion of our bodies and the actions that result from that motion. This does not mean that internal actions work in the same way: "Hence some metaphysicians have been led unwarily, but exceeding absurdly, to suppose the same concerning volition itself, that that also must be determined by the Will" (p. 192). This is an understanding of causality that Eric Piaget (1930) says is produced when one is a young child, where we see actions coming about by a simple relation of cause and effect. It is as a result of this unsophisticated understanding of causality, which tends to persist throughout one's life, that the difficulty in understanding the more complex causal mechanisms of the mind exist (Freeman, 1999).

Essential to understanding Edwards' conception of human freedom is his distinction between natural and moral necessity. The difference between them is this: "By *natural necessity*, as applied to men, I mean such Necessity as men are under through the force of natural causes; and distinguished from what are called moral causes, such as habits and dispositions of the heart, and moral motives and inducements" (p. 27). Natural necessity includes such things as mathematical facts, that we feel pain when we are injured, and scientific facts like gravity.

When determining whether an action is to be praised or blamed, it matters whether it was done under natural necessity or moral necessity. Actions committed by natural necessity or not worthy or praise or blame, but actions committed by moral necessity are. In point of fact, the more an action was done by moral necessity, the more it should be worthy of praise: "And the stronger the inclination [to do good] is, and the nearer it approaches to necessity in this respect; or to impossibility of neglecting the virtuous act, or of doing a vicious one; still the more virtuous, and worthy of higher commendation" (p. 202). On the contrary, if good inclinations have little to do with one's actions, then they should not be considered as highly.

To illustrate the difference between these two necessities, Edwards offers the example of two different men in prison. In the case of the first man, he is told he can be free if he goes and makes a humble apology to his ruler. Though he is willing to do so, he cannot, because he is locked in his cell and cannot get out. This is what natural necessity is like. In the case of the second man, the same offer is made, but the ruler comes and opens the cell, making his offer of forgiveness in person. The prisoner refuses because of his strong pride and ill-natured disposition. This is what moral necessity is like. Edwards argues that though there are these two necessities, and that the second man's "evil dispositions may be as strong and immoveable as the bars of a castle" (p. 205) the second man should be blamed for his actions while the first man should not.

Edwards' view of necessity not denying freedom (at least not moral necessity) places him fairly clearly in the compatibilist category. This distinction between moral necessity and natural necessity adds nuance to the typical compatibilist account. The primary relevance of Edwards to this project is twofold. First, he offers further arguments against the idea that for there to be human freedom, the will must be uncaused. Second, he offers a view of causality that bears some resemblance to Freeman's theory of circular causality. Also interesting is the fact that he sees freedom as existing on the level of the entire individual, rather than in the will itself.

William James

At the outset of this work, William James was quoted as saying that if we are free, the first act of our freedom should be to assert that freedom. For James, this freedom exists as a kind of indeterminism. He sees the universe as belonging "to a plurality of semi-independent forces, each one of which may help or hinder, and be helped or hindered by the operations of the rest" (James, 1977, p. 605). The only way this is possible is in an indeterministic way (which James also defines as chance), where both right and wrong ways of acting are possible. This is the only that we can feel regret, because we know we have acted badly when we could have done something good. Therefore, James' theory of free will is based on the judgment of regret, which "represents that world as vulnerable, and liable to be injured by certain of its parts if they act wrong... It gives us a pluralistic, restless universe, in which no single point of view can ever take in the whole scene" (p. 606).

The way the will works is that action begins with objects and thoughts of objects. The course of an action taken is then modified by the pleasures and pains it brings. In other words, "*what holds attention determines action*" (p. 708). This can also be thought of as interest. The will, therefore, is a relationship between the mind and the ideas it has: "T*he essential achievement of the will, in short, when it is most 'voluntary,' is to attend to a difficult object and hold it fast before the mind... Effort of attention is thus the essential phenomenon of the will*" (p. 709). As we consider an idea, and hold it in our minds with focused attention, eventually it will come to "call up its own congeners and associates and ends by changing the disposition of the man's consciousness altogether" (p. 711).

For the will to be considered free, this focused attention must actually matter. It must make a difference in the complex process that occurs as we make our decisions: "The effort appears, in other words, not as a fixed reaction on our part which the object that resists us necessarily calls forth, but as what the mathematicians call an 'independent variable' amongst the fixed data of the case, our motives, character, etc." (p. 713). This is similar to the criterion that John Searle requires for free will. Our conscious deliberation, or focused attention, has to be more than simply an illusion or an interpretation of our actions after we have made them.

Doyle (n.d.) argues that the theory of free will James outlines is what can be called a two-stage model. The chance of random alternatives leads to the choice which changes an uncertain future into a fixed past. What we have with this is first the freedom of alternatives, followed by the willing action of choosing one of them. In the first step, we are free, in the second step, we have what Doyle calls "adequate determinism," where the cause and effect of our actions can essentially be explained.

James' ideas are included here because of this idea of chance, or indeterminate alternatives being possible. This is an idea similar to what has formulated from attempting to explain free will in terms of quantum physics (this idea will be considered in chapter five). His two-stage model of free will provides an example of how such a theory can be constructed.

James is also important because of his explanation of free will existing in the ability to focus our attention as we make decisions and have that attention matter. This is also an idea that has been utilized by those who try to use quantum physics to explain free will. It can be seen particularly in the work of Jeffrey Schwartz (1999, 2002) and Henry Stapp (1999).

Maurice Merleau-Ponty

Maurice Merleau-Ponty was a 20th century philosopher who is usually classified as a phenomenologist. Merleau-Ponty (1962) explains that freedom exists, but it is a freedom that exists in relationship with the world. It is impossible to determine with precision what part of our actions is determined by the situation, and what part determined by freedom. There is never absolute determinism and never absolute freedom to choose: "I am never a thing, and never bare consciousness. In fact, even our own pieces of initiative, even the situations which we have chosen, bear us on, once they have been entered upon by virtue of a state rather than an act" (p. 527).

Merleau-Ponty uses the example of a prisoner who has been captured and is being tortured to explain how this works. The prisoner does not want to reveal the information he is being tortured for because he feels loyalty to his comrades and the cause they are fighting for. He may also have imagined being in this situation, and promised himself that he will not break under the pressure:

> These motives do not cancel out freedom, but at least ensure that it does not go unbuttoned in being... in so far as he has committed himself to this action... it is because the historical situation, the comrades, the world around him seemed to him to expect this conduct from him... we choose the world and the world chooses us" (p. 527).

Our involvement with the world places us in "an inextricable tangle" (p. 528) which means we cannot say where determinism by the outside world begins and our internal freedom ends. Far from limiting our freedom, however, it is this relationship with the world that makes us free:

> The fact remains that I am free, not in spite of, or on the hither side of, these motivations, but by means of them. For this significant life, this certain significance of nature and history which I am, does not limit my access to the world, but on the contrary is my means of entering into communication with it. It is by being unrestrictedly and unreservedly what I am at present that I have a chance of moving forward; it is by living my time

> that I am able to understand other times, by plunging into the present and the world, by taking on deliberately what I am fortuitously, by willing what I will an doing what I do, that I can go further. I can miss being free only if I try to bypass my natural and social situation by refusing to take it up, the first place, instead of assuming it in order to join up with the natural and human world (pp. 529-530).

Our actions occur in the world by the process that Merleau-Ponty calls the "intentional arc," which "projects round about us our future, our human setting, our physical, ideological and moral situation, or rather which results in our being situated in all these respects. It is this intentional arc which brings about the unity of the senses, of intelligence, of sensibility and motility" (p. 157). The intentional arc is a dialectical relationship between our actions and the world. It is meant to "capture the idea that, in learning, past experience is projected into the perceptual world of the learner and shows up as affordances or solicitations to further action" (Dreyfus, 2005, p. 132).

This idea of the intentional arc has been taken up by neurodynamic theory. The intentional arc means to act in "repeated cycles of action and perception" (Freeman, 1999, p. 120) where we focus our attention to "achieve optimal assimilation of the self to an object. The self adapts to that object and learns about it by reshaping the body, and also reshaping or repositioning the object" (p. 121). Our actions occur as a result of a "disequilibrium between ourselves and the world." In neurodynamic terms, this disequilibrium is "an endogenous instability that puts the brain onto an itinerant trajectory, that is, a pathway through a chain of preferred states, which are learned basins of attractions" (p. 121). Precisely what that all means will hopefully become clearer in the next chapter.

In this process, Freeman is reminded of St. Thomas' explanation of assimilation, where "the imagination [phantasms]... allows us to generalize and abstract the internal structures with which we act and understand" (p. 120). This focusing of attention also calls to mind William James' definition of free will. If effort of attention is the "essential phenomenon of free will," then this theory would seem to allow for such a possibility, and, at least in part, explain how the process might work.

Hans Jonas

Hans Jonas was a 20th century philosopher who was a student of Martin Heidegger. He tried to integrate a scientific account of life into his thought (Levy, 2002). Jonas (1966) sees human freedom as more of a quality of our condition as living organisms rather than as a property of the will:

> "Freedom" must denote an objectively discernible mode of being, i.e., a manner of executing existence, distinctive of the organic *per se* and thus shared by all members but by no nonmembers of the class: an ontologically descriptive term which can apply to mere physical evidence at first" (p. 3).

In Jonas' thought, the contradiction between freedom and necessity and autonomy and dependence can be traced back to the most primitive forms of life. The dynamics of organisms are more than mere static existence, and the purposive dynamics that propel an organism's development, or its becoming actual, "must be included in the concept of physical causality" (p. 2). The mind is "prefigured in the organic from the beginning," speaking of its most primitive form and therefore so is freedom (p. 3). Each level of development has brought more freedom, but it has also brought us more peril. In each higher level of development there is still a dependency on the lower levels, and everything of the lower is retained. This development began with the separation of the organism from the world, by "assuming a position of hazardous independence from the very matter which is yet indispensable to its being" (p. 4). The contradiction in relationships like that between freedom and necessity exists because our lives are dependent on conditions in the outside world over which we have no control:

> Thus dependent on the propitiousness or unpropitiousness of outer reality, it is exposed to the world from which it has seceded, and by means of which it must yet maintain itself. Opposing in its internal autonomy the entropy rule of general causality, it is yet subject to it. Emancipated from the identity with matter, it is yet in need of it; free, yet under the whip of necessity; isolated, yet in indispensable contact; seeking contact, yet in danger of being destroyed by it, and threated no less by its want (p. 5).

Our freedom in the world exists because of the process of metabolism. By the process of metabolizing, we sustain ourselves by having foreign material pass through our systems. A metabolizing being is "never the same materially and yet persists as its same self, *by* not remaining the same matter" (p. 76). This is a way in which we are separated from the world and yet dependent on it. This creates a double natured relationship between the organism and the materials it uses from the world: "Dependent on their availability as materials, it is independent of their sameness as these; its own, functional identity, passingly incorporating theirs, is of a different order. In a word, the organic form stands in a dialectical relation of *needful freedom* to matter" (p. 80).

We understand the world in which we have become, in a sense, separated, or at least somewhat autonomous from, by acting in it: "The experience of living force, one's own namely, in the acting of the body, is the experiential basis for the abstractions of the general concepts of action and causation" (p. 23). From this action we experience a dynamic image of the world. This dynamic process modifies us and allows us to achieve what Jonas calls transcendence. Thompson (2007) notes that Jonas is using an idea from Heidegger, who saw transcendence as "the always-already-surpassing or being-projected-beyond oneself in the world that is proper to human existence (*Dasein),*" but expands it by extending the idea "down to the very roots of life" (p. 157). In a way this seem to be similar to what Merleau-Ponty meant when he said that our actions emerge into the world as a result of disequilibrium.

The uniquely human aspect of our freedom comes in our ability to manipulate images and ideas with the imagination. We can hold an idea in our minds and modify if we so choose. The example Jonas gives is of making a drawing. When we decide to draw, we can reproduce an accurate likeness of what we see, but this is only one possibility:

> The first intentionally drawn line unlocks that dimension of freedom in which faithfulness to the original, or to any model, is only one decision: transcending actual reality as a whole, it offers its range of infinite variation as a realm of the *possible*, to be made true by man at his choice (p. 172).

This capacity, which shows that we can "wrestle form from fact" is enough to suffice as "evidence of human freedom" (p. 174). In a way this seems to resemble Aquinas's idea of how we understand the world. Just as for St. Thomas we understand the world on the basis of the forms of the world being modified by the phantasms in our mind to become representational mental forms (which is not the same as the form in the world), for Jonas there is a process occurring where the dynamic interaction of world and our understanding can create, by imagination[13], new forms. Another example Jonas uses to illustrate this process is

[13] It is also possible that Aquinas's phantasms can be thought of as the faculty of imagination (Stump, 2003)

Adam's naming of the animals in the book of Genesis. He sees this act of namely as the first truly human act, as a new meaning is created by Adam's labeling of the animals. Giving of names is like image-making, and:

> Image-making each time re-enacts the creative act that is hidden in the residual name: the symbolic making-over-again of the world. It exhibits what the use of names takes for granted: the availability of *eidos* [form] as an identity over and above the particulars, for human apprehension, imagination, and discourse (p. 173).

The primary importance of all this is to further illustrate the idea that it is by acting into the world that we create meaning. This theory can be traced back to Aristotle, Augustine, and Aquinas. Jonas adds to this understanding by providing a more scientific basis for understanding the nature of our dynamic relationship with the world.

Conclusion

A few key points emerge as a result of this survey of the history of free will. The first is that free will probably does not need to consist in the ability to do otherwise. At best, the ability to do otherwise is, as Descartes put it, a defect. The second point is that our freedom exists as a result of our interaction with the world and emerges from that interaction. This is a dynamic process, and as we will see in the next chapter, just as this relationship exists between the self and the world, it also exists in the way that our brains function. We can also see that perhaps it is better to understand freedom as a property of the whole person rather than existing in the will.

CHAPTER FOUR

A NEURODYNAMIC THEORY OF FREE WILL

From a neurodynamic perspective, the reason the free will question is difficult is that:

> A voluntary action is clearly intentional, but it must proceed with awareness. Yet when a person has chosen an action, he or she may not be the first to become aware of the choice. Instead, observers of the action may be the first to become aware of it and attribute agency to the actor. Both the observers and the actor may understand the self-organizing dynamics of brain as the source of intentional actions, but, even if they do not grasp this, they attribute the action correctly to the actor, although in all modesty the actor may demur, and rightfully so, that the observed action may not have been what he or she consciously intended (Freeman, 1999, p. 137).

Determining causality can often be quite confusing. Daniel Wegner (1999) uses this as evidence that our conscious wills are illusory. We are so bad at determining our causality because we are deluded in attributing causality to our consciousness. Really our actions are determined by the mechanical processes that occur during the functioning of our brains (this idea will be discussed in greater detail in chapter five).

The idea that there is a question of free will versus universal determinism reflects a mistaken belief about how causality works. However, in the neurodynamic theory explained in his book *How Brains Make Up Their Minds* (1999), Walter Freeman claims that the dynamics of the brain do support the "biological capacity to choose," and that the power to choose is "an essential and unalienable property of human life" (pp. 4-5). He lays out three conditions that must be met to show that choice exists. First, we need to comprehend how the brain works to construct the options we choose from; second, we need to explain what is happening in the brain at the moment of choice; and third, we must explain the mechanism of awareness and how linked sequences of awareness form our consciousness.

Essential to this process is intention, an idea that Freeman has borrowed from St. Thomas Aquinas, and which is explained here in chapter two. Freeman defines intention as "the directing of an action towards some future goal that is defined and chosen by the actor. It differs from a motive, which is the reason and explanation of the action, and from a desire, which is the awareness and experience stemming from the intent" (p. 8).

There are three main properties of intentionality. The first is that our brains and bodies are unified. Our actions into the world and our perceptions of it are "unified across all our senses at rates faster than we can perceive" (p. 18).

The second is that the whole of our life's experience is important for every action we take. For example, if I were to take a break from writing this and play the piano, not only would the decades I've practiced be important, but also every other experience I've had, be it the deadened attention span I have from hours a day spent staring at the television to the general physical condition of my body.

The third property is that intentionality has a purpose. As Aristotle wrote so many years ago, we are teleological creatures. There is an end, or goal, to our actions, whether we are aware of that end or not. The fact is that we need not be aware of it. Consciousness is not necessary for there to be intentionality, "perception is a continuous and mostly unconscious process that is sampled and marked intermittently by awareness, and what we remember are the samples, not the process" (p. 18). When I play the piano, if I have practiced a song enough times, I do not need to be entirely conscious of what I am doing. In fact, I will play the piece better if I am not focused on where my fingers need to be and what the notes are. The habits that have formed from the time I spent practicing will be a much better guide, and work much faster, then if I had to be conscious of every part of the performance.

It is by intention that we are able to form meaning. Much as it is described in the philosophical systems of meaning contained in the past two chapters, meaning comes about by our acting into the world. We act into the world, and then assimilate ourselves to what we perceive as a result of that act. This is a unidirectional process, because, as Aquinas said, the forms of matter are incompatible with the forms of the intellect: "The world is infinitely beyond our limited powers of creating forms, and its particulars are inaccessible and useless to us... Our unidirectional perceptual system is our best asset in matching our limited capacities to the infinite world" (p. 28).

In the brain, the process of meaning arises due to the interaction of our neurons. The interaction of individual neurons with each other is important, but so too is the interaction of neuron populations, which are made up of frequently interacting neurons. These neuron populations have specific properties that "cannot be reduced to the level of the single neuron" (p. 21). It is the neural populations of the limbic system that are essential to the formation of intention:

> In each hemisphere, the sensory cortex receives input, the motor cortex implements action, and the hippocampus provides multisensory integration and orientation in space and time. Each part has reciprocal connections to the others. The entire hemisphere constructs goal states through its interactive neural activity patterns (p. 33).

To understand how this works, it is necessary to go into more detail explaining neurons and neural populations. The most basic element of the mind is the neuron. Each neuron can receive multiple signals from other neurons through its dendrites, but it can only output one signal at a time through its axons. When the axon of one neuron connects to the dendrite of another, a synapse is formed. One neuron will be connected to many others, and their connections can resemble a treelike structure. Every time something new is learned, new synaptic connections are made.

The activity of neurons can take several forms. A neuron may be at rest, it may be exciting or inhibiting activity in other neurons, or it may be in the process of changing, because it is adapting to a new experience. These possible forms are referred to as the "state space" of the neuron by Freeman. Though neurons do change, certain tendencies may be seen in a particular neuron and this is referred to as an "itinerary."

The neuron will receive pulses from other neurons through its dendrites, and translate those pulses into waves that are outputted by its axons. Each neuron will have at least ten thousand input and output connections, and: "The dense network of axons, dendrites, synapses, and interconnected neurons, along with the blood vessels and the branches of supporting cells called glia, form the tissue we call neuropil... This is what the gray matter of the brain and spinal cord is made of" (p. 47).

Close groupings of neurons connected together are referred to as neural populations. These neural populations somewhat resemble a neighborhood. They are also sometimes called a "cortical column" (p. 50). These neighborhoods receive pulses and waves at "collective trigger zones", but the relationship between pulses and waves is different in communication between populations than it is between single neurons. The communication between individual neurons is a linear relationship, and can change rapidly, but the communication at the population level is nonlinear and does not change as quickly. This level of activity in the brain is said to be mesoscopic, and is the first step by which patterns of activity are created. This mesoscopic level is in between the microscopic level of individual neurons and the macroscopic level of global brain states.

At the mesoscopic level of neural populations, the activity of individual neurons is determined by the community and not individually, though this is only a portion of a neuron's overall activity. Freeman points out that this process in someway resembles the way people function in societies. Just as most of the time individuals will be working for themselves and dealing with their own concerns, some times they will also participate in the activity of their community. In the same way, part of an individual neuron is contributing to the overall activity of the community, but the majority of its activity is autonomous (pp. 59-60).

There are four requirements for mesoscopic populations to form: there must be "many semiautonomous, independent elements" (the neuron); there must be weak interactions with many other neurons; there must be a nonlinear relationship between the input and output these neurons receive; and there "must be a huge source of matter and energy" (p. 52). This is true of not just brains, but all complex systems, like "ecological systems, weather systems such as hurricanes and tornadoes, and even galaxies" (p. 52).

As populations of neurons in the brain interact with each other, stable operating points, called point attractors are formed: "They are essential for the oscillations of excitatory and inhibitory neural populations that are synaptically coupled in negative feedback loops... The positive and negative feedback loops of neural populations are responsible for this ability to create behavior freshly with each new movement," and is how "behavior is generated in the brain," rather than externally (p. 62). This oscillation of excitatory and inhibitory neural populations occurs because of sensory or electrical stimulus. Normally, these populations tend towards stability, and even with the introduction of stimulus, eventually these populations tend to move back to a steady level of activity. This process is "regulated by neurochemicals secreted by the brain stem under limbic control" (p. 55).

The neurons of the brain receive stimuli from its sensory receptors, such as skin, eyes, ears, and nose. These receptors are much like other neurons, only their dendrite ends are specialized to the input they are intended to receive (p. 66). For example, in the case of scent, "the brain generalizes by forming a mesoscopic pattern of activity that includes the small number of neurons bringing the activity into the brain, a lot more neurons besides." It seems to be the case that every neuron involved with the reception of information from the sensory receptors is "involved with the perception of every odor." In the case of scent, this would be the bulbar neurons (p. 70).

The information received from the sensory receptors is ultimately sent to the brain via a wave that is distinguished by its variance in amplitude (amplitude modulation or AM). This amplitudinal pattern is determined not by the stimulus itself, but rather by the synaptic connections sending the information. This is judged to be so because AM levels differ consistently and recognizably in individuals. These AM patterns are "dependent on context, history, and significance - in a word, meaning" (p. 77).

With scent, Freeman has found through animal experiments that with new sensory inputs AM wave patterns are not initially sent out. "This is a state of 'I don't know what it is, but it may be important." If this does prove to be the case a new AM pattern is formed, and all other patterns already known are slightly changed (p. 79). In this process, the neurons in the sensory receptor make themselves "similar to the form of a stimulus in the world, and so perform the process of assimilation" (p. 81). In each instance of a new input, the brain creates "an individualized pattern that is compatible with the history and goals of the organism... The construction revealed by the AM patterns result from what Aquinas called the imagination [phantasms], and what I call the nonlinear dynamics of neuron populations" (p. 82).

An AM wave that is sent from a sensory input will be slightly different each time it is sent, even when that sensory neuron has received the same input. This introduces chaos to the system. Chaos is useful for the brain because this keeps it somewhat flexible: "Chaos generates the disorder needed for creating new trials in trial-and-error learning... It gives an optimal balance between flexibility and stability, adaptiveness and dependability" (p. 87). One might also state that it is this chaos that makes it difficult to find any kind of true linear causality in the human will. At the very least, on the basis of this formulation, the cause for our action would be in ourselves rather than external to us, and this is what Freeman argues.

One should not conclude from this that the external world is irrelevant to our development. Freeman appears to take an approach along the lines of what Matt Ridley (2003) explains in *Nature via Nuture*, where both genetics and environment play a role in one's development and thus his or her decisions: "Each of us is born with genetic and cytoplasmic endowments that establish some general limits to the directions and extent of growth in striving for the wholeness of intentionality" (Freeman, 1999, p. 115) The meaning we create may come from ourselves, but we cannot create any meaning we want. Our choices are limited by the content of the world. The world is as we perceive it, but only to a point. As we saw in Aquinas, the forms in the intellect were not the same as the material forms that they represent, but there had to be a material form to be represented.

This explanation still does not explain how we make our decisions.

In this process, goals emerge. Once a goal has emerged, then we act, and observe the results of our actions, constructing meaning in the process. Finally, our brains are modified by what we have learned. As these states occur, they are accompanied by "dynamic processes in the brain and body that prepare the body for forthcoming action and enable it to carry them out." These processes are experienced as emotions, which often have a physical manifestation, such as sweating during a stressful situation (pp. 91-92). Freeman's view of emotions also resembles that of David Hume, in that there is no contrast between reason and emotion. Both rational actions, which are traditionally thought to be reasonable ones, and irrational ones, which are thought to be the more emotional ones, are done by emotion. They are simply different kinds of emotions: "In my view, both kinds of actions are emotional and intentional, in that both emerge from within the individual and are directed to short- or long-term goals, but clearly they are different" (p. 95).

These emotions are essential to intentionality, but subordinate to "intentionality, perception, and the construction of meaning" (p. 92). The degree of emotion we experience can be attributed to the "intensities of chaotic fluctuations in the populations of the forebrain, which are regulated by the neuromodulators secreted by brainstem nuclei under limbic control. We can regard reason as expressing a high degree of assimilation to the world" (p. 136). To elaborate further, Freeman states:

> The full meaning of a stimulus for the organism emerges... only at the global level. Meaning depends on the entire history of an animal, which is embedded in the neuropil by synaptic modifications during learning. The meaning is shaped by present context, which is provided by the senses of the body and the world under limbic control, and it includes the states of emotion and

> affect that are implemented by the neuromodulatory nuclei of the brainstem, also under limbic control, in preparation for the execution of intended actions, particularly social interactions that require nonverbal signaling (pg. 114).

As described in the last chapter, this process is like Merleau-Ponty's intentional arc. What must be stressed again in all of this is that these decisions are being made by the individual and are not the result of external necessity, at least not necessarily so.

If one accepts this conception of how the mind works, causation within the mind is not explained, and cannot be explained by linear causality. It is fairly clear how decisions happen, but not why. Freeman makes the case that how causation is explained depends on how it is defined. The first meaning of causality mentioned is that of making, or movement, which in Aristotelian terms is the efficient cause of an action. This is what linear causality can be explained as, and would answer the question of why. The second meaning for cause is that which explains, rationalizes, or blames. This is known as the formal cause of an action. This is how circular causality can be thought of. A third meaning of cause is that we know causes as the result of "constant conjunctions of events in sequence" (p. 126). The comprehension of these events is a property of the mind, and not imposed by the world external to oneself. This is similar to how Hume conceived of causation.

Freeman argues that the first of these meanings is the most difficult to use, especially in attempting to explain the relationship between individual neurons and the populations that they are a part of. It is the neurons themselves that are creating these neural populations, but they are simultaneously being constrained by them: "Simultaneity violates the requirement that effects must follow causes, and the distributed nonlinear feedback makes a mockery of any attempt to determine which neuron caused which to fire or not to fire" (p. 129). The idea of cause and effect should truly only be understood as a way in which minds create meaning. We see that if we hold our hand too close to the fire it gets burned, and we learn this should not be done unless we wish to burn our hands. However, just because this relationship holds true for simple actions, this does not mean that it can be used to create systems of complex meaning. Complex systems do not allow for such simple explanations (p. 138).

Freeman uses the research of Eric Piaget to explain why this occurs. Essentially, this is a process learned early in life, well before we have "the language and logic to describe and question it" (p. 130). The attribution of all things having a definitive cause and effect would thus seem to be a flaw in our thinking. It creates a false dichotomy between free will and determinism, "free will and universal determinism are irreconcilable boxes to which linear causality leads, and the only ways out of them are to deny freedom by assuming randomness, or to deny cause except as an action in a social context by an intentional being" (p. 139). A neurodynamic approach solves this problem by providing a "new and enlarged conceptual framework, in which interrelations among parts creating wholes can be described without a need for causal agents" (p. 131).

Our awareness and consciousness do play a role in all of this. Awareness is seen as the faculty that regulates the global AM patterns that emerge from the interaction of various neural populations. Awareness acts to coordinate and constrain these populations, though this is not so much a localized faculty as it is a feature that emerges from this interaction:

> Awareness, then, is a distributed event that integrates the component subsystems and minimizes the likelihood of renegade state transitions within them. Consciousness is the process that makes a sequence of global states of awareness. It is a state variable that constrains the chaotic activities of the parts by quenching local fluctuations. It is an order parameter and an operator that comes into play in the action-perception cycle as an action is being concluded and as the learning phase of perception begins (p. 135).

Similarly, Bressler (2007) describes consciousness as a

> threshold that is crossed in terms of the number of coordinated areas, the overall strength of coordination, or the recruitment of particular areas. If, then, global neurocognitive assessment states are a prerequisite for consciousness, the latter should be considered an emergent property of the interaction of distributed cortical areas (p. 68).

In sum, Freeman believes our decisions are made out of the chaotic dynamics that occur within the relationships of individual neurons to their communities and the interactions between those communities. It is in this dynamic that intention exists: "Our intentional actions continually flow into the world, changing the world and the relations of our bodies to it. The dynamic system is the self in each of us. It is the agency in charge, not our awareness, which is constantly trying to catch up with what we do" (p. 139).

Subsequent research has expanded upon this theory. Evan Thompson (2007) agrees with Freeman, but thinks his ideas can be enhanced by adding Fransisco Varela's neurophenomenology of time-consciousness. In chapter one, the 1 and 1/10 time scales of consciousness were mentioned. At the 1 time scale, there is large-scale integration of neural populations. At the 1/10 time scale, there are "elementary sensorimotor and neural events" (p. 331). In succession of these time intervals, there is "competition between different neural assemblies[14]: when a

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[14] This seems to be the terminology Varela uses to explain what Freeman refers

neural assembly is ignited from one or more of its smaller subsets, it either reaches coherence or is swamped by the competing activations of other overlapping assemblies" (p. 331). What this adds is the notion that the precise time an individual neuron fires makes a difference in its participation in a neural population. One way of describing this is phase synchrony. After an action has occurred, the next action must be prepared for, which involves the disintegration or "phase scattering" of the neural synchronization that had caused the preceding activity (p. 333). In this period of transition, there must be a relaxation time for the neurons. What we experience as a moment in time is actually made up of the 1 scale time frame, but since it can also be thought of as comprising a collection of shorter moments on the 1/10 time scale, "this relaxation time (which is variable and dynamically dependent on a host of factors, such as fatigue, interest, motivation, affect, age, and so on) defines a temporal frame or window of simultaneity, such that whatever falls within this window counts as happening 'now'" (p. 334).

One of the consequences of this flow of time is the introduction of the concept of metastability:

> Metastability means that the surface or manifold over which a trajectory courses is unstable, so that the trajectory shifts constantly from one unstable region to another without ever settling down to any one of them. In a metastable system, noise and dynamic instability are intrinsic to the system, so that the system can switch autonomously from one behavior to another, without any external instruction (pp. 337-338).

This seems to mean that the fact that there is variability in the relaxation period of neurons is a vital part of the dynamic brain system, because out of that instability a kind of stability is formed.

Metastability arises because of the tendency of "specialized brain

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to as neural populations.

areas to express their autonomy, with the tendencies for those regions to work together as a synergy" (Kelso & Tognoli, 2007, p. 40). This is important, because if each part of the brain has too much autonomy, then they could not work together, and if they were too coordinated, then "the system gets stuck, global flexibility is lost" (p. 43). One of the main advantages to this approach is that it favors neither the extreme of the brain being entirely integrated or the extreme of the brain being entirely segregated: "For example, it makes no sense to say that brain is 60% segregated and 40% integrated. Rather, metastability is an expression of the full complexity of the brain" (p. 44). The authors speculate that the process of making a decision might arise as a result of the "transition between the metastable and the monostable" (p. 49) activities in the brain.

Out of this idea of metastability, the authors present what they call "metastable coordination dynamics." Though this theory is similar to Freeman's, it differs in the fact that, unlike Freeman, "a subtle blend of integrative and segregative *tendencies*" (p. 52) is emphasized, whereas Freeman makes the process seem a little more definite.

Conclusions

If we considers this theory of will in light of Walter's criteria laid out in chapter one, we see that this conception of will definitely meets his latter two criteria. It is clear that we are the originator of our actions (Freeman is fairly insistent on this point) and that our actions are done for understandable reasons (at least fairly understandable). With the first criterion, it is unclear if we can say for a particular action that we could have acted otherwise. This is the question that Freeman claims it is virtually impossible to answer. One might say that given the chaos that exists in the decision-making process, which means that there is no guarantee that a person will act in the same way under the same circumstances every time, that this criterion could be met, because the actions occurring in these separate instances can be different.

On the other hand, the fact that they are separate instances means that they are not entirely the same, no matter how similar they may be. Time has passed between these instances, and one's preferences may have changed without one realizing it. For example, one may choose the same food every time he goes to the same restaurant, but in between the last trip and the coming one, he may have been to a different restaurant and had a new kind of food. If the understanding of Freeman is correct here, that experience of food would not only have created a new pattern of memory for the new food, it would have slightly altered all the others. If the new food was enjoyed, perhaps that food will be ordered instead upon the next trip to the old restaurant. If the new food was disliked, perhaps that will taint the enjoyment of the old dish because they may have similar smells or textures.

If the explanations given in this example were to be true, the alternative chosen may have been chosen necessarily, and there was no real chance that it would have been otherwise, despite the chaotic nature of the system by which decisions are made.

This theory does meet the criteria suggested by John Searle. His idea was that if "you had an account of brain processes that explained how the brain produced the unified field of consciousness, together with the experience of acting, and in addition how the brain produced conscious thought processes" (2008, p. 72) you would have an explanation of how the self could exist. Having a self, you could then be considered free. It seems to me that Freeman's theory has accomplished this.

In terms of three categories of free will mentioned in chapter one, Freeman seems to resist categorization, as he sees the debate between free will and determinism as a false dichotomy. We cannot say either is true because this reflects a poor understanding of causality. This probably makes him a compatibilist.

What this theory of freedom shows is also the benefits of the interaction that can be had between philosophy and science. Walter Freeman explains his theory with frequent reference to St. Thomas Aquinas. Reading both Freeman and Aquinas helps to understand them both better. This is like the dynamic process itself. The cause of understanding the one better is the other, which is modified by the first in turn. In reading Freeman, Aquinas's concept of the relationship between intellect and will is more understandable, and in reading Aquinas, Freeman's concept of intentionality is more understandable.

CHAPTER FIVE

OTHER NEUROSCIENTIFICALLY INFLUENCED THEORIES OF FREE WILL

The neurodynamic theory described in the last chapter is, of course, one of many neuroscientific theories that try to explain how we make decisions. Though there are similarities in how all these theories explain how the brain works, there are also many differences, both in what is emphasized and in how the various areas of the brain work together. This should not be a surprise given that comparatively speaking, the science of the brain is still a young one. It can be quite difficult to decide which of these theories is superior to the others, especially when one is not a specialist in the field. While the contention here is that a neurodynamic theory seems to be a very promising way of explaining human behavior, it is not my intent to criticize too harshly any other theory, especially on the basis of its scientific merit. Part of what the process of assimilation shows us is that each of us will have a unique perspective on the reality of the world. While this is certainly not to say that everyone is right, as there is a truth in the world that we must assimilate our ideas to, it does mean that more than one of these theories may be correct, especially within the context of its focus.

In this chapter we will look at a selection of these other neuroscientific theories (those of Henrik Walter, Gerald Edelman, Patricia Churchland, Daniel Dennett, and Michael Gazzaniga), as well as the argument made for free will from quantum physics, which can be understood in terms of neuroscience. We will also consider Daniel Wegner's view of the conscious will being epiphenomenal, which is partially explained using neuroscientific evidence.

Henrik Walter

Like Freeman, Walter emphasizes the chaos of the decision-making process that emerges from a neurological perspective of the brain: "Not only can chaotic processes help explain quasi indeterministic capacities to act otherwise and flexibility; under certain conditions they also produce stable and predictable behavior" (Walter, 2001, pg. 296). This idea of order emerging out of chaos is explained in the aptly named chaos theory. Individual variations in a system exist and cannot be predicted, but general predictions can be made. An example commonly used is weather systems (pg. 166). It is difficult to predict the weather more than a few days in advance, but we also know that it is very likely to be cold and snowy in Illinois during the winter. We do not know on which days it will snow, nor which days will be warmer than others, but we can be fairly certain that we will not have the same kind of weather as we have in summer.

In Walter's theory of natural autonomy mentioned earlier, chaos helps to explain the difficulty of determining causation. Walter uses the example of a nonlinear equation in mathematics, where "terms are repeatedly multiplied with themselves." Doing this causes the result of one equation to enter in to the result of another in a "nonlinear way" (pg. 299). The brain does much the same thing in its processes, as there is a "high degree of positive and negative feedback" (pg. 299). When reflection is occurring, several feedback loops may be working out the consequences of a decision. When we are consciously aware of the process that is going on, we might think of that as rationality (pg.

299).

Walter does not seem to agree with Freeman's theory that causation is solely the result of internal processes in the brain:

> Having meanings depends on using meaningful structures for specific purposes, as well as on interaction with the system in which a thing is embedded. In the same way, active externalism says that the way a brain currently behaves depends on external matters. Even the evaluation loop providing authenticity is connected to actions and the experience that results from them (pg. 300).

In other words, it seems that what Walter means is that we cannot have meaning without the world around us. Theoretically, if a mind existed in a space devoid of anything to sense, it could not construct meaning, because there would be no meaning to construct. The meaning we receive from the world we see would seem to have an influence on the way we make decisions, and the way that this sensory input is perceived may not be an entirely internal process.

In response to this, Freeman might respond that while we could not have meaning without external objects to perceive, the actual process by which this meaning is constructed is entirely internal to our minds. The things we experience do serve to make us different than we would have been had we not experienced them, but it was an internal process that drove us to experience them, and it is an internal process that causes any adaptation to the experience and assigns meaning to it.

Based on his summary of natural autonomy stated above, it does seem that Walter's theory matches all three criteria for a definition of free will. Even though he does not believe it exist, his theory at least closely approximates the criteria he laid out, though perhaps the third criterion, of a man being the originator of his own actions, is more ambiguous, given the discussion immediately preceding this paragraph. As with Freeman, the first criterion is also somewhat questionable, but perhaps the fact that it is difficult to give an account of how choice is free in a particular situation given chaos does not invalidate the overall principle. As stated in the discussion on Freeman's position, just because we cannot know for certain whether we would have acted otherwise does not mean we would not have. Walter's natural autonomy also seems to meet the criteria of John Searle. Like Freeman, he is probably a compatibilist.

Patricia Churchland

Patricia Churchland (2002) takes what seems to be a generally dynamic approach to the understand of neuroscience. Churchland bases her neuroscientific philosophy of free will on three main points. The first is that feelings are essential to reasoning about what choices we will make (citing David Hume). The power of these emotions depends on the circumstances:

> An agent's decision to change the television station may be more unconstrained than his decision to pay for his child's college tuition, which may be more unconstrained than his decision to marry his wife, which may be more unconstrained than his decision to turn off the alarm clock. Some desires or fears may be very powerful, others less so, and we may have more self-control in some circumstances than in others (p. 212).

Consequently, we should conceive of the control we have over our decisions existing in degrees rather than in an either/or fashion. This is not so much a spectrum of control as it is a parameter space, "where the dimensions of the parameter space reflect the primary determinants of in-control behavior" (p. 212). This view is also somewhat similar to what Rene Descartes wrote about the ability to do otherwise. The times in which we have the most choice in our actions are the times when they are least important. Descartes called this a defect of the will, because it is when we do not know what we want that we have the most choice.

In a previous chapter, we considered evidence from neuroscientist Antonio Damasio that shows that when a person has a disconnect between their emotional faculties and their judgment, they tend to make worse choices. Churchland considers the same evidence. She points out the implications this has for rational choice theory, which assumes that people are rational actors. It seems that when people are at what we might call their most rational (when they have no access to their emotions), they have trouble deciding, because they have no guidance from their emotions telling them which choice to prefer. A better way to think of the decision-making process is to see it as a competition of our desires in which the strongest wins out. As well as Hume, Jonathan Edwards conceived of decision-making in this way.

Churchland's second main point is that the way we become more moral is through the life experiences we have which develop our habits (citing Aristotle). She states the formation of habits in dynamic terms, where the formation of a habit is "interpreted as contouring the terrain of the neuronal state space so that behaviorally appropriate trajectories are 'well grooved' or strongly attractive" (p. 230). We learn about what choices to make by "being exposed to prototypical examples of rational actions, as well as foolish or unwise or irrational actions" (p. 230). We learn to recognize patterns of behavior that we can match to a particular situation. We slowly learn to "generalize to novel but relevantly similar situations" (p. 230). In most cases, these patterns tend to resemble the standards of the community we live in.

The third main point is that assuming we are responsible for our actions is necessary for us to learn how to evaluate the consequences of them, "the basic idea is that *feeling* the social consequences of one's choices is a crucial part of socialization – of learning to be in the give-and-take of the group" (p. 234). This is part of how we acquire the habits we have. This will "contour the parameter-space landscape in an appropriate way" (p. 234). We learn to be responsible, in part, by being taught the need for responsibility.

These first two points have already been discussed in detail in previous chapters. The third point has not yet been mentioned, although it is very closely related to the other two. The idea that we need to learn to be responsible for our actions so we can evaluate the consequences of them is simply an area of virtue that we need to be habituated into. The fact that responsibility must be taught seems to me to be evidence that it is not, strictly speaking, a determined feature of our characters. If this were the case, then there would be no need for teaching about responsibility, we would already have this virtue. As there are many irresponsible people in the world, this is clearly not the case.

As Churchland does not see the will as being uncaused, but does believe we have some control over our decisions, she would seem to fall within the compatibilist category. Though she does not base her conclusions on the same research as Walter Freeman does, her conclusions do not seem incompatible with the neurodynamic viewpoint presented in the last chapter.

Quantum Physics

Patricia Churchland (2002, 2006) has criticized the idea that quantum physics has relevance to the free will question. As she states the thesis:

> Stripped to essentials, the hypothesis claims that although an agent may have the relevant desires, beliefs, etc., he can still make a choice that is truly independent of *all* antecedent causal conditions. On this view, the agent, not the agent's brain or his desires or his emotions, freely chooses between cappuccino and latte, for example. It is at the moment of deciding that the indeterminacy or non-causality or the break in the causal nexus – whatever on wants to call it – occurs (2002, p. 206).

In a way, the argument for quantum physics can, at times, seem to good to be true. As the view is stated by Doyle (n.d), where quantum physics enters into the decision-making process is in the formation of our possible choices. Quantum indeterminacy exists as the noise (perhaps this could be understood as chaos) within our brains that "is inherent in any information storage and communication system." This generation of random possibilities breaks the chain of determinism, but it is not the only element involved in decision-making. Doyle sees the process as a two-stage model, where in the first stage, these random possibilities are generated, and in the second, we select between the possibilities. Our reasons for choosing are not random. They are the result of the preferences that come from our past experiences. In other words, the actions we take are not random (for that would deny responsibility as much as absolute determinism would). We have reasons for them. This process should be thought of as being a continuous one. Possibilities are constantly being generated, and on occasion we act to choose between these possibilities.

Doyle explains that a key difference between these two stages of the decision-making process is on the scale of the brain where they are formed. The quantum indeterminacy that generates alternative possibilities exists on a micro level:

> These are drawn from our memory of past thoughts and actions, but randomly varied by unpredictable negations, associations of a part of one idea with a part or all of another, and by substitutions of words, images, feelings, and actions drawn from our experience. In information communication terms, there is cross-talk and noise in our neural circuitry.

The possibilities are chosen from by a macro level brain process. At this level, quantum indeterminacy has little to no effect. This works to:

> reduce the noise to levels required for an adequate determinism. Our decisions are then in principle predictable, given knowledge of all our past actions and given the randomly generated possibilities in the instant before decision. However, only we know the contents of our minds, and they exist only within our minds. Thus we can feel fully responsible for our choices, morally and legally.

Doyle believes that this generation of possibilities also explains how imagination works, where at times ideas seem to spontaneously arise, seemingly out of nowhere:

> The strong feeling that sometimes "we don't know what we think until we hear what we say" reflects our capability for original and creative thoughts, different from anything we have consciously learned. Something as simple as substituting a synonymous word, or more complex replacements with associated words (metonyms) or wild leaps of fancy (metaphor) are examples of building unpredictable thoughts.

This view of the imagination is rather similar to Freeman's view of imagination, which is as a result of the "nonlinear dynamics of neural populations" (199, p. 82), since part of the nonlinear process occurring in the brain is as a result of chaos. The AM patterns that are formed in the brain are never exactly alike as a result of this chaos, but they do seem more ordered than in Doyle's explanation, because the variance in them (unless modified by experience) is not that great.

What this creates is a view where "Compatibilists and Determinists were right about the Will, but wrong about Freedom. Libertarians were right about Freedom, but wrong about the Will." Of all the views presented here, this one seems best able to meet all of the criteria that Walter says is necessary for free will. The weakness in this view is that the quantum indeterminacy that Doyle basis it on is very difficult to show evidence of. He admits that: "Nothing physically localized is likely to be found. The randomness of the Micro Mind is simply the result of ever-present noise, both thermal and quantum noise, that is inherent in any information storage and communication system."

Another theory on the way quantum physics interacts with the will can be found in the work of Jeffrey Schwartz and Henry Stapp. Jeffrey Schwartz is a psychologist who works with patients who have obsessive-compulsive disorder. He found that an effective way of treating the disorder was with mindfulness meditation, which included focusing on the control of the will. As Schwartz explains it, there is a way of seeing the world that is brought about by meditation where one can "observe their sensations and thoughts with the calm clarity of an external witness: through mindful awareness, you can stand outside your own mind as if you are watching what is happening to another rather than experiencing it yourself" (Schwartz & Begley, 2002, p. 11). Schwartz likens this to Adam Smith's "impartial and well-informed spectator" (p. 11). In a similar way, with OCD sufferers, Schwartz observed that a small part of his patients' minds were able to observe their behavior, allowing them to observe and reflect "insightfully on their sheer bizarreness" (p. 13).

Schwartz implemented a mindfulness meditation approach so that his patients could use the impartially observing part of their minds to make changes to the rest of it. By focusing their attention, this treatment would give them the power to "strengthen and utilize the healthy parts of their brain in order to resist their compulsions and quiet the anxieties and fears caused by their obsessions" (p. 13). The reason this is possible is because of the neuroplasticity of the brain. Even adults are capable of having their neurons form new connections and assume new roles.

In part, this ability of consciousness to redirect the way the brain functions can be explained by quantum physics: "What we know now of quantum physics gives us reason to believe that conscious thoughts and volitions can, and do, play a powerful causal role in the world, including influencing the activity of the brain. Mind and matter, in other words, *can* interact" (pp. 16-17). Schwartz compares this idea to the belief of William James that the will is most free when it focuses on a thought and concentrates on it. Quantum physics suggests that, "there *is* no fully formed brain state until attention is focused" (p. 18).

One of the things that quantum physics shows is the uncertain nature of causality. Rather than seeing causality as existing in a linear relationship between a cause and an effect, what has been shown is "the fundamental role played by the observer in choosing which of a plenitude of possible realities will leave the realm of the possible and become actual" (p. 263). In other words, what happens seems to be partially dependent on what is observed. This was found as a result of an experiment done in 1801 by Thomas Young. Young was attempting to measure whether light consisted of particles or waves. To do this, he set up a black curtain with two vertical slits cut closely together and shined a light at the curtain with a screen set up on the wall opposite to it. On the screen, the light created a pattern of "alternating dark and light vertical bands" (p. 265). What was happening was: "where crests o flight waves from one slit met crests of waves from the other, the waves reinforced each other, producing the bright bands. Where the crest of a wave from one slit met the trough of a wave from the other, they canceled each other, producing the dark bands" (pp. 265-266). Later experiments were done so that the light source would only emit a single photon of light at a time: "As hundreds and then thousands of photons make the journey (this experiment was conducted by physicists in Paris in the mid 1980s), the pattern they create is a wonder to behold. Instead of the two broad patches of light, after enough photons have made the trip you see zebra stripes" (p. 267). What happens is that light is "vacillating between a particlelike existence and a wavelike one" (p. 267).

The behavior of the light particles was explained in quantum physics as being a wave of probability, because the light was only measured at two points, at its release, and at its end. This is the basis of the Schrödinger wave equation, "the equation modestly settles for describing the probability that those traits [of the wave] will have a particular value" (p. 269). When a particle is unobserved, its behavior is fairly determined by probability, but it is not certain, and has no definite location. When it is observed, "the wave function seems to undergo an abrupt chance: the location of the particle it describes is almost definite... Before the observation, the system had a range of possibilities; afterward, it has a single actuality" (p. 269). This is known as the "collapse of the wave function" (p. 269). One way of explaining this is that it is the very act of observation that causes the location of the particle in the wave function to become definite.

Another famous experiment that illustrates this principle is the case of Schrödinger's cat. A cat, a radioactive atom, and a poisonous pellet are placed in a sealed box. The pellet will be released if the atom decays. After a certain amount of time, there will be an equal chance that the cat is alive or dead. At this point, according to quantum physics, the cat is both alive and dead. The only way of knowing is to open the box, at which point the probabilities of the situation become actual. It is only by observation that this can happen.

Schwartz quotes a conversation he had with physicist Henry Stapp where Stapp explains this as: "In quantum theory, experience is the essential reality, and matter is viewed as a representation of the primary reality, which is experience" (p. 278). This thought is not incompatible with the ideas that have been presented in previous chapters, where we understand the world as a result of our actions into it. What seems to differ is the belief here that the primary reality is our experience. Schwartz recognizes that even in quantum physics there is disagreement over this. For example, Werner Heisenberg believed that in the case of Schrödinger's cat, "the infamous cat was indeed either alive or dead, even before an observer looked and collapsed the wave function; it is nature herself who collapses the wave function" (p. 281). This interpretation would be more in line with the other arguments presented in this work.

Nevertheless, the role of the observer is very important to the process. Schwartz and Stapp see a three part process occurring in quantum physics: "the evolution of the wave equation as described by the

Schrödinger equation, the choice of which question to pose..., and nature's statistical choice of which answer to give" (p. 282). Our freedom exists in our choice of which question to pose.

Schwartz applies this theory to the brain:

> once the brains of observers are included in the quantum system,the wave function describing the state of the brain of any observer collapses to the form corresponding to new knowledge... The collapse occurs in conjunction with the conscious act of experiencing the outcome of the observation (pp. 284-85).

This idea also does not seem too dissimilar from Walter Freeman's explanation for the formation of knowledge from experience, where the AM patterns in our brains that constitute meaning are modified by our actions into the world. Schwartz believes that the brain state "evolves deterministically until a conscious observation occurs." Just before that observation, there are many possible ways that our thoughts might branch, but "when an observation registers in the mind of the observer, the branches are brutally pruned: only the branches compatible with the observer's experience remain" (p. 285).

This makes consciousness rather important. By directing our consciousness, we can influence our actions. They are not simply mechanically determined: "Quantum theory demonstrates how mental effort can have, through the process of willfully focusing attention, dynamical consequences that cannot be deduced or predicted from, and that are not the automatic results of, cerebral mechanisms acting alone" (p. 319). Schwartz recognizes that this power is not absolute, however. There are both active and passive aspects of consciousness, and our passive thoughts cannot always be overcome: "Sometimes the power of those passive, unbidden, and unwanted brain processes – the voices the schizophrenic hears, the despair a depressive feels – is simply too great for mental force to overcome" (p. 321).

This view of free will does well in meeting the criterion of John Searle. Our conscious thoughts not only way a role in our decision-making, with effort they can play a very important role. It also seems to meet all three criteria of Henrik Walter, even the first, as our will seems to allow us to do otherwise (at least sometimes). This does depend on how strong our wills are, so perhaps under this criterion, only the strong willed would be considered to be free. In a way, this is not dissimilar to our interpretation of St. Augustine, where only those who have faith in God can be considered truly free. In fact, Schwartz and Stapp argue that without some kind of belief in freedom (one way of which is through God), morality is destroyed:

> When medieval science connected man directly to his Creator, man saw himself as a child of the divine imbued with a will free to choose between good and evil. When the scientific revolution converted human beings from the sparks of divine creation into not particularly special cogs in a giant impersonal machine, it eroded any rational basis for the notion of responsibility for one's actions (p. 276).

As already noted, this view of quantum physics does not seem incompatible with what we have discussed in previous chapters. The idea of mindfulness meditation being able to strengthen the power we have to actively direct our actions in the way we want them to go is particularly interesting, and will be considered further in the next chapter. What is problematic about this view, and others similar to it such as that hypothesized by Stuart Hameroff (2006) is that it may stretch the relevance of quantum physics further than it rightly can go. The difficulty here is in how these quantum events that happen on a very small scale affect the scale on which we observe behavior. Certainly, quantum physics does seem rather uncertain. Because of this, I would argue that the neurodynamic approach seems a more promising one for the free will argument. Regardless of whether this argument from quantum physics is correct, I would argue that there is still relevance to Schwartz's findings about mindfulness meditation.

Benjamin Libet

Benjamin Libet's studies are frequently cited in neuroscientific articles concerning free will. On the basis of his experiment concerning freely voluntary actions, he found that the brain began to work 550 msec before an action took place, but awareness of the decision to act took place only 150-200 msec before the action. There is an electrical charge that occurs in the brain before a voluntary action takes place. This can be recorded at an area of the scalp located at the top of the head (Libet, 2004, p. 124). Because of this knowledge, Libet was able to design an experiment that tested conscious intent to act. In the experiment, the subject would sit 2.3 meters from a clock face. A cathode ray oscilloscope was set up to project a ray of light that revolved around the clock face. The subject was asked to perform a random, voluntary action such as sudden movement of the hand, and to notice his first thought to perform this action and remember where the light was on the clock face at that time.

Because the conscious awareness of the intention to act still does occur approximately 150 msec before the action intended is actually performed, Libet also found that it is possible to change one's mind, or veto that action for up to 100 msec before the act occurs. However, there were no recorded instances of a spontaneously vetoed action, only actions that were planned to be set into motion at a predetermined time.

Libet concludes on the basis of this, that, "free will does not *initiate* a volitional process; but it can control the outcome by actively vetoing the volitional process and aborting the act itself, or allowing (or triggering) the act to occur" (p. 143).

The implication of this is that while we may not have the freedom to decide to act, we do have the freedom to decide not to act. Libet points out that because of this, it is difficult to blame someone for having the wrong thoughts, since these thoughts can arise outside of one's control, but one can be blamed for acting on those wrong thoughts, since he can choose not to. He also points out that this is contrary to the Christian doctrine that states a man is guilty not only for his wrong actions, but also for his wrong thoughts. However, he recognizes that this may form a "physiological basis for original sin" (p. 151). In other word, it is only because of original sin that these wrong thoughts arise.

Libet states that what he cannot prove is whether our will is moved by deterministic or non-deterministic causes. It is not something that studies like his can prove. As of yet, there are no studies that have proved this one way or the other. Yet he does point out that because of this, there is no scientific reason to conclude that our will is determined solely by mechanistic processes. It is a speculative belief that cannot be definitively proved because it is not falsifiable. He believes that, if anything, this research points more towards the possibility of will not being completely determined. There is a desire in man to believe that he has free will, at least over some of his actions, and this is the basis for our views on human nature. Given the speculative nature of both theories, it is better to believe the one that presents a more flattering picture of man.

A recently published study in *Nature Neuroscience* (2008) has performed similar experiments to the ones Libet has, except with an fMRI machine. This study has found that the unconscious impulse towards action occurs even earlier than when Libet found it to: "Taken together, two specific regions in the frontal and parietal cortex of the human brain had considerable information that predicted the outcome of a motor decision the subject had not yet consciously made. This suggests that when the subject’s decision reached awareness it had been influenced by unconscious brain activity for up to 10 s" (Soon et al, 2008, p. 1).

Daniel Dennett (2004) offers a strong criticism of Libet's experiments. He argues that as a result of the nature of Libet's experiment, the subject has already pre-committed to cooperate with the experimenters. The subject's actions cannot really be random, because they know at some point they will act to move their hand. What these experiments really show is that "conscious decision-making takes time" (p. 239). Our actions are the result of our deliberation over time. At the actual time of our acting, our past actions, reflections on those actions, and anticipation of future actions play a role in determining what we do. The action we take may have arisen unconsciously before we were aware of what we would do, but, at least, in part, that unconscious decision was influenced by our conscious thought. This is particularly true with habits. At one time we had to learn how to do something like shoot a basketball, and think about the process of our shots, but as we become more experienced, it becomes second nature to us, and much less thought goes into our act of shooting.

Daniel Dennett

Daniel Dennett considers free will to be an evolved concept. It exists as the result of our creating it, and there is no guaranty that it will necessarily survive as we continue to develop: "It is an evolved creation of human activity and beliefs, and it is just as real as such other human creations as music and money" (2004, p. 13). We can best understand the nature of our freedom by a cooperative effort between philosophy and science. As such, Dennett presents what he calls a "naturalist" solution to the problem of free will.

A key to understanding Dennett's position lies in his criticism of Libet's experiments. To understand how we make decisions, we need to be aware that they are the result of a series of past thoughts. What exists in our minds is a competition of ideas vying for control of our behavior. The thought that is dominant is the one that comes into our consciousness, "for a mental content to become influential is for it to get into position to drive the language-using parts of the controls. All this has to happen in the arena of the brain, in 'central processing,' but not under the direction of anything" (p. 254). We have what Dennett refers to as a "user-illusion" of our selves in the brain which helps us to see our past intentions. This gives us "a means of interfacing" with our selves. This is similar to the position of Daniel Wegner described below.

Citing David Hume, Dennett argues that there exist in us certain natural motives, like affection for children and the desire for self preservation. As a result of the process described above, which allows us to reflect on our reasons for doing things and therefore "change them into entirely different reasons," we are able to move beyond these natural motives:

> Hume saw ethics as a kind of human technology, and he saw reflection as the tool we get from nature that permits us to revise our natural instincts, enhancing them with prosthetic elaborations whose rationales (free-floating until Hume and others pinned them down and represented them) are, in fact, aimed at still more freedom, consistent with security from harm. Eyeglasses for the soul, you might say (p. 260).

Dennett goes on to explain the evolution of morality as a kind of "mimetic engineering" where we have tried to construct ideas of morality for living in communities. He sees the first systematic efforts at this originating in Plato and Aristotle.

We become more or less free as the result of the kind of systems of morality we design. Especially important is the kind of education we receive. If we are raised to be the kind of people who look for and give reasons, we will have a better grasp on how to develop the right kind of morality. Like Aristotle argued, Dennett sees morality arising from taking the right kind of actions, so we need opportunities to take those actions, so that they will become easier to perform the more frequently we do them: "The self is a system that is *given* responsibility, over time, so that is can reliably be there to *take* responsibility, so that there is somebody home to answer when questions of accountability arise" (p. 287).

In this view, our degree of freedom is a result of what we have been taught and the actions we take into the world. This is not unlike what we have seen in the position of Schwartz and Stapp and St. Augustine noted above. Our freedom is dependent on the context of our lives and what we believe about our lives. Dennett points out that this makes our freedom very fragile. As we know more and more about what factors influence our decisions, it becomes easier to manipulate those decisions, and we have more doubt about whether they are truly free. Since our freedom is dependent on what we believe about it, the fact that we find it worth wanting "can be used to anchor our conception of free will in a way metaphysical myths fail to do" (p. 297).

This last quotation shows that though they have similar ideas about how belief can lead to freedom, St. Augustine, Schwartz and Stapp, and Dennett all have very different beliefs about which belief it is that will lead to that freedom. Dennett would argue that St. Augustine's faith in God has great potential to diminish freedom (see chapter six).

Dennett should probably be classified as a kind of compatibilist. His theory of free will fails to meet the first of Henrik Walter's criteria, but meets the other two. As noted in chapter one, Dennett does not think the ability to do otherwise is important to free will. Though Dennett does see consciousness as a kind of illusion, it does seem to play a role on the decisions we make, and therefore, his theory meets John Searle's criterion. However, this idea of consciousness and conscious will being illusory is problematic. As Daniel Wegner advocates a similar view in more detail, this problem will be addressed below.

Daniel Wegner

Daniel Wegner (2002) wrote a book called the *Illusion of Conscious Will*. That title should give some clue as to the position that he holds – but only some clue. Wegner's conception of what it means for the conscious will to be illusory is perhaps an indication that illusion was a poor choice of words. Wegner states that the conscious will is an illusion in the sense that "consciously willing an action is not a direct indication that the conscious thought has caused the action" (p. 2). We confuse the experience of willing with the actual forces that cause our actions. A better way of understanding the will is to only think of it as an experience of our actions, but not inherent in those actions. Under certain circumstances, we could have acted without this experience. This can be thought of a phenomenal will. The will can also be understood empirically, where it is measured by "examining the actual degree of constant conjunction between the person's self-reported conscious thought and the person's action" (p. 14). This idea of causation being a constant conjunction of action should be familiar from Hume. Using Hume, Wegner explains causation as being "an event, not a thing or characteristic or attribute of an object" (p. 13).

The confusion in our willing arises because we view causation in two ways, as mechanical and mental, or as physical and psychological. Mechanical causation is typically concerned with objects. We do not need to think about the intentions or desires of an object acting, like a computer fan moving its blades. It simply happens as a result of the causal forces acting on it. Mental causation is typically concerned with subjects. In this case, we do concentrate on the intentions and desires that caused an action. These two systems seem to develop separately from each other. It has been shown in the research of Eric Piaget that sometimes children under-attribute intentions to people, and attribute them to objects. Because we view the world in both ways, this has "created two largely incompatible ways of thinking. When we apply mental explanations to our own behavior-causation mechanisms, we fall prey to the impression that our conscious will causes our actions" (p. 26). What really happens in our decision-making is more complicated than what we perceive, and many causes are not perceived. Wegner likens this to being fooled by a stage magician:

> The real causal sequence underlying human behavior involves a massively complicated set of mechanisms. Everything the psychology studies can come into play to predict and explain even the most innocuous wink of an eye. Each of our actions is really the culmination of an intricate set of physical and mental processes, including psychological mechanisms that correspond to the traditional concept of will, in that they involve linkages between our thoughts and our actions. This is empirical will. However, we don't see this. Instead, we readily accept a far easier explanation of our behavior: We intended to do it, so we did it (p. 27).

Wegner goes on to cite Spinoza, who said that our idea of freedom is just our ignorance of the causes of our actions.

On the basis of research done by Wilder Penfield and Jose Delgado, Wegner believes that there is a part of the brain that provides the experience of will, but that it is separate from the structures in the brain that produce actions. Wegner uses the research of Libet to show that the system that produces action is prior to our awareness of it, because there is brain activity before we are consciously aware of our desire to move. However, if the arguments about Libet's research above are correct, this does not necessarily allow one to separate the two as easily as Wegner does. Wegner also draws from the work of V.S. Ramachandran (see Ramachandran & Blackeslee, 1996) which deals with phantom limb syndrome, where subjects who have lost limbs still feel those limbs and think they can move them. One of Ramachandran's experiments created a situation where it would appear to subjects that they still had their arm, by tricking their mind into thinking that an experimenter's arm was their own. In this case, the subjects still experienced the sensation of moving the arm. From this, Wegner concludes that "willful movement can be experienced merely by watching *any* body move where one's own body ought to be" (p. 44). This is probably the result of mirror neurons in the brain, which are activated both by our own movements and the movements of others.

This research at least shows how confused we can often be about the cause of our own actions, and Wegner cites many other examples, such as how people are fooled by Ouija boards and hypnotists as further evidence. When we are confronted with evidence that shows we have been mistaken about what we thought was causing things we will revise our interpretation, but we will not always admit that we were wrong. Instead, we will claim that this was always the way things were. This is especially the case when the cause of our actions turns out to be radically different than what we first perceived it to be. Often this is as a result of neurological disorders (Ramachandran & Blackesless and Damasio, 1999 provide many examples of this).

One particular example that Wegner uses to illustrate his point is amusing enough that it bears repeating here. In early 20th century Berlin, there was a horse named Clever Hans whose trainer, Wilhelm von Osten, believed he had taught his horse to be able to read, spell, tell time, and give answers to mathematical problems. The answers were given based on the number of times the horse tapped his hoof. He could also shake his head to indicate no. This could even be done outside of the trainer's presence. It seemed that the horse was very intelligent. Eventually, due to the investigations of Oskar Pfungst, the real explanation was discovered. It turned out that the horse did not know the answer to the questions, but did seem to have picked up on a subtle mannerism in his trainer and most questioners. Most of the time when someone asked a question, he would incline his head slightly forward to see Han's hoof tapping, and then straighten when the correct answer had been reached. Hans would start tapping his hoof when the questioners leaned forward, and stop when they straightened up. When this was pointed out and demonstrated to the trainer, he continued to insist that the horse was actually answering the questions: "Despite this evidence, von Osten claimed that this was just one of Han's little foibles - the horse was often tricky and stubborn – and maintained that with further trained the problem could be eliminated. It never was" (p. 192).

This phenomenon occurs as a result of what Wegner calls "action projection." Ownership of our actions can not only be wrongly attributed to ourselves, but it can also be wrongly attributed to others, like Clever Hans. A similar phenomenon can occur in group behavior, like in mobs, where the mob of people seems to take on a mind of its own, and the individuals in the mob are moved by the actions of the group. It can seem like no one person is actually in control of what is going on.

Though it is an illusion, Wegner does see a purpose for conscious will. It is what acts as the mind's compass:

> Conscious will is particularly useful, then, as a guide to ourselves. It tells us which events around us seem to be attributable to our own authorship. This allows us to develop a sense of who we are and are not. It also allows us to set aside our achievements from the things that we cannot do. And perhaps most important for the sake of the operation of society, the sense of conscious will also allows us to maintain the sense of responsibility for our actions that serves as a basis for morality (p. 328).

For something that is an illusion, this is a faculty that seems to play an important role in determining our actions. Wegner even admits that it is probably better to think that we are in control of our actions, even if we really are not. It is a "positive illusion," at least most of the time: "Even if conscious will isn't an infallible sign of our own causation, it is a fairly good sign and so gives us a rough-and-ready guide to what part of experience is conjured by us and what part is bedrock reality" (p. 333). In the end, this is a non-trivial illusion: "On the contrary, the illusions piled atop apparent mental causation are the building blocks of human psychology and social life" (p. 342).

It is difficult to conceive of how something that is apparently so important can not be real (unless Wegner has some other meaning for the word illusion that is not apparent to me). Perhaps the distinction can be made that consciousness itself is real, and the illusion is the belief that our conscious thoughts are what cause our decisions. Going back to Wegner's original definition, the key is the phrase that our conscious willing is not a "direct indication that conscious thought has caused an action" (p. 2). It seems to me that though there may not be a direct indication, at times our conscious will does in fact work to cause us to act. What Wegner's evidence shows is that we are oftentimes bad at identifying those occasions. This does not mean that our conscious will does not, at times, have an influence over what we do. Given the importance that Wegner attributes to conscious will, he may not even deny this. Nevertheless, I do not understand the need to say this faculty is an illusion when it is simply capable of being mistaken.

It is difficult to classify Wegner's position. Of all the philosophies surveyed thus far, his is probably the one that comes closest to determinism, even though he does think the illusion of the will is important to our actions (like most other philosophies here, ultimately it should probably be considered a compatibilist one). It is also unclear whether he meets either Walter or Searle's criteria for free will. When addressing the question of free will, Wegner states that conscious willing is a "humble estimation of the causal efficacy of the person's thoughts in producing the action," but that is not the only or even best source of what has caused an action (p. 336). In other words, though not solely determinative, conscious will is important to determining responsibility for our actions. Because of this, despite what Wegner says, it is probably not an illusion, and the fact that he calls it such is simply a poor choice of words.

Michael Gazzaniga

Michael Gazzaniga believes that "brains are automatic, but people are free" (2006, p. 99). He takes a view of causation that states that all our actions have definable causes. The fact that we cannot yet explain the causes of our actions is simply a shortcoming of our current scientific knowledge. Presumably, this is something that can eventually be overcome. When we try to address the question of free will by ideas like neurodynamics or quantum physics, we seem to miss a fundamental point, that: "Our freedom is found in the interaction of the social world" (p. 99). Our actions ought to be considered free if they come from ourselves, as long as we are not impaired by some kind of disorder or external source.

We determine whether someone is responsible for their actions as a result as a result of what a society has collectively decided makes them responsible: "Responsibility is a human construct that exists only in the social world, where there is more than one person. It is a socially constructed rule that exists only in the context of human interaction. No pixel in a brain scan will ever be able to show culpability or nonculpability" (p. 100). Therefore, it is impossible to find neural correlates of responsibility in brains: "In neuroscientific terms, no person is more or less responsible than any other for actions. We are all part of a deterministic system that some day, in theory, we will completely understand" (p. 102).

This view of responsibility and brains would seem to be one Hacker and Bennett would look on favorably. It avoids the mereological fallacy of attributing human behavior to an organ that is only part of our selves. It is also similar to the idea seen in Jonas and Merleau-Ponty where we only experience freedom in relation to the world. Nonetheless, Gazzaniga's account seems less satisfying than a neurodynamic one. His view fails to explain how it is that the interactions between humans that lead to the formation of standards of freedom and responsibility are possible. How can what seems on his view to be completely determined, in what seems to ultimately be a linearly causal relationship, create freedom? What Gazzaniga does not see, that someone who expressed a similar view like Hans Jonas did, is that freedom can be traced to the most simply functions of our being. Though we certain need to avoid the mereological fallacy, and Gazzaniga's understanding of freedom certainly does so, looking at causality from a more circular perspective better avoids this problem. It is difficult to classify Gazzaniga's position using the criteria we have established. It seems to be fairly determinist, but since he believes that determinism is somehow compatible with freedom, he must be a compatibilist.

Gerald Edelman

To understand Gerald Edelman's theory of free will, it is first necessary to explain his theory of consciousness. This is because Edelman includes the feeling of will among the possible conscious states we might have. Underlying this theory are three assumptions: one, that "only conventional physical processes are required for a satisfactory explanation of consciousness" (Edelman & Tononi, 2000, p. 14); two, that consciousness developed by the process of evolution; and three, that the qualitative nature of consciousness makes describing it much different than experiencing it. Like many other theories described here, Edelman believes that the act or experience of being conscious precedes our understanding of it.

There are many kinds of conscious experience. All these share the qualities of privateness, unity, and informativeness. Key among these qualities is that of unity. Our consciousness is more than the sum of its parts. This unity and coherence is "tied to consciousness's so-called *capacity limitation* - the fact that we cannot keep in mind more than a few things at a time" (p. 26). This is because there is a limit on how many things we can keep in mind before the coherence of a single conscious state is disintegrated. This limited capacity and the "serial succession of conscious states is the price paid for their integration - for the fact that they are not reducible to a simple sum of independent components" (p. 27). The strength of this integration of consciousness is how Edelman explains the problem of missing limbs that Wegner mentioned. We find it preferable to believe the missing limb is still there than to acknowledge its absence: "Apparently, the feeling of an absence is far less tolerable than the absence of a feeling" (p. 29).

In Edelman's theory there are three systems in the brain essential to understanding its global functioning. The first is the thalamocortical system. In this system there are many subdivided groupings of neurons, each with a particular function. These areas are reciprocally connected with each other. This provides "a major structural basis for *reentry*, a process of signally back and forth among reciprocal connections" (p. 44). This process of reentry is the basis for the integration of perceptual and motor processes, "it alters selective events and correlations of signals among areas and is essential for the synchronization and coordination of areas' mutual function" (p. 48). The second system is a "set of parallel, unidirectional chains that link the cortex to a set of its appendages, each with a special structure - the cerebellum, the basal ganglia, and the hippocampus" (p. 45). The third system is what Edelman calls the "value system," which is a collection of a small number of neural groupings in the brain stem and hypothalamus that release neuromodulators capable of influencing neural activity and neural plasticity (p. 47). Central to all of this is the fact that the interaction happens in a nonlinear fashion:

> All these features of the brain – connectivity, variability, plasticity, ability to categorize, dependence on value, and the dynamics of reentry – operate heterogeneously to coordinate behavior... the nonlinear aspects of the brain, the body, and various parallel signals from the environment must be categorized together to understand the processes of perceptual categorization, movement, and memory that underlie consciousness (pp. 49-50).

Though the basis for the nonlinear nature of brain dynamics are different than the way Freeman describes them, it seems that similar principles are at work here.

At the heart of Edelman's theory of consciousness is the idea of the dynamic core. The dynamic core consists of groups of neurons that are highly integrated with each other through the reentry process that occurs in interaction with the thalamocortical system. These groups are widely distributed throughout the brain. Neuronal groups are not part of the core if their functional interactions are strictly unidirectional. This core is constantly changing as the result of the experiences we have. Any conscious experience is the result of the activity of this entire dynamic core.

The function of the dynamic core is a process that exists and should not be thought of as causal: "only transactions at the level of matter or energy can be causal. So it is the activity of the thalamocortical core that is causal, not the experience that it entails" (2006, pp. 91-92). This makes consciousness an illusion in the way it is seen by Daniel Wegner. It is epiphenomenal, but it does "inform us of our brain states and is thus central to our understanding (p. 92). What this means for free will is that "one must conclude that core states as physical events are determined. Nonetheless, when not physically bound or in prison or in the throes of neural disaster, we can honestly claim the ability to do 'as we like' or 'see fit' illusory or not" (p. 94). This is the case because we should not base our normative concerns on the functions of our brains. We cannot derive an "ought" from an "is." Although the value systems in the brain do provide a foundation for building a system of morality, they do not "directly determine them" (p. 95). This naturalistic fallacy will be considered in more detail in chapter seven.

The fact that we can be free while being determined means that Edelman should probably be classified as a compatibilist. His idea of consciousness being epiphenomenal is problematic for the same reasons as it is in Wegner. On this basis, he would seem to fail to meet the criterion of John Searle, and in that respect, he may be seen as a determinist. Though there are similarities between Edelman's theory and Freeman's, Freeman's theory seems to provide a sounder basis for explaining how freedom is possible.

Conclusion

So why should a neurodynamic approach be preferable to any of these others? Furthermore, given that I am writing this as a political scientist, am I qualified to judge which of these ideas is better than the others? What I would emphasize is that many of these ideas are not so different from one another. Points of emphasis may differ, but Henrik Walter, Gerald Edelman, and Patricia Churchland all describe what seems to be a dynamic approach to understanding the brain (yet they do all differ on the question of free will). Though I do not have much background in neuroscience, there are two main reasons I would give for supporting a neurodynamic view (beyond the fact that it is endorsed by many neuroscientists).

The first is the way that the foundations of a neurodynamic approach can be found in the annals of philosophy. As the preceding chapters have shown, the ideas of Aristotle, David Hume, William James, Maurice Merleau-Ponty, Jonathan Edwards, and especially St. Thomas Aquinas have been helpful in understanding these neuroscientific ideas.

The second is the way that this theory resembles the way other complex systems in the world work. There is an underlying logic and order to the world, even if it is not particularly comprehensible, and though there is much chaos present within it. It makes sense to understand the brain, as a very complex system within the world, in these terms. Furthermore, understanding one of these complex systems better can help to understand the others. This is a reason why the political implications of this view will be considered in chapter seven.

CHAPTER SIX

NEUROSCIENCE, RELIGION, AND THE SOUL

In chapter two it was pointed out that the problem of free will emerged from Christianity. One of the reasons that neuroscientists have argued that there is no free will is because it has traditionally been thought of as being a faculty of the soul, and these scientists do not believe that the soul exists. Mele, Vohs, and Baumeister (2010) point out that often times when neuroscientists are arguing against free will, what they are really arguing against is the idea that it is a "magical, nonphysical, nonevolved power" (p. 2). The argument goes that neuroscientific research eliminates the possibility of the soul being a physical part of the body, and because of this, the soul does not exist. For example, in *Brain-Wise*, Patricia Churchland states that: "The weight of evidence now implies that it is the *brain*, rather than some nonphysical stuff, that feels, thinks, and decides. That means there is no soul to fall in love" (Churchland, 2002, p. 1). Francis Crick's book, *The Astonishing Hypothesis* (1995) purports to be a scientific search for the soul, but what is found is that the brain determine everything. This leaves no place for a soul.

This dualistic way of understanding the problem seems to have developed in large part as a result of the influence of Rene Descartes. We have discussed this in more detail in chapter three. On this view, the body and soul are separate entities, only connected at one point in the brain. This is not the only way of understanding the relation between soul and body, and it is far from the best way.

Another way of understanding the body-soul relationship goes back to St. Thomas Aquinas. Aquinas' conception of this relationship can be characterized as halfway between dualism and materialism (Hasker, 1999, p. 161). For Aquinas, the soul is not a complete substance, but instead "a constituent of the psychophysical whole which is the human being" (p. 166). This is an improvement over Descartes, but Hasker finds this view problematic because it is based on Aquinas' understanding of form, which he finds to be ambiguous. Hasker does see promise in an insight from Eleonore Stump, who suggests that, given a modern understanding of neuroscience, Aquinas' view might lend itself to seeing the mind or soul as emerging from the function of the brain (p. 170).

Drawing from this idea, Hasker elaborates on what such an emergence might look like. He first defines different kinds of emergence, using the categories of John Searle (1992). The most basic form of emergence is one where the emergent feature cannot be explained simply from the "composition of elements and environmental relations" but rather by "the causal interaction among the elements" (p. 173).

A slightly more complex definition states that a feature emergent not only when it has to be explained by the causal interaction of its more basic elements, but also that the newly emergent property will play a causal role in the interaction of the elements that have produced its creation:

> So if (for example) consciousness is emergent in this sense, the behavior of the physical components of the brain (neurons, and substructures within neurons) will be *different*, in virtue of the causal influence of consciousness, than it would be without this

> property; the ordinary causal laws that govern the operation of such structures apart from the effects of consciousness will no longer suffice (p. 174).

In this process both downward and upward causation are occurring. This is another way of explaining what Searle feels is necessary for free will to exist. In fact, Hasker claims that free will could be an emergent feature: "If there is an indeterministic element in the operation of consciousness, so that consciousness can cause things it isn't caused to cause... then it might well be the case that 'consciousness could cause things that could not be explained by the causal behavior of neurons'" (p. 177).

The final definition Hasker offers is one where a newly emergent property not only plays a role in the interaction between itself and the parts that make up its emergence, but also that it can cause things that cannot be explained by the behaviors of its component parts. Once something like consciousness has emerged, "it then has a life of its own" (p. 172).

This idea of downward causation is only a metaphor for thinking about consciousness. These levels are not concrete and capable of causing things on their own. Instead, this process should be thought of as being one where "the only causal influences are those of the ultimate constituents in their interactions with each other, and the only way the "higher levels" can make a difference is by *altering or superseding the laws* according to which the elements interact" (p. 176).

This understanding of emergences creates the possibility of an indeterminist free will (Hasker refers to this as libertarian free will). In stating his own view on the subject, Hasker believes that the emergent consciousness exists in some form that is like the second and third definitions offered above. It is by this third definition of emergent consciousness that an indeterminate free will becomes possible. Our mental properties "involve emergent causal powers that are not in evidence in the absence of consciousness" (p. 190). This creates what Hasker calls an "emergent individual," which is a "entity which comes into being as a result of a certain functional configuration of the material constituents of the brain and nervous system" (p. 190). This emergent individual is a unity, and cannot be reduced to its separate parts. Thinking of people in such a way is helpful to avoid the mereological fallacy.

Hasker compares this emergence to the kind that occurs in the production of magnetic or gravitational fields. In such production, the field is distinct from what produces it. A magnetic field is generated "by the magnet in virtue of the fact that the magnet's material constituents are arranged in a certain way... But once generated, the field exerts a causality of its own, on the magnet itself as well as on other objects in the vicinity" (p. 190). As such, consciousness also may also be some kind of field. It may even be the case that this field is a "spatial entity" (p. 192). Where the analogy fails is that we know in detail how magnetic fields are generated, but we know very little about how consciousness emerges (p. 191). Furthermore, the kind of emergence occurring in these other fields is much weaker than "the strong sense required for properties of the mind" (p. 192).

Hasker believes the advantage this theory offers is that it avoids the problems of Cartesian dualism by explaining a close and natural connection between the mind and the brain. As noted, the mind itself may be some kind of entity that exists in a volume of space "sufficient to encompass those parts of the brain with which the mind interacts" (p. 192). This can account for the fact that consciousness may be divided as a result of damage to the brain. It also allows consciousness to be more than epiphenomenal.

This theory proves advantages over a more materialist view because it "recognizes the necessity of recognizing both teleology and intentionality as basic-level phenomena" (p. 193) instead of seeing them as observations of processes that do not have either of these properties. This theory does require a kind of materialism in that the "potentiality for conscious life and experience really does exist in the nature of matter itself" (p. 194).

Hasker sees this as potentially being a cost of his theory. This is the case because, since he wishes to formulate a theory that is compatible with Christianity, it seems this understanding makes survival after death less likely. On the other hand, this may be an understanding closer to that of the writers of the Bible, who used the terms "soul" and "spirit" to refer to the human person as a whole rather than as "discrete metaphysically separable parts" (p. 206). Such an understanding comes from ancient Greek philosophy. However, one must still explain what happens to the soul between death and resurrection. This would seem to be more easily answered by a traditional dualist conception of the soul.

In the case of emergent dualism, the way Hasker explains this is to suggest that the field of consciousness can survive even without a body to generate it. After the body has died, the field of consciousness is then sustained by God until the resurrection. Although Hasker finds the comparison somewhat troubling, he does point out that a parallel theory exists in the survival of black holes. Black holes are able to survive as self-sustaining fields even after the body that generated it is gone. In the end, Hasker concludes that this is not a question that can be answered by science, but that nothing in the theory of emergent dualism prevents resurrection from being possible.

One problem with this view is that Hasker believes his theory allows for the possibility of an indeterminist free will. Though he feels brain-mind and mind-brain interactions should be seen as deterministic, the interaction between the causal processes of the mind may have the causal indeterminacy essential to a libertarian free will. As we have discussed at length earlier, this indeterminacy is in fact not necessary to free will, and may actually be harmful to the theory. By this explanation people can be seen as acting without reason. However, this view is not essential to emergent dualism, though perhaps it would be better to define it using only Searle's second definition of emergence and not the third. Not only is this third definition the one which provides the foundation for indeterminate free will, it is the one that seems least plausible, as it can be seen to explain consciousness as being independent of neurological processes once they have generated it, rather than simply playing a causal role in decision-making. This seems to be going too far, and it is somewhat unclear to what extent Hasker believes this to be true. For the most part, he focuses on the second definition of emergence.

Another problem with this theory is that it might really just do the same thing as other theories of dualism, in that it makes the body and soul two different kinds of substance. Kevin Corcoran states that this theory is problematic because it means the soul is not dependent on the body for its continued existence, "in both this life and the next" (Corcoran, 2006, p. 43). It is true that the soul is not dependent on the body in the next life in Hasker's theory. It is less clear whether the soul remains dependent on the body while they are both alive. Inasmuch as this is true, this would certainly be a problem. It would seem that this is more likely to be the case under the third definition of emergence, but less so in the second. It is arguable that so long as we confine ourselves to the second definition of emergence, the soul would be dependent on the body because the causal forces at work would be acting in both ways – from the body to the soul and the soul to the body.

Corcoran also criticizes this theory because the emergent soul is still a different kind of substance from the body. The soul, though it may be spatial, is still immaterial, just in a different way. This opens up this theory to some of the same criticisms that other forms of dualism face. However, while this may be true, given the nature of emergent dualism, it seems possible that advances in science could at least be able to prove or disprove it, as the nature of consciousness becomes better understood. This is an advance over something like Cartesian dualism, where the soul cannot be proven to exist in any way but through introspection. The concept of an emergent soul in this theory is appealing, but it would be better to find a way to explain this process that maintains the unity of body and mind to make it consistent with the neurodynamic theory presented in chapter four.

As an alternative, Corcoran offers a way of explaining the soul in a more materialist way. This is done by looking at our natures from what Corcoran calls a constitutional (rather than reductivist) view of materialism. By this understanding, we are "*constituted by* our bodies without being identical with the bodies that constitute us" (pp. 65-66). This is true even though a person and a body share the same matter. They are what Corcoran calls "spatially coincident" (p. 66). As examples of other things that can be understood in this way, Corcoran uses dollar bills and college diplomas. These things have meaning beyond the mere paper that they are made out of. The paper of dollar bills and diplomas are not identical to the dollar bills and diplomas because the former lack properties the latter have. The paper of a college diploma lacks the meaning inherent in the idea of a college diploma. It has little value as a mere piece of paper. The meaning it signifies can only exist by means of the college that issued the diploma. If the college closes, or loses its accreditation, the meaning of the diploma will change, even though the paper itself does not.

In this way, we can also distinguish persons from bodies. Persons are "minimally, beings with a capacity for intentional states" (p. 67). We have beliefs and desires and are able to see ourselves from a first-person perspective. This idea of personhood seems to be Corcoran's explanation of what the soul really is. Our bodies are simple physical organisms. The difference between the two is that there is no need for physical organisms to have intentional states to be considered physical organisms, whereas it is necessary for persons to have intentional states to be considered persons. Because of these different properties, persons are not identical with bodies. Despite this, persons are essentially constituted by their bodies. If the body dies, then the person also ceases to exist. The survival of the body is a necessary condition for the persistence of the person.

In order to have a first-person perspective on our selves, the body is also necessary, because it provides something for which the self to refer to. More than this, it is also necessary for other embodied persons to exist. This is something that is apparent in the very nature of God in Christianity. God exists as a relationship of the three persons of the trinity. Furthermore, in Genesis, man begin his existence in relation to others, even before the creation of woman: "It is no exaggeration to say, therefore, that human persons are always – from the beginning of the Christian narrative to its very end – *persons-in-relation*" (p. 74). This is also compatible with the notion explored in this work that meaning is created through our acting in the world.

Corcoran believes it is a strength of the constitutive perspective that it accounts for the necessity of these relationships. It is also able to meet the insight of dualism, that we are not identical to our bodies, while still meeting the demands of materialism, in that we are essentially physical creatures.

As Corcoran also wants to explain things from a Christian perspective, this still leaves him with the problem of life after death. If persons can only survive with the existence of a body, then how can there be life after death? He proposes two possibilities that might answer this. The first is that there is a gap in our existence. The body dies, and then is recreated at the resurrection, with no existence in between. The second is that there could be an "intermediate state of consciousness" (p. 138) where we continue to exist embodied in heaven, but not glorified until the end of time. Corcoran concludes that this resolution occurs by a miracle, whether one is a dualist or materialist.

Both of these theories provide promising alternatives to the traditional dualist view that has so often been criticized by science. The strength of Hasker's position is that it offers a scientific explanation of the way the soul might emerge from the body by its physical processes. In some ways, this explanation duplicates other emergent properties that have been found in nature, although it is much more complex than any of those, and not nearly as well understood. Hasker's theory does not seem incompatible with the neuroscientific perspectives that have been presented earlier. This theory becomes stronger if the weaknesses noted above are corrected.

The strength of Corcoran's position, though light on science, is that it provides an account that can explain the difference between persons and bodies in physical terms. The self is a psychological construct that exists in relationship to the body and the other selfs in the world. As noted, this is compatible with other concepts we have explored. Corcoran's explanation also avoids the mereological fallacy. As he states:

> Just as brains are not the thinkers of thoughts, but people are, it is likewise true that physical organisms are not the thinkers of thoughts. Human persons think thoughts in virtue of being constituted by physical organisms of a certain physical complexity. It is wrong to suggest that one's physical organism is a good candidate for thinking thoughts just as it is wrong to suggest that my body is a good candidate for the blameworthiness of some action (p. 79).

What does create some trouble in this account is the thought that this psychological understanding of the self may be epiphenomenal. While we may think that our selves are thinking thoughts, this phenomenon is an illusion. Explaining the emergence of consciousness in more physical terms does a better job of avoiding this concern.

But even if we can find a way to explain the soul that is more compatible with science, broader criticisms of religion on scientific grounds remain. There have been many books written by the so-called "new atheists" such as Daniel Dennett (2006), Sam Harris (2005) and Richard Dawkins (2006) that criticize religious belief in just this way. They seem to think that science makes religion unnecessary and harmful. Since the idea that understanding begins with belief is a central tenant of this work, it is important to examine their arguments.

In Daniel Dennett's book *Breaking the Spell* (2006), he tries to make the case that belief is very dangerous. Once people make a commitment to their beliefs, they are unlikely to change them. Furthermore, as time passes, a strange process occurs where their original commitment gets buried under layers of defensive reactions. This is true not only of religion, but of other beliefs, among them belief in free will: "Belief in free will is another vigorously protected vision... those whose investigations seem to others to jeopardize it are sometimes deliberately misrepresented in order to discredit what they see as a dangerous trend" (p. 202).

An article by Nyhan and Reifler (2010) illustrates how this process can work in politics. They found that those who had the most committed ideological views were least likely to change their opinions when confronted with facts that disagreed with those views: "corrections fail to reduce misperceptions for the most committed participants. Even worse, they actually *strengthen* misperceptions among ideological subgroups in several cases" (p. 30). Research from Westen et al. (2006) has looked at brain activity when devout supporters of a political candidate are using this "motivated reasoning" (p. 1947). During such reasoning, there is neural activity in different parts of the brain than when someone is reasoning about issues on which he or she has no strong attachment. This led the researchers to conclude that there are greater levels of emotion involved during this process of justifying one's beliefs. Though this may seem a fairly obvious conclusion, it may not seem so to the partisans engaged in the justification of their beliefs. It may seem that they are reaching the conclusions they do by a quite logical process. This goes back to what Wegner wrote about the conscious will. Often times we do come up with very bad justifications for the actions we take, ones that completely ignore the true causal forces at work in our decision. An action can seem to be quite reasonable, without actually being so.

It can be argued that belief is unavoidable, however. Andrew Newberg (2006) points out that there is no alternative to belief. We can never see the world from a perspective outside ourselves, and so we must make assumptions about the world. It seems that our brains are "naturally calibrated" to embrace spiritual beliefs in particular. Dennett would probably say in response that these beliefs still need to be justified and closely examined.

He argues that this not the case with religion. In fact, sometimes the more incomprehensible a doctrine is the better. This is for the same reason that our conscious wills are often mistaken about the cause of things, or we refuse to adapt our strong political beliefs in the face of opposing evidence: we will always try to form a consistent narrative that allows us to make sense of the world. Therefore, if a doctrine is particularly incomprehensible, it requires a great deal of effort to make sense of it. As Dennett puts it: "The anomalies give the host brains something to gnaw on, like an unresolved musical cadence, and hence something to rehearse, and rehearse again, and baffle themselves deliciously about" (2006, p. 230).

Once we reach an understanding of a difficult religious doctrine, we may become quite attached to it because of our effort. This is, of course, true in all difficult endeavors. As I write this work, it is quite unlikely that I am going to change my conclusions unless faced with overwhelming evidence to the contrary (and if the above research is correct, maybe not even then). Not only have I spent a fair amount of time developing this thesis, but enacting a radical change in my thinking would mean doing a lot more work.

The difference between this project and religion, and what Dennett finds so dangerous about religion, is that I will probably not be willing to risk my life to defend my work here (at least not physically), even though I do believe what I have written. Many people are willing to risk their lives when their religious beliefs come under threat. For Dennett, the difference between scientific and religious beliefs is that there is detailed evidence to support belief in science (experimental data and such), and only anecdotal evidence at best to support religion. Therefore, it would be foolish to die for scientific belief, because believing in it or not will not change its truth. There is more than "mere professing" that can be done to offer evidence of scientific truth. With religious belief, we find it very difficult to interpret the beliefs of others because of this idea: "When it comes to interpreting religious avowals of others, *everybody is an outsider*. Why? Because religious avowals concern matters that are beyond observation, beyond meaningful test, so the only thing *anybody* can go on is religious behavior (p. 239).

Given that, unlike with science, there is a lack of available evidence to support the existence of God, Dennett sees no reason why he ought to believe in him: "it could all just be nature, and, no, the fine tuning of the laws of physics can be explained without postulating an intelligent designer" (p. 243). Dennett believes in the "glory of nature" and that there is no need for a "middleman" to have put it in place (p. 244). Of course, this does not necessarily mean that God does not exist, merely that Dennett sees no place for him in the world, and finds no benefit that can be gained from belief in him.

Of course, one need not reach the same conclusion that Dennett has. In *A Fine-Tuned Universe*, Alistair McGrath (2009a) lays out a theory of natural theology which has its basis in Christianity. Though McGrath admits that the existence of God cannot be proved from nature, he finds that there is a "consonance" between nature and the trinitarian God of Christianity. To see God in the universe is to look at it in a certain way, a result of one's viewpoint towards the world. Nature itself simply exists. Its meaning is revealed by being identified by the observer, who "must construct a schema, a mental map, in order to make sense of what is observed" (p. 4). One way of understanding the world is by a method of natural theology, which seeks to "find common ground for dialogue between the Christian faith and human culture, based on a proposed link between the everyday world of our experience and a transcendent realm" (p. 5). McGrath sees nature as a socially constructed entity, not an objective reality as the Enlightenment treated it.

This understanding of the way meaning is constructed is also similar to what we have seen from the neurodynamic approach to the human brain. Here too, our perception of the world exists as a result of the actions we take into it, assimilating the world to the knowledge that we already possess. Thus, no two people will see the world exactly alike, though many times this difference may not be too distinct. This does not mean that everyone's formation of meaning is equally correct. If this were so, there would have been little point for McGrath to have written his book. McGrath argues that the Christian faith provides the best explanation for understanding nature: "the Christian vision of reality possesses an internal coherence and consistency which is at least matched by its remarkable ability to make sense of what we observe and experience" (p. 22). McGrath recognizes that nature does not change, but our understanding of it does. We should look on the way we perceive nature as an iterative process: "Theology is here conceived as a dynamic process of exploration, based on iterations. The outcomes of a given iteration is then fed into the next iterations as its presuppositions" (p. 33). This process is much like what Freeman described as circular causality.

Past attempts at natural theology have proven to be unsuccessful at explaining the way the world works. McGrath's proposal suggests several important theological elements that should be observed to improve on these attempts:

The first is that the concept of nature should be seen as an "interpreted entity" in contrast to the view of the Enlightenment mentioned earlier. This means we are free to see nature as we wish, "forcing us to identify the best way of beholding the natural world" (p. 27).

The second is that we should not try to prove the existence of God by looking for evidence in nature. Nature should be viewed from a Christian perspective, which provides "distinct notions of God, nature, and human agency" (p. 27).

The third is that no key elements of faith can be proved from nature either.

The fourth is that the question of how human perception takes place is a very important and theologically significant one. It is important to see this perception as a dynamic process. We can see that faith is "the transformation of the human mind to see things in a certain manner, involving the acquisition of certain habits of thinking and perception" (p. 39).

The fifth is that the process of perceiving the world should be thought of as more than a sense making exercise. We should be sure to consider the "rational, aesthetic, and moral dimensions of the human engagement with nature" (p. 28).

The sixth is to emphasize the importance of natural theology as provided a way of interaction between the Christian church and secular culture. It can provide a "navigable channel from human interest in the beauty of nature or the notion of the 'transcendent' to the 'God and Father of our Lord Jesus Christ'" (p. 28).

This way of understanding the world holds much appeal, especially as it expresses many insights similar to what have been discussed earlier in this work. It seems unlikely that such a view would be persuasive to someone like Daniel Dennett, however. Even if God and nature can be thought of as being consonant, this is not necessarily so. As such, the argument would seem to be about whether an atheist or natural theological viewpoint provides the best way for understanding the world. Because of the many flaws Dennett sees in Christianity (more of which are explored below), he would argue that it does not. In counter to this, McGrath would probably point out that these flaws are not the flaws of Christianity itself, but the flaws of the imperfect humans who practice Christianity. If these people are genuine Christians, their faith has made them better than they would have been otherwise.

Though certainly differing from orthodox Christianity, Jean Jacques Rousseau explains another kind of natural theology in the "Confession of Faith of the Savoyard Vicar," a selection from his work on education, *Emile* (1762). McGrath argues that to try and understand the relationship between God and nature as the Enlightenment did leads to "a vision of God that is at best Deist" (p. 67), and this can be seen from Rousseau. In Rousseau's story, the Savoyard Vicar had a crisis of faith after he had been unable to maintain his vow of chastity. After much soul searching, what the Vicar found to be the best guide for belief was his conscience: "Let us consult the inner light; it will lead me astray less than they [the church] lead me astray; or at least my error will be my own, and I will deprave myself less in following my own illusions than in yielding to their lies" (p. 269).

Operating in this way, the Vicar first reasoned that there must have been a first cause in nature, to have set things in motion. He believed this because, by his senses, he saw that the matter which is outside himself was naturally at rest, though frequently in motion. Therefore, what was in motion must have been caused by something else. This could be reducible to a first cause. This was one of the central articles of faith that the Vicar found for his natural religion.

From the particulars of the motion of the world, the Vicar observed that there was an order to it, which implied that the first motion of the world was caused by an intelligent being. This was another of the Vicar's articles of faith. It was not something that could have come about by chance, though the end to which it moved cannot be determined. The world is well ordered, and this implies that the intelligence behind it is wise, powerful and good. It is this intelligent mover we call God. Much as his ends cannot be determined, his particular nature is also a mystery: "I perceive God everywhere in His works. I sense him in me; I see Him all around me. But as soon as I want to contemplate Him in Himself, as soon as I want to find out where He is, what He is, what His substance is, He escapes me" (p. 277). Of course, Dennett and other atheists would find neither of these two declarations persuasive.

The Vicar also spoke on organized religion, and found it wanting. Reason gives a clearer picture of God than revelation, which leads to contradictions and conflicts. Because there are many different revelations, there cannot be the same consensus that reason brings. Each prophet's claim of having received a revelation from God would seem to be a revelation only of what is in harmony with the prophet's own inclinations. All that is necessary to please God is to want to please him, to be sincere in praise, to revere him in "spirit and truth[15]."

[15] This "New Religion of Sincerity" is talked about by Arthur Melzer (1996), who states that because this natural religion puts sincerity in place of truth, it provides less ground for intolerance, since it makes religion a very personal determination: "The only heresy is to follow others... Each individual must save himself and only himself" (p. 356). It also weakens the authority of organized religion. This sincerity bring to mind a prayer of Thomas Merton, which contains the words, "the fact that I think I am following your will does not mean that I am actually doing so. But I believe that the desire to please you does in fact please you." Melzer

Still, the Vicar neither rejects nor embraces revelation, but did state that it is the responsibility of the individual to not depend on the judgment of others. To give adequate study to all of the different revelations and determine which of them is true requires a lifetime of study, and is an occupation unsuited for common man. Natural religion resolves this difficulty with its simplicity.

Following the Vicar's speech, Rousseau writes of just what a transformative effect this will have on his fictional pupil Emile:

> ...what new holds we have given ourselves over our pupil. How many new means we have for speaking to his heart! It is only then that he finds his true interest in being good, in doing good far from the sight of men and without being forced by the laws, in being just between God and himself, in fulfilling his duty, even at the expense of his life, and in carrying virtue in his heart. He does this not only for the love of order, to which each of us always prefers love of self, but for the love of the Author of his being - a love which is confounded with that same love of self - and, finally, for the enjoyment of that durable happiness which the repose of a good conscience and the contemplation of this Supreme Being promise him in the other life after he has spent this one well (314).

This religion, properly integrated into one's education, seems essential to the development of a good citizen.

Even this minimal version of religion would probably seem objectionable to Dennett. The influence that Rousseau claims for his religion does not necessarily exist, or may exist in spite of religion. Dennett argues that, notwithstanding claims to the contrary, there is no need to ground morality on the basis of religion. Believers seem to be no more or less moral than others: "The prison population in the United States shows Catholics, Protestants, Jews, Muslims, and others – including those with no religious affiliation – represented about as they are in the general population" (2006, p. 279). God may have been

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rightly notes in his article that Rousseau's re-imagining of religion has been influential, though not completely.

necessary for the founding of a society, but once trust has been established between its citizens, that religion is no longer needed. By remaining, religion provides an outlet for zealots, because it speaks in moral certitudes.

In short, religion, since it seems to provide no real benefits, is too dangerous to exist. People care about it too much, and are willing to fight to the death over it. If religion must remain, we should treat it like our allegiance to a sports team.[16] If we must have religion, we should also work to make it a religion that has a greater basis in reason. One should not blindly accept the creed of one's faith, but seek to understand it and question it where it does not make sense:

> Love is not enough. That's why those who have an *unquestioning* faith in the correctness of the moral teachings of their religion are a problem: if they themselves haven't conscientiously considered, on their own, whether their pastors or priests or rabbis or imams are worthy of this delegated authority over their own lives, then they are in fact taking a personally *immoral* stand (p. 295).

Religion also needs to work harder to prevent zealotry. Moderate believers have a responsibility to condemn the actions of the zealots of their religion and work to make it so their religions have doctrines which are less absolute. They must engage in the:

> unpleasant and even dangerous work of desanctifying the excesses in each tradition *from the inside*. Any religious person who is not actively and publicly involved in that effort is shirking a duty... Until the priests and rabbis and imams and their flocks explicitly condemns by name the dangerous individuals and congregations within their ranks, they are *all* complicit (p. 301).

This needs to be done so that zealots do not benefit from the good work done by these moderate believers.

In some ways, what Dennett wants is what religion should want. It

[16] There are at least fewer deaths as a result of conflict between sports fans, though violence is not entirely absent.

should be expected that believers do understand their creeds. They should question them in order to understand them better. When Augustine said that belief precedes understanding, he did not mean that we should blindly accept the doctrines of the church, but that we should let our efforts to understand God and his mysteries be guided by our faith. Understanding is necessary. So long as believers question the doctrines of the church in this way, then they should do as Dennett suggests.

Believers should also be willing to condemn the acts of those who perform actions that are contrary to their religious doctrines, especially by those who are a part of their church. Even so, if this condemnation does not always happen, it seems too zealous a requirement to insist that all believers are complicit in the acts of zealots when they are not condemned. However, the real problem here lies in which acts ought to be condemned. It is usually rather easy to speak against someone who blows up an abortion clinic or acts as a suicide bomber on a crowded street. What creates the problem is when Dennett calls for the desanctification of the excesses of each tradition. Here there is much more room for conflict, in that what should be considered an excessive doctrine in a religious tradition is likely to vary widely.

In this way, the natural religion of Rousseau is possibly about all the doctrine someone like Dennett would feel safe with. In Rousseau's religion, there is very little to fight over, as there is very little doctrine to begin with. Though Dennett would still not adhere to such a religion himself (and it is not clear Rousseau actually believed in this natural religion), such a religion would probably be sufficiently tamed to do little harm.

But what of Dennett's claim that religion fails to improve the morality of its adherents? It would seem that the issue is not whether Christians or Jews or Muslims are more moral than atheists, the issue is whether a particular Christian, Jew, or Muslim would be more or less moral if in different circumstances they had not been raised in a religious family or a family of a different religion. It seems impossible to answer this question. Dennett does admit that religion is helpful for the founding of a society, so shouldn't religion still be necessary after that founding to preserve the unity that was first created? Dennett says it is no longer necessary after trust has been established, but if religion can help establish that trust should it not also be able to help preserve it?

Perhaps the problem is that everyone does not have the same religion. This is the argument of Thomas Hobbes in the *Leviathan* (1651). Hobbes believes that there will be little conflict about religion in a particular society if all citizens publicly confess to the religion of their country. Belief in this religion is not even necessary, only public loyalty to it. The importance of this religion is more for the unity of a society than it is its spiritual salvation. Considering the conflict and schism found in the history of religion, this does not seem a very realistic possibility. The power of the state necessary to accomplish it also is very unappealing. It is also contrary to our desire to seek the truth that Dennett rightly encourages.

The alternative to this was the effort made by John Locke (1689, 1695) and other enlightenment thinkers (such as the aforementioned Rousseau) to increase tolerance amongst Christians. They did this by trying to emphasize the minimum beliefs necessary for faith and by showing that one should not rely on any interpretation of scripture but one's own. Because it is very difficult to understand some of the doctrines of Christianity, believers should be temperate in what claims they make about God and his doctrines and how forcefully they make them. The hope was that this would result in a world where no one church was able to dominate, and thus no church would be able to impose its beliefs on government.

This is an effort that seems largely to have been successful. Since the time of the enlightenment, there have been far fewer violent conflicts between the various denominations of Christianity over matters of doctrine. This success led James Madison in the 51st Federalist paper to state that the same principles can apply to political liberty: "In a free government the security for civil rights must be the same as that for religious rights. It consists in the one case in the multiplicity of interests, and in the other in the multiplicity of sects. The degree of security in both cases will depend on the number of interests and sects."

It is probably not a coincidence that the rise of the new atheists has coincided with the violent struggle between the predominantly Christian West and the forces of radical Islam. Christopher Hitchens has especially become known for his strong criticism of the Islamic faith, although other new atheists have made similar claims. In a 2007 book review, Hitchens quotes a passage from Sam Harris (2006), which states that:

> The same failure of liberalism is evident in Western Europe, where the dogma of multiculturalism has left a secular Europe very slow to address the looming problem of religious extremism among its immigrants. The people who speak most sensibly about the threat that Islam poses to Europe are actually fascists. To say that this does not bode well for liberalism is an understatement: It does

not bode well for the future of civilization.

If this is correct, it would seem that, while liberalism was successful in making Christianity more tolerant, this tolerance came at the price of becoming tolerant of intolerance. What would seem to be necessary for peace between these religions would be a similar liberalization of Islam. This idea is alluded to in a recent article by Hitchens (2010) about the building of the Cordoba Mosque in New York where he argues that tolerance should be a "two-way street" between the West and Islam.

Of course, the liberalization of Islam seems like a process that is very difficult. Though early in its history there was more of an interaction between religion and philosophy, this relationship was not well maintained, especially concerning natural philosophy:

> Since about 1500, most Muslim theologians have followed the general approach of Al-Ghazali, who regarded both natural philosophy and theology as posing significant threats to Islamic orthodoxy. On this view, nature is held to be incapable of disclosing anything reliable about God, and it might mislead the faithful into making idolatrous or blasphemous judgments" (McGrath, 2009a, pp.66-67).

Lacking a similar recent tradition of interaction between religion and science would seems to mean that such efforts at liberalization would be met with greater resistance than they were in Christianity (which was far from an easy process itself).

The point here is not to argue for any particular cause for the conflict between the West and radical Islam[17], but rather to suggest that this conflict has created an increased concern about the effect religion has on the world and has influenced this new atheist movement.

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[17] This issue is far more complicated than what is presented here, and involves many more factors than religious conflict. One interesting, and not often considered factor is suggested in a paper by Thayer and Hudson (2010), which presents evidence that polygyny contributes to increased violence in the cultures where it is common (as it is in many Islamic nations).

It does seem that there is some scientific evidence that religion can have a positive effect on one's life, particularly when it comes to controlling our wills. Perhaps Augustine was right when he wrote that belief in God can make the will more free. At the very least, the evidence suggests that a certain kind of religious belief, concentrating on the contemplative practice of meditation, seems to have positive effects on brain, including possibly increasing the power of will. This is the argument we discussed in the last chapter from Schwartz and Begley. B. Allen Wallace (2009a) makes a similar argument on how to achieve free will from a Buddhist perspective. Wallace sees free will as a spectrum, at times we are free and at times we are not. This freedom is something that can be cultivated and increased, and this can be done by the process of contemplation,

> when one plumbs the depths of awareness, one discovers a dimension that is not sullied, not contaminated, not shrouded by these mental afflictions or defilements. It is, by nature, pure. If this is true, then it suggests that freedom might be something that can be achieved by purifying the mind of its afflictive tendencies and cultivating greater insight. It also suggests that freedom is something that might be discovered by penetrating the veils of the ordinary functioning of our psyche to a deeper dimension (p. 57).

On this view, it seems to be within our power to make our lives better.

Anthony Newberg (2006) has done studies on the neural activity of Catholic nuns and Buddhist monks engaged in contemplative practice. He has found that engaging in such practices over many years "disrupt the normal processes of the brain – perceptually, emotionally, and linguistically – in ways that make the experience indescribable, awe-inspiring, unifying, and indelibly real" (p. 168). Meditation practitioners end up with a "vivid sense of reality" that makes their beliefs come to be closely held "personal truths" (p. 183).

Wallace (2009b) cites other studies done that show how meditative practices have reduced stress and improved the immune systems of workers in a Wisconsin biotech firm, and improved the attention levels of a group of children between the ages of four to six (pp. 31-32). A study of the Hawn Foundation's "Mindfulness Education" program has shown positive benefits in "attentional control, aggression, behavioral disregulation, and social competence" (p. 4) compared to the children who did not participate in the program.

Again this can be a problem for the atheists as it is a matter of faith. Hebrews 11:1 (KJV) states that "faith is the substance of things hoped for, the evidence of things not seen." However, if faith is accepted as evidence on what basis can one argue against it? Patricia Churchland (2002) argues that:

> Once you have backed into the faith corner, you have no recourse against terror and repression in the name of religion, no recourse against bigotry, demagoguery, misogyny, or abuse posing as religion. You have no basis for criticism of cruel religion. This is precisely because faith is not a matter of evidence and analysis, not a matter of argument and criticism. It is belief *independent* of those things. If the faith option works well for decent folks, it works every bit as well for scoundrels... Faith has been used not only by the charitable and the kind, but also by those who insist on their divine right to unquestioned rule or their divine right to destroy another tribe or enslave women. How can we reason with any of these persons if they claim faith, and faith alone, as the basis of their reasons? (p. 390).

This is why efforts like Alistair McGrath's are so important for religious belief. He attempts to show that faith and reason are not incompatible, and that Christianity provides a view of the universe that is consonant with the scientific evidence.

Conclusion

This chapter began by suggesting that neuroscientists deny the existence of free will because they deny the existence of the soul. However, since this work has argued that from a neuroscientific perspective free will does seems to exist, what does this mean for religion? If neuroscientific evidence shows we do have some kind of free will, or at least allows the possibility, does this answer the question in a satisfactory way for Christianity? I think the answer is that it can. This chapter has considered two suggestions for how the idea of the soul might be understood in a way that can be more compatible with science. This shows the possibility that scientific discoveries need not result in a denial of Christian doctrines.

Overall, this work has argued that a neurodynamic approach offers the best way for understanding how the brain works, and that this understanding allows for a compatibilist free will. This understanding is drawn primarily from the work of Walter Freeman, aided by St. Thomas Aquinas. This is an example of how religious belief and scientific evidence can be consonant with each other. Aquinas, writing from a Christian perspective in the 13th century, was able to understand the mind in a way similar to what has been discovered by at least one neuroscientific theory.

Legal scholar David Opderbeck (2010) argues for a similar conclusion. He writes that though neuroscience and religion would seem to be in conflict concerning free will, there is common ground between them. Because neuroscience presents a view of the will that makes an indeterminist free will impossible, at best seeing consciousness and the self as an emergent property of the brain that allows us to choose between a selection of evolutionary strategies, it does raise some of the same questions about free will as Christianity. We have seen thoughts on free will from St. Augustine, St. Aquinas, and Jonathan Edwards in chapters two and three. None of these theologians argue that man possesses a free will of the kind that an indeterminist would say we have. Citing Martin Luther and the catechism of the Catholic Church, Opderbeck points out that, for Christians: "'Freedom,' then, is not libertarian freedom - the freedom to do anything at all - but the increasing flourishing of the human person who pursues the good" (p. 23). It is by the grace of God that man has this freedom.

The conclusions that Christian theology draws from the idea that we do not have an indeterminate free will are obviously much different than what a typical neuroscientific theory will indicate, but "conflict is unnecessary if we can agree that neurobiology as a natural science properly investigates only one aspect of reality that is at least potentially thicker than any causally reductionistic account would suggest" (p. 24). Opderbeck proposes looking at reality as stratified into three layers (using the work of Roy Bhaskar). These layers are the "empirical (observable by human), actual (existing in time and space), and real ('transfactual and enduring more than our perception of it') (p. 24). Quoting from Michael Polanyi, Opderbeck observes that each of these layers of reality is under the "marginal control" of the layers above it. These layers are not independent of each other, and the higher layers are not reducible to the lower.

When arguing from a Christian perspective, one must make an appeal to the higher, spiritual, layer of reality. What neuroscience tells us is true, but it does not reveal the whole picture:

> The neurobiological portrait of human ontology is true within its own layer of meaning. Human agency and will is constrained and perhaps determined by the brain, and function of the human brain is constrained and perhaps determined by evolutionary biology.

> Christian theology insists, however, that there are yet more fundamental layers of reality, which might not be accessible to empirical methods (p. 27).

When investigating the nature of reality, one must use the methods suitable to the layer of reality one is investigating. Natural science is an appropriate method for investigating created reality. "Theological science" is an appropriate method for reflecting on the nature of God. Natural science fails when it tries to claim for itself the entire nature of reality. This is because the positivist nature of science "depends on an idiomatic structure that is neither verifiable nor self-evident" (p. 25). The foundation on which it stands cannot support it.

So long as science is seen in this way, then science and religion both become valuable tools for understanding the nature of reality. Science then does not make the existence of a soul impossible, though it may change the way we choose to explain what a soul is.

CHAPTER SEVEN

NEUROSCIENCE AND POLITICS

This chapter considers the political implication of neuroscientific research. It looks at the relevance of a neurodynamic understanding of the brain for the study of politics. Such a view is then contrasted with the political conclusions drawn from neuroscience found in the work of George Lakoff, Drew Westen, William Connolly, and Leslie Paul Thiele.

At the very outset of trying to derive any political lessons from neuroscientific research, one is confronted with the issue of the naturalistic fallacy. The concern here is that we should not use what we see in nature to determine what we ought to do. In other words, we cannot determine an "ought" from an "is". David Hume is often cited in explaining why this can be a problem. Hume notes that moralists often make a jump in their writings from explaining what is to what ought to be, and that they provide little justification for this jump:

> For this *ought*, or *ought not*, expresses some new relation or affirmation, 'tis necessary that it shou'd be observ'd and explain'd; and at the same time that a reason should be given, for what seems altogether inconceivable, how this new relation can be a deduction from others, which are entirely different from it. But as authors do not commonly use this precaution, I shall presume to recommend it to the readers; and am persuaded, that this small attention wou'd subvert all the vulgar systems of morality, and let us see, that the distinction of vice and virtue is not founded merely on the relations of objects, nor is perceiv'd by reason (Hume, 1739, pp. 469-470).

John Searle (1964) describes Hume's position in more contemporary terms when he states that: "no set of descriptive statements can entail an evaluative statement without the addition of at least one evaluative premise" (p. 43).

For Hume, our morality comes from how we feel about something rather than a rational system of judgment, whatever source it might be derived from. William Casebeer (2003) argues that this is all Hume is saying, especially since elsewhere in his work he provides a form of naturalized ethics:

> Hume is arguing that moral judgments (as it were) arise not from reason but from our passions. We should not look to reason for the wellspring of morality... it does not motivate us; rather our *passions*, which are not ratiocinative, move us to act, and therefore only they can adequately ground morality" (p. 18).

Because of this, there is no tension between Hume's naturalistic ethics and the naturalistic fallacy.

Considering Hume in this way, perhaps we might say that looking at morality from a neuroscientific perspective is a very Hume-like way of approaching the issue. Neuroscientific research seems to confirm the primacy of emotions in our decision-making. This research has shown that emotion is central to our decision-making processes, and that without it, we reason very poorly (see Damasio, 1999). Walter Freeman's (1999) research has shown that reason cannot be distinguished from emotion. What we think of as reason is instead a kind of emotion, and what we traditionally think of as emotion is simply another form of it.

While neuroscientific research cannot tell us how to be moral, it can explain the process by which we make moral decisions. Understanding this process is valuable in understanding morality itself.

A neurodynamic approach to the formation of morality and our interactions with society is one that applies the dynamics of the way the brain works to our interactions with the world. The meaning we find in the world is transmitted to others when they can perceive and assimilate that meaning in similar ways. This can then form the basis for collective action. At each level of interaction, be it between neurons and neural populations or between groups of people, a similar process is occurring:

> In each level, the individual retains autonomy but accepts constraint in respect to the embedding surround. Failure of individual autonomy leads to apathy and stagnation. Failure of constraint leads to anarchy and disintegration. The archetypal experience is with the committee, struggling for consensus in the face of boredom, obstinacy, and conflicting purposes, dissolving pent-up resentments with banter and laughter (Freeman, 1999, p. 142)

As in our selves, in the world we must have freedom, or there is little incentive to act, but that freedom cannot be without restraint, or there will be insufficient order in which to act.

In the individual, meaning emerges as a result of the interactions occurring in the brain. For a society, meaning emerges as a result of our interactions with each other:

> Humans, as social beings, have assimilated meanings from one another. Each member of a society constantly learns through intentional action and learning, so that his or her behaviors become increasingly predictable, intelligible, and mutually supportive for others. We adopt these forms of meaning through educational, religious, and military institutions in which many people participate and that are tools for activity in concert (p. 143).

The difference between individuals and society lies in the source meaning is derived from, but not what that meaning contains. In both cases, these meanings exist as patterns in the brain. In the case of the meanings we assimilate from others, this is done by creating within ourselves,

> assimilated states with uninterrupted accumulations of episodic memories of others by observing and interpreting their actions, which we experience as empathy. We confirm our similarity by shared gestures and words of joy, condolence, threat, and invitations to cooperate. Our isolation provides the gift of privacy but also the curse of loneliness. Most of us have an intense desire to understand others and to be understood. We respond by making ourselves similar to each other; this is assimilation" (p. 144).

This shows the importance of mirror neurons in the brain. As implied in the name, mirror neurons mirror the behavior of others when we see them act. They then evoke a response similar to if we had acted so ourselves. This is something that has been stressed as being an important political revelation from neuroscientific research (see Lakoff and Westen below).

This works so long as society is relatively stable. Though what is important for both an individual and society can change over time, so long as this happens gradually, these new needs can be assimilated into the meaning of our selves and our society. Radical change alters the process significantly. On a small scale, such a radical change may be the death of a parent, or becoming a parent oneself. On a large scale, this change can be things like an attack on one's country or a major natural disaster. With these traumatic events, a great deal of unlearning can occur and may result in radically new formations of meaning:

> The variety and ubiquity of neuromodulators suggests that unlearning, which is a prelude to learning new assimilated meanings through cooperative action, is brought about by the release of one or more neuromodulators in the brain. It seems to me that the action of such chemicals can loosen the synaptic fabric of the neuropil and open the way to the dissolution of beliefs and their replacement by new ones. We experience the process as a religious or political conversion and remembering the light that came, or as falling in love and remembering the song they played (p. 151).

Sometimes this unlearning can be brought about intentionally, and this explains the development of the techniques used by religions to bring about conversions (trance states, speaking in tongues, music), militaries to develop unit cohesion (basic training), and even fraternities and sororities to integrate new members (hazing). The goal is to create an intense emotional experience shared with others that will develop a deep trust between the members of a group.

Leonid Perlovksy (2007) explains the process of meaning formation using dynamic logic. In a dynamically logical system, one begins with uncertainty and finds meaning through learning, as our knowledge instinct drives us to see similarity: "In the process of learning, as models become more accurate, and the similarity measure more crisp, the value of similarity increases" (p. 82). This is an "evolution from vague, uncertain, fuzzy, unconscious states to more crisp, certain, conscious states" (p. 84). This allows for differentiation to occur. As this differentiation occurs, the results become accessible to our consciousness, though the early stages of the process are not.

There is a hierarchy of meaning in the mind[18], which becomes progressively uncertain as we reach its top. There is a synthesis that occurs at the highest levels of the mind, as the more certain perceptions of the lower levels are integrated into as coherent a whole as they can possibly be made into. This synthesis is "the feel of overall meaning and purpose of knowledge," and is "related to the meaning and purpose of life" (p. 91). Synthesis works to connect "language and cognition, concepts and emotions, conscious and

[18] A similar hierarchy can be found in Freeman's work, with meaning emerging from the microscopic level of the neurons to the mesoscopic level of neural populations to the macroscopic level of global brain states. Since there is feedback between these levels, they are not strictly hierarchical. This is the case with any dynamically hierarchical system, as Perlovsky alludes to in his article.

unconscious" (p. 92). By this process, we form our core values such as religious and political beliefs. These values are so important to us that it becomes increasingly difficult to objectively evaluate them. The more attached we become, the more this is so. At a certain point, the process of differentiation can no longer occur. Though the processes of synthesis and differentiation can be cooperative, they can also exist in an antagonistic relationship with each other. The process described above is a negative one, but synthesis can also inspire creativity and stimulate differentiation.

Perlovsky believes this process can also explain the rise and fall of civilizations, and has developed mathematical equations that attempt to test this hypothesis. What his work seems to show is that the higher the level of differentiation in a society is, the more knowledge that society has accumulated. The higher the level of synthesis, the more meaning is found in that accumulation of knowledge. When the level of synthesis in a culture falls below a certain level, "knowledge loses any value, culture disintegrates, and differentiation reverses its course" (p. 94). In other words, a society ends up in a period like Europe's Dark Ages, where a great deal of knowledge was lost. When a culture has a much higher level of synthesis, there is less differentiation, and that culture stagnates. When a stagnate culture and a more dynamic one (with more differentiation and less stability) interact, Perlovsky has found that the stagnant culture becomes more dynamic. The price of this greater level of differentiation is a decrease in the culture's stability, but the interaction that occurs does seem to be "beneficial in the long run: both cultures evolve with more stability" (p. 102).

While differentiation is important for a culture's progress, it would seem that synthesis is the more important element. Without differentiation a society stagnates, without synthesis a society collapses. Perlovsky sees the key to preserving synthesis in art and religion:

> Since time immemorial, art and religion have connected conceptual knowledge with emotions and values, and these provided cultural means for maintaining synthesis along with differentiation. A particularly important role in this process belongs to music, since music directly appeals to the emotions (p. 103).

This is similar to Freeman's emphasis on the importance of ritual and shared intense emotional experience in developing the ties that bind a group of people in a society.

In evaluating the modern world, Perlovsky finds that the West currently lacks a high level of synthesis:

> The current cultural neurodynamics in Western culture are characterized by the predominance of scientific differentiation and the lack of synthesis. More and more people have difficulty connecting scientific highly-differentiated concepts to their instinctual needs. Many turn to psychiatrists and take medications to compensate for a lack of synthesis. The stability of Western hierarchical values is precarious, and during the next downswing of synthesis hierarchy may begin to disintegrate, leading to cultural collapse. Many think that this process is already happening, more so in Europe than in the US (p. 100).

This is an ambitious extrapolation of a cultural theory from the dynamics of the way the brain works. We have previously argued that one of the strengths of a neurodynamic theory is that it is similar to the dynamics of other complex systems. Perlovsky makes an interesting attempt to show how such a comparison might be made. However, it seems very difficult to quantify what the levels of a culture's synthesis and differentiation are. Perlovksy claims above that Western culture has much more differentiation than it has synthesis, and while this is probably true, he offers little real evidence to support his claim.

Perlovsky does admit this difficulty. While he has equations to explain the relationship between the levels of synthesis and differentiation in a culture: "We had no data with which we could select scientifically correct values for the parameters. It will be up to sociologists and cultural historians to decide upon appropriate parameter values" (p. 96). This is a significant obstacle to any scientific study of history or politics. As Aristotle wrote at the outset of *The Nicomachean Ethics*, the ability to understand politics does not allow for much exactness. We can only achieve a "rough outline" of the truth and speak in generalities (1094b3-25). The values that Perlovsky seeks cannot be stated with sufficient precision to make any predictive claims about a particular culture. There is some value in the general principles it espouses however. This does seem to be a reasonable explanation generally for why cultures change.

Drew Weston

Drew Weston (2007) emphasizes the importance of emotions in politics. He states that, contrary to common belief, emotions are more important than ideas in politics. Weston contrasts the presidential election campaigns of Bill Clinton with George Bush and George W. Bush with Al Gore and John Kerry. In each case the candidate who made the best emotional appeal was the winner. With the exception of Clinton, Weston observes that the Democratic party in America has failed in recent elections (before 2008) because they have appealed to "intellect and expertise" rather than emotion. In fact, the Democrats have exhibited an "*irrational emotional commitment to rationality* – one that renders them, ironically, impervious to both scientific evidence on how the political mind and brain work and to an accurate diagnosis of why their campaigns fail" (p. 15).

Weston cites the work of neuroscientist Jeffrey Gray to explain how emotion works in the brain. Gray has found that the brain has two emotional systems, one for positive emotions and one for negative emotions. In these different systems, different neurotransmitters are released. In the positive emotional system, dopamine is released, in the negative emotional system, norepinephrine is released. When one negative or positive emotion is experienced, others tend to follow. People who experience sadness may also experience anxiety at the same time, for example. Also important to the process of emotions are the associations that are formed while emotions are experienced, a process that seems to happen unconsciously. Priming effects can be used to manipulate this process. For example, a study was done in which word pairs were presented to research subjects. Among these words pairs were ocean and moon. Subjects presented with this word pair were later asked to name a laundry detergent. They were much more likely to name *Tide* laundry detergent than those who did not see the word pair. This is because there were overlapping associations between the thoughts of tide as an effect of the moon on water and *Tide* as a laundry detergent.

The political implications of this suggest that "the choice of words, images, sounds, music, backdrop, tone of voice, and a host of other factors is likely to be as significant to the electoral success of a campaign as its content" (p. 87). The process of association can be manipulated in political advertisement. One example of this is from George Bush's presidential campaign against Michael Dukakis. The Bush campaign ran an ad against Dukakis about Willie Horton, a criminal who had committed additional crimes while given a weekend furlough by the state of Massachusetts while Dukakis was governor. The explicit intent of the ad was to show that Dukakis was soft on crime, but, as Weston puts it, "a second, more emotionally powerful network – about scary black men – was also activated" (p. 92).

It would seem that the more that is learned about how emotions influence behavior, the better politicians will be able to manipulate their voters. This is particularly the case with weakly held beliefs. When someone attempts to change strongly held beliefs, they are much less likely to succeed. When confronted with evidence that contradicts strongly held beliefs, a person is able to find a way to reason around it. Not only are the negative emotions that it brings defeated, but they are replaced by positive emotions. Several experiments have also found that "people evaluate evidence that disconfirms their cherished beliefs much more critically than evidence that supports their values, attitudes, religious beliefs, or scientific theories" (p. 100). If this is the case, then it would seem the greatest possibility for manipulating these people would be in providing one's political supporters false information about an opponent, because they are less likely to question it.

Though Weston's neuroscientific evidence is different than what a neurodynamic approach shows, his conclusion about the importance of emotions is similar to what a neurodynamic approach shows. His evidence does seem to minimize the role of the conscious will in decision-making, as the associations made happen unconsciously and greatly influence our decisions. Weston does point out that many of the conclusions reached in the studies he cites are the result of decisions made quickly. When confronted with evidence over a longer period of time, people may be more likely to change their mind, even if this evidence goes against one's position. He uses the gradual change in opinion about America's war against Iraq as an example of this. In 2003 many were in favor of the war, but as Iraq was believed to have no weapons of mass destruction, support for the war fell. We can change our minds on the issues we feel most strongly about, but usually we cannot change our minds on these issues quickly.

George Lakoff

George Lakoff (2008) cites Weston as an important influence on his work, and there are many similarities between their ideas. Lakoff focuses on the discovery of mirror neurons as he uses neuroscience to explain political behavior. Mirror neurons are important to his views about politics because he attributes to them the empathetic ability we have to see things from the perspective of someone else. This function of the brain helps us to place other's actions in the context of a narrative, which is the same thing we do when understanding our own actions. Like Weston, this idea of the importance of narrative in our lives is central to Lakoff's theory. He refers to these narratives as "frames" (p. 22). These frames tend to structure a large amount of our thoughts, and even our basic actions have frame structure at the neuronal level. From simple frames, we go on to form more complex ones:

> Simple narratives have the form of frame-based scenarios, but with extra structure. There is a Protagonist, the person whose point of view is being taken. The events are good and bad things that happen. And there are appropriate emotions that fit certain kinds of events in the scenarios. In a simple Rags-to-Riches scenario, for example, the initial state of the Protagonist is poverty, where the appropriate emotion is sadness; then there are intermediate states of hard work with varying emotions of

> frustration and satisfaction; and finally a state of wealth, with emotions of joy and pride (p. 23).

By the empathy that is at least in part affected by our mirror neurons, we understand public figures by fitting them into these narrative frames. For example, Lakoff points out that former President George W. Bush's life could be seen as fitting a "Redemption narrative." Bush was an alcoholic and a failed businessman who eventually gave up alcohol, became a born-again Christian, and then engaged in a political career. Skilled politicians recognize the need to craft stories like this about themselves to make themselves more appealing to the voters.

Oftentimes placing a particular event or person into a narrative is an unconscious process. Lakoff cites a figure that states that 98 percent of the thinking that goes on in our brain is unconscious. The problem he sees is that most people do not think in this way. They operate on the assumption that reason is supreme and that good facts and ideas have the power to persuade more so than appeals to emotion. Lakoff believes that in America the liberals in politics have particularly fallen prey to this error. Since Lakoff is writing from a liberal perspective, part of his purpose is to correct this misapprehension.

Lakoff explains liberals and conservatives in America as possessing two different modes of thought about the world. The liberal focus is on empathy. They want to use government to create protection and empowerment, which will lead to more freedom. The conservative focus is on obedience to authority. This obedience will lead to more freedom and success. The problem with liberals is that they have too often presented their policy proposals using "frameless facts" (p. 53). They do not place their arguments within the proper narrative. This is something that conservatives have done much better, and so arguments such as that over the "war on terror" are done within the conservative framework.

In an individual person, both of these competing views of the world tend to exist, but in "nonoverlapping areas of life" (p. 70). Someone can have a more liberal view on the need to provide universal health care coverage to the country, but have a conservative view about the need to launch preemptive strikes against America's enemies. Lakoff terms this idea "biconceptualism" (p. 70). At times, the various positions on the individual issues someone takes can seem contradictory when viewed as a whole. A person will see him or her self as liberal or conservative despite these contradictory views depending on which side they have identified with more. The use of one view tends to inhibit the use of the other. The way to change minds in politics is to point out these contradictions. If you are trying to make someone more liberal, concentrate on the areas that you already agree and show how these are liberal ideas. One should cast the figures on his or her side as heroes and the figures on the opposite side as villains. The goal should be to make the unconscious factors that go into political belief more conscious.

In particular, this can be done through metaphor. Lakoff argues that morality is founded in metaphor. These metaphors arise when we experience two different kinds of experience at the same time: "As it turns out our experiences of well-being and ill-being correlate regularly, especially in childhood, with many kinds of other experiences" (p. 94). For example, we feel good when we eat good food and bad when we eat rotten food. This creates the metaphor that purity is moral and rottenness is immoral: "The result is that we learn an extensive system of mostly unconscious primary metaphors for morality and immorality just by living normally in the everyday world, within a culture and a family. We just live. Our brains do the work" (p. 94). Mirror neurons play an important role in this process, as they help to apply these metaphors to others.

Lakoff believes that for liberals to be successful, they need to activate peoples' capacity for empathy. For conservatives to be successful, they need to activate peoples' capacity for fear. The more frequently these emotions are activated and the stronger they are, the stronger the connection people make between them and a particular issue will be. This is why politicians tend to repeat themselves so often.

Lakoff hopes that by revealing the way that the brain acts in political debate and how it can be manipulated by use of framing, at the very least it will be more difficult to do so in those who are aware of this new knowledge. We have seen earlier that strongly held beliefs are unlikely to change when confronted with contradictory evidence. The question is, how does this process work when the contradiction is within oneself? It would seem that one must come up with a new narrative that incorporates these contradictory ideas. This new story might be assumed to be biased towards whichever of the ideas one is more emotionally attached to. Lakoff's suggestions would seem to be ways that could be effective in helping to make this process more favorable to whatever ideology one adheres to.

In reading Lakoff's book, it must be observed that he does the very thing he is advocating. Conservatives are presented in very negative terms, as needing fear and obedience to authority to succeed, and liberals are presented as advocating understanding and hope. By presenting the one side as villains and the other as heroes he would seem to be trying to provoke the same kind of emotional response in his readers that he suggests they do when trying to persuade others. Lakoff, like Westen, is correct in identifying the power of the emotional response in political arguments (although at the end of his book he claims this is a new way of understanding what it means to be human. In reality it is a understanding that can be traced back as far as Aristotle's *Rhetoric*). While he does recognize this, and attempts to explain its neurological basis, he seems overly reliant on the discovery of mirror neurons. A neurodynamic theory presents a much more comprehensive view of how this process occurs. It must be admitted, however, that since Lakoff's appeal is largely an emotional one, this weakness is not necessarily detrimental to his claims. He does not seem as interested in presenting an accurate picture of the world so much as he does in presenting a persuasive and emotionally powerful one. He points out that past appeals to mere facts have failed liberals, so they need to change their ways.

Leslie Paul Thiele

The importance of narrative is also central to Leslie Paul Thiele's neuroscience-based theory of politics. It is by narrative that we are able to understand the world and cultivate our faculties of practical judgment. This faculty of practical judgment is essential for living well. If citizens do not have good judgment, it will be very difficult for them to maintain their freedom. Thiele quotes from Merleau-Ponty, who wrote that judgment is what is necessary to make the sensations we experience into perceptions, and V.S. Ramachandran, who points out that these perceptions are often times, by the act of judgment, taking "significant liberties" with the raw sensations we receive from our environments (p. 3). In this way, judgment resembles the process of assimilating ourselves to the world. Judgment would seem to act in the same capacity that phantasms do for Aquinas and the nonlinear dynamics of neural populations do for Freeman.

The foundation for the development of practical judgment is laid down by habit. Thiele traces the origin of this insight to Aristotle, who wrote of virtue being cultivated by habit. As we noted in chapter two, for Aristotle, virtue must be developed in a person by performing right actions to the point that they become habituated. This habit "orientates, but does not determine behavior" because of the habit of reflective deliberation (p. 24).

This is shown in neuroscientific research as well. Thiele frequently cites the work of Jeffrey Schwartz on neural plasticity. It is this plasticity that allows our interactions with the world to change our behavior:

> Our interactions with the world, as well as our minds' interactions with itself, reshape our brains. This neural sculpting alters the habits of our thoughts and changes the ways we grapple with existence. Just as we become competent speakers by being exposed to a supportive linguistic environment, so we become competent judges by exposure to an environment that challenges us to assess, evaluate, and choose while providing opportunities for correction... Through experiential learning we cultivate the skills of perception and assessment that yield accurate correspondence judgments as well as the social aptitudes that facilitate ethico-political judgment. Formal instruction is not much involved in the process (p. 89).

Experience is not enough, however. This experience must be paid attention to and reflected upon if it is to result in practical judgment, "experience enhances the skills of judgment only by way of the exercise of judgment" (p. 95). This has been shown in the work of neuroscientists Michael Merzenich and R. Christopher deCharms, who have observed that "physical change in the structure and future functioning of the nervous system" can only occur when attention is paid to what we experience (p. 94).

This process is still somewhat limited. Thiele argues that our genetic make up does determine which lessons we are predisposed to learn. Part of the development of a practical judgment is to learn the limits of what we can do and learn how to act within those constraints. For example, citing the work of Gerd Gegerenzer, Thiele points out that we tend to judge best when we "employ skills that were selected for in the natural environment" (p. 85). Oftentimes "fast and frugal heuristics" that are more natural prove to be a better guide than the "more sophisticated, deliberative, and rational models of decision-making developed in recent decades" (p. 83). Sometimes the quality of our judgment can even be worsened by searching for rational justification for things (p. 140).

This is not to say that reflection and deliberation are not helpful when making decisions, but that unconscious factors are also integral to the process. If we want to become better at making judgments, we need to work to cultivate the unconscious processes that occur during decision-making. Though intuition, which arises from unconscious processes, is important and helpful in making decisions, this is much more the case for those who are experts in a particular field. Intuitions must arise from experience for them to be useful. These intuitions can tell us where to direct our powers of reasoning:

> Reflection and rational analysis are as indispensable to good judgment as the tacit knowledge and skills that precede and

> inform deliberative efforts. The better part of intuition may well be discerning when, where, and how the power of reason should be dutifully engaged (p. 142).

Judgment should be done with reasoning, but this reasoning cannot be fully explained.

In order to educate the unconscious processes occurring during decision-making, we need to exercise the areas of the brain that are "inaccessible to direct intervention" (p. 155). One way of doing this is by the use of "implementation intentions." Research has shown that subjects who state that when a particular action happens they will either do or not do something in response to it are more likely to achieve their goal and have greater skill:

> Indeed, the use of implementation intentions has been shown to enhance perceptual and motor abilities, facilitate effective operation in the midst of distractions, reduce stereotyped or prejudicial reactions, attenuate the negative effects of emotions and moods, aid the shunning of temptations that thwart goal achievement, and offset disruptive priming effects (p. 156).

This can even lead to more innovative and imaginative solutions.

Another effective strategy is the use of a "devil's advocate" in thinking. This can help to mitigate the effects of poor intuitive notions that we have developed. This works because it forces the right hemisphere of the brain, which is more receptive to novelty, to be active, and counter the work of the left hemisphere of the brain, which tends towards rationalization and conformity. This is just one way of achieving "whole-brain judgment," which creates a product that is "linguistic and imagistic, symbolic and concrete, habituating and inventive, calculative and intuitive, explicit and tacit" (p. 160). The various processes of thought must be brought into proper harmony with each other. Thiele points out that while being ruled by rationality may produce greater control by the consciousness, it will be a tyranny, and tyrannies do not tend to produce better judgments.

Though Thiele frequently cites Jeffrey Schwartz's work throughout his book, he does not mention meditation techniques as a way of improving one's mindfulness as Schwartz does. This too would seem to be a helpful method for educating the unconscious and helping us be aware of it.

The most important tool for educating practical judgment is narrative. It is by narrative that we are able to make sense of our selves. In fact, narrative seems to be what gives rise to the self: "An individual becomes a self... by abstracting a persona from its narrative passage through space-time. The self is the character-in-development distilled from the multiple, variegated, spatio-temporal sequences of events" that make up a person's life (p. 203). It is how we make sense of our lives.

These narratives are what give us our reasons for the actions that we take, and they also allow us to "assess, evaluate, and choose these reasons" (p. 223). Using the work of Alasdair MacIntyre, Thiele suggests that when we have achieved the right mix of "authorial presence and empathetic extension, the development of a self capable of moral relationship is in progress" (p. 225).

The narratives of others, in the form of plays, novels, movies, histories, and other forms of expression also help to develop our judgments. These stories provide us with a "workshop where skills and sensibilities may be finely honed" (p. 238). An understanding of narrative is what can guide practical judgment:

> The questions one asks of fiction – what is the example of a particular character's action an example of – is the same

> question the moral and political judge asks of life's experiences. He seeks to discover the narrative nest that captures the most important lessons to be learned from any particular situation. He then assesses how the plot might (best) unfold (p. 268).

We seek meaning, and make sense of events, by placing them in the context of a story. This is why Lakoff speaks above of how politicians look to control their own stories and place themselves into a narrative that will appeal to people. C.S. Lewis is quoted as saying that the expression of a parable is more than a literary device; it is a "fundamental component of understanding and a basic structure of the human mind" because it expresses one story within another (p. 268).

Narrative also teaches us that there are multiple sides to the same story. This means that for practical judgment, the "assimilation of multiple perspectives is a prerequisite" for our judgments to be sound (p. 273). This helps us to see the complexity of the world:

> The practical judge affirms that we cannot attain a firm grasp of a multi-dimensional, dynamic world through a reductive, analytical application of epistemological or ethical rules... the assessment and evaluation of ethico-political life remains grounded in the intuitive capacities, tacit skills, and affective imagination that allow insight into a deeply complex realm of interaction and emergence (p. 279).

In the cultivation of this judgment we find our freedom. By being skilled at the art of practical judgment we develop the ability to direct our own path in life. We are able to understand what we are capable of and how to direct our attention to the cultivation of moral habits and intuitions. We are able to find the balance between reason and emotion that allows this to occur. Thiele states that this is a lesson democracies should learn: "Here the spoils of widely cultivated practical wisdom would be the fine balance between liberty and law, autonomy and obligation, innovation and order, creativity and custom that defines the best of democratic life" (p. 291).

There is much in Thiele's thought that is similar to a neurodynamic approach. His suggestions for the cultivation of a better practical wisdom would also seem to be the antidote for the ways of manipulating the electorate in democracies that Westen and Lakoff warn of and advocate for. As Lakoff notes in his work, one way of avoiding manipulation by these tactics is simply to be more aware of them.

The problem with Thiele's work is that he provides little basis for determining what makes for good habits. As Ivan Kenneally (2008), in a review of Thiele's work, points out, Thiele argues that there is "'no trans-historical, culturally universal, non-contingent principle of right.' Rather, we are left with 'more or less persuasive stories'... Indeed, practical judgment turns out to be a kind of reader's instinct" (p. 106). Any narrative that becomes a "meta-narrative" and claims an authoritative status over others on the basis of anything more than its ability to persuade is taking things too far. There is no universal morality. Kenneally sees this as leaving little ground on which to base morality, and fails to recognize the "real possibility that some stories have proven more attractive because *they are more true*" (p. 107).

Thiele quotes from C.S. Lewis about the power of parables. Lewis also believed in the power of stories, but it was this belief that led him to accept the fact that there was one story that was superior to the others. In *Surprised by Joy* (1955), Lewis explains that one of the reasons why he eventually became a Christian was that he came to see that the story of Christianity was the fulfillment of all other myths. It was a myth or story that was so powerful because it happened to be true. It is the "summing up and actuality of them all" (p. 128).

Thiele is right to stress the importance of practical judgment and recognize the role that narrative plays in this process. The fact that he does not believe that any kind of universal grounds for morality can exist is disappointing, but does not mean his other conclusions are invalid. One can set aside this weakness in his position, and build from the other insights that he has. In particular, the suggestions he has for avoiding unconscious manipulation would seem to be valuable.

William Connolly

William Connolly (2002) uses neuroscientific evidence to argue for political pluralism, particularly a political pluralism that is "appropriate to the acceleration of speed, compression of distance, and multidimensional diversity marking contemporary life" (p. 2). Connolly shows, by using examples from V.S. Ramachandran and Antonio Damasio, that neuroscientific evidence supports the "intersubjective character of thinking and judgment," which supports the need for a pluralistic view of the world (p. 6). This research "can show us how the inwardization of culture, replete with resistances and ambivalences, is installed at several levels of being, with each level both interacting with the others and marked by different speeds, capacities, and degrees of linguistic sophistication" (p. 7).

Perhaps we can understand what Connolly means here as being the importance of the interaction between the conscious and unconscious thought that goes into decision-making. These processes are different, as consciousness is an emergent result of unconscious processes, and occur at different speeds (conscious thought tends to be much slower than unconscious thought). Culture is influential on both of them.

One way we can see the difference in speed on levels of perception is in the process of memory. Connolly makes a distinction between "virtual memory" and our conscious perceptions of memory. Virtual memory is what we perceive before we are conscious of it, and is processed with an explicit image of the memory. This form of memory is rapid:

> Perception is quick, as it must be to inform action. The human capacity of explicit image recall is far too slow to keep up with the operational pace of perception as we walk, ramble, and run through action-orientated contexts. So virtual memories are called up rapidly, but their vital role in perception is lost the perceiver sunk in the middle of the action. Perception thus seems pure and unmediated to us. But it is not. It is a double-entry activity guided by the concerns of possible action, not by a spectatorial quest to represent an object in all its complexity (p. 26).

This process can be done in a parallel fashion in our minds, with many objects processed simultaneously, in contrast to the slow, serial process that occurs in our conscious mind. These two processes have an intersubjective effect on one another.

A second idea that is key to understanding this process is that of somatic markers, drawn from the work of Antonio Damasio. A somatic marker is a "culturally mobilized, corporeal disposition through which affect-imbued, preliminary orientations to perception and judgment scale down the material factored into cost-benefit analysis, principled judgments, and reflective experiments" (p. 35). This disposition works so that "it mixes culture and nature into perception, thinking and judgment; and it folds gut feelings into these mixtures" (p. 35). These markers also help us perceive things faster than we would have if things were left to the slow process of our conscious thought.

Connolly uses the film *Citizen Kane* to show a third process of memory, where the "effects of the past on the present cannot take the form of explicit recollection, even when time is available" (p. 38). *Citizen Kane* shows the life of Charles Foster Kane through a series of flashbacks, in stories told about him by others, as a journalist tries to find the meaning of Kane's last words. This prevents an objective picture of his life from being formed, as we are dependent on others to explain Kane's life to us. In the end, the viewer, but not the characters of the film, is given a hint as to the meaning of these last words, but even the meaning of this is ambiguous. How we interpret it depends on how we have perceived the scenes preceding it. The subtle foreshadowing that occurs in these scenes may change our observation of the end of the movie if we pick up on them, but they may not be explicitly recalled. This shows the third kind of process of memory.

All processes of memory are at work interacting in one's conscious perception. This complexity of perception should discourage one from coming to any kind of conclusion that commits the "hubris of deep, authoritative interpretation" (p. 41). The complex interplay of these factors means that the process of memory will be a very subjective process. This supports Connolly's general advocacy for a pluralistic approach that is respectful of multiple points of view. It would seem, however, that this theory suffers from the same weakness that Thiele's does. This would seem to provide a weak ground on which to base morality, because it does makes it more difficult to find a basis on which one can claim that one way of life is superior to another. Connolly argues that his pluralism is the "lifeblood of democratic politics." It folds "creativity and generosity into intracultural negotiations over issues unsusceptible to settlement through preexisting procedure, principle, or interest aggregation alone" (p. 138). One could say that what this theory (and Thiele's) shows is the need for freedom, much as a neurodynamic one does, because the complexity of how we perceive the world and the intersubjectivity of that world makes agreement difficult and shows us that all complex issues can inevitably seen from multiple perspectives. Connolly argues that with a "deep pluralism" it is possible for diverse cultures of coexist with each other:

> Negotiation of an ethos of deep pluralism is the most promising way to extend freedom in a state already inhabited by multiple faiths; it is also the best way to promote generous cross-state and cross-civilizational relations in a diverse world that is smaller than heretofore because ti spins faster" (p. 129).

One must ask then, if this need for freedom is a sufficient basis for the formation of morality?

Conclusion

What is central to many neuroscientifically grounded theories of politics is the need for freedom. A neurodynamic understanding of the formation of meaning shows that meaning best emerges in freedom. It also illustrates the often observed paradox of freedom. Freedom is necessary for meanings to be formed, which are necessarily for a society to thrive, but to have too much freedom makes thriving impossible. In Chapter Six, a quote from Sam Harris was mentioned that blamed the tolerance of the Western world for the spread of radical Islam. The need to give all groups freedom was held to be more important than working to curtail the freedom of groups that could threaten it. Regardless of whether Harris is right in this particular case, in general this is a real concern for a society. Freedom must be balanced with the need preserve it. As noted earlier, this neurodynamic theory also recognizes that too little freedom can also be dangerous to a society (or mind) because it results in an excess of stagnation and rigidity.

This shows that freedom is important, but it is not an end in itself. It is a necessary, but not a sufficient condition for having a good society. Beyond showing the necessity of a certain level of freedom, any political lessons derived from this theory cannot say what makes for a good society. The question is whether this is a sufficient foundation for a good society.

It would seem to be the foundation of our modern, liberal democracies. The strength and weakness of modern, liberal democracy is that, in essence, it does not try to answer much. At its heart, a liberal democracy exists to provide the security of property and the freedom to enjoy that property. It keeps individuals safe by seeking to prevent one individual from dominating, and also preventing the majority from dominating. As it says in *Federalist* 51, "Whilst all authority in it will be derived and dependent on the society, the society itself will be broken into so many parts, interests, and classes of citizen, that the rights of individuals, or of the minority, will be in little danger from the interested combinations of the majority." A multitude of competing interests prevent domination by any of them.

Such a government does not inculcate the virtues that many of the philosophers discussed earlier, such as Aristotle, St. Augustine, and St. Thomas Aquinas praises most highly, but it is possible that it does create a kind of virtue. In his essay, "Ethics and Politics: The American Way," Martin Diamond (1986) writes of this virtue. He states: "When Madison's 'policy of opposite and rivals interests' is understood in the light of this distinction between avarice and acquisitiveness, we can begin to see the grounds for some of the excellences we all know to be characteristic of American life" (p. 101). The virtues created are, "...frugality, economy, moderation, labor, prudence, tranquility, order, and rule" (p. 101), and "independence, initiative, a capacity for cooperation and patriotism" (p. 107). Certainly moderation is an Aristotelian virtue.

It is a matter for debate whether these virtues are enough to promote a good society. Some of these virtues can easily become vices. In his article, "The Founding Fathers: an Age of Realism," Richard Hofstadter (1986) offers the criticism that in seeking to control the vices of men, and by institutionalizing this attempt, the founders of America may have created a government that allowed the pursuit of vicious mercantile ends to be conducted with greater efficiency and in the guise of legitimacy (pp. 73-74).

In *Centesimus annus*, Pope John Paul II (1991) also address the danger of economic concerns overwhelming everything else:

> All of this can be summed up by repeating once more that economic freedom is only one element of human freedom. When it becomes autonomous, when man is seen more as a producer or consumer of goods than as a subject who produces and consumes goods in order to live, then economic freedom loses its necessary relationship to the human person and ends up by alienating and oppressing him.

The great disparity in wealth that can be created by concentrating solely on the economic, with little to no regard for anything else, can be a problem. An excess of equality is a bad thing, but for there to be freedom to pursue one's own ends, there must be a certain amount of equality of opportunity. If someone chooses to pursue wealth, then he or she can be allowed to do so only so much as it does not keep other individuals from having the ability to pursue their own ends. Thus, to a certain extent, government must be involved in the regulation of the economy, to insure that the power that wealth can bring will not be overwhelming. In light of recent economic events, perhaps, for example, it is necessary for governments to prevent the creation of private institutions that could be considered "too big too fail."

It also seems that these democratic virtues are weaker and less robust. Martin Diamond refers to them as bourgeois virtues. But perhaps they are the best that can practically be achieved with government. In the *Politics*, Aristotle outlines the ideal regime, but he considers all regimes. He also writes of the best regime given particular circumstances, "the best regime and best way of life for most cities and most human beings..." (1295a28-29). Among the three regimes to be considered legitimate is the polity or mixed regime. It is this type of regime that modern liberal democracy might be classified as. Characteristic of this regime is a large middle class, which will act to moderate the tension between rich and poor (1295b23-27). It is a regime lacking in the excellence that his ideal would provide, but it is a regime that allows for more people to take a larger part in it.

Of course, even if the government does not provide much meaning for life, other forces will seek to fill the void. Sometimes these forces will try to remake the government in their own image, looking to deprive people of freedom in the belief that they are in possession of the knowledge of the proper ends for all humanity. Frequently, these forces can be religious, as the world sees now with radical Islamic groups, but they need not be. Communism would seek to fill this role just as vigorously. It is in this that the need for multiple groups is seen, to moderate the effects of the others. We have discussed this somewhat in chapter six.

This also brings to light the need for a respect for the rule of law. If these groups seek to make their changes within the rules of legitimate government, then they can accept their defeats when they happen and work to fight another day. But where there is no law established, or the laws are ineffectual, then things become dangerous, because then the ends of these groups may be pursued with violence.

Furthermore, when these groups (particularly religion) are in their proper place in society, they can contribute to providing the virtues that the state itself does not seek to provide. Martin Diamond points out that it was the American founders' expectation that the virtues the state did not instill itself would still come from "religion, education, family upbringing, and simply out of the natural yearnings of human nature". So long as these institutions function to do this, without seeking to control the state, it is not necessary for the state to provide them itself. Given the lack of agreement on what is proper virtue, it is better that it does not.

However, when religion and the other institutions fail to fulfill this capacity, then great confusion emerges. Perlovsky alludes to this in the above discussion of the tension between synthesis and differentiation in the mind. In his book *Lost in the Cosmos*, Walker Percy (1983) observes that humans have great difficulty in understanding themselves. Traditionally the understanding of one's self has been done in relationship to nature or to God, and as stated, the expectation of the American founders was that religion would continue to do this. However, in what Percy terms a "post-religious technological society"

(p. 113), the society in which we now live, these options are not available. It is possible that such a society could only have emerged because of the freedom that democracy and technology have brought. What remains is self seen as immanent, "a consumer of the techniques, goods, and services of society," or self as transcendent, "a member of the transcending community of science and art" (p. 113). These seem to be problematic ways of understanding the world. The immanent self is impoverished because it has lost, "its sovereignty to... the transcending scientists and experts of society... the self sees its only recourse as an endless round of work, diversion, and consumption of goods and services" (p. 122). The transcendent self gets pleasure from the loss of self, "when the self explored the world through signs before falling into self-consciousness" (p. 124). The difficult is when this person must reenter into the world he has transcended, and it then seems a lesser place. We cannot flourish in such a state of confusion.

These problems seem similar to what Perlovsky was describing when he mentioned the lack of synthesis in the West. As the influence of religion falters, no substitute has adequately arisen to take its place. Perhaps there needs to be more songs about science. A better alternative would seem to be showing how religion and science can be consonant with each other as Alistair McGrath has attempted to do. This provides a way of understanding science that preserves the important role of religion in the cultivation of morality.

CHAPTER EIGHT

FREE WILL AND THE LAW

Michael Gazzaniga (2005) contends that one day the issue of free will and neuroscience will dominate the legal system. This is because recent neuroscientific discoveries can make it seem like we are not in control of our own actions. We have already seen Gazzaniga's own solution to the problem, that brains are determined, but personhood is in many ways a social construction, and therefore, not determined. Though my solution and Gazzaniga's are different, we both agree that neuroscience should not diminish the responsibility we have for the actions we take. One's answer to the question of free will has its most immediate impact when it comes to questions of law. The more one believes that the brain determines our actions, the more one will argue that the law needs to be changed. This chapter looks at arguments both for and against changing the legal system on the basis of belief in free will, and at how neuroscientific methods of collecting evidence have already begun to change it.

Steven Morse (2004) argues that neuroscience, as we currently understand it, has limited "normative implications for law and society" (p. 158). To understand why this is the case, it is first necessary to understand the law's concept of the person. Under the law, which is defined as being rules and institutions that "are created and enforced by the state" (p. 159), the assumption is that people are practical reasoners. That is to say that they are "capable of acting for and consistently with their reasons for action and who are generally capable of minimal rationality according to mostly conventional, socially constructed standards of rationality" (p. 164). Because of this, those who are legally responsible are those who are able to "grasp and be guided by good reason in particular legal contexts, who are generally capable of properly using the rules as premises in practical reasoning" (p. 165)

This is in contrast to a view of action that sees it in mechanistic terms. Under the most extreme application, this view would reduce our actions to biological causes (similar to the way Wegner's position is described in chapter five). Because we have yet to entirely understand how this process might work, this approach is not yet a legally relevant one, but could be some day (as Greene and Cohen argue below). Some try to take an assimilative approach (much as I have tried to do with this work), where "desires, beliefs, and intentions are genuine causes" (p. 161) integrated with a mechanistic framework. This question cannot be resolved until the relationship between mind and body is better understood.

Morse points out that seeing our behavior in a more mechanistic way is a view that existed long before neuroscientific evidence was considered. These views have usually been considered very controversial. There is no need at the moment to accept such viewpoints as correct:

> At present and for the foreseeable future, no one can demonstrate irrefutably that we are 'merely' ultracomplicated biological machines. We have no convincing conceptual reason from the philosophy of mind, even when it is completely informed about the most recent neuroscience, to abandon our view of ourselves as creatures with causally efficacious mental states" (p. 167).

Even if we are not aware of all the causes of our actions, or mistaken about them, this does not necessarily mean that we are acting without intentionality and consciousness. It is already recognized in the law that responsibility is diminished when consciousness is diminished. At best all neuroscience can show thus far is that when there is damage to the brain that there is a diminished capacity to understand one's actions.

Morse addresses Libet's experiments, and argues that all they show is that nonconscious neural activity precedes conscious awareness. This does not mean that consciousness lacks a causal role. Additionally, the idea that all our actions are mechanistically determined is an incoherent one. Assuming one accepts this idea, the consequences of it do not seem relevant. Even if we think that we have no control over our actions, in practice we will continue acting as if we did. If we really have no control over our actions, it might be impossible to stop acting in such a way.

In looking at the question of free will and the law, Morse contends that is only a compatibilist position that is capable of justifying our current legal doctrines. Such a position "insulates responsibility from the challenges that scientific understanding seems to pose" (p. 173). This is because compatibilism "treats responsibility practices as human constructions concerning human action and asks if they are consistent with facts about human beings that we justifiably believe and moral theories we justifiably endorse" (p. 177). The other two possibilities, determinism and indeterminism both believe that the will must be uncaused to be free. Morse, for reasons similar to those that have been presented in this work, does not believe this is the case. An action can be caused and still not be excused, "because causation per se has nothing to do with responsibility" (p. 177).

Because for the law, what is necessary for responsibility is rationality and self control, neuroscientific discoveries show little need to radically change the law unless they can show us that we have no true sense of rationality at all. Neuroscience can help us to evaluate whether or not a person can act rationally, but it cannot tell us how much rational capability we need to have to be considered responsible. Such a question is "normative, moral, and, ultimately, legal" (p. 179). Though brain defects often lead some to question whether those afflicted are legally responsible, this can only be the case if such a defect lowers rationality to a point past what the law says is necessary for it. To think otherwise is what Morse calls a "psycholegal error" (p. 180).

What neuroscience can possibly assist the law in doing is refining how, for example, it is able to detect conscious lying or how much faith to place in the memories of witnesses. This will only be the case if neuroscientific techniques to aid in these areas prove to be very reliable. The law is fundamentally conservative and resistant to change. Assuming at some point, neuroscience does produce reliable ways of lie detection or other behavioral predictions, this could pose a threat to civil liberties. It is unclear how the law might react to this. If criminal behavior can be predicted before it happens, then the those who could commit crimes in the future might be forced into treatment against their wills. Perhaps they might be required to take behavior altering drugs to curtail potentially violent impulses. It seems likely that if such a thing is possible, then it will be instituted by the law. But this is far from where the science is now.

For neuroscientific evidence to be admissible in the United States legal system, two approaches are used: "the *Frye* 'general acceptance' standard, and the Federal Rules standard as interpreted in *Daubert v. Merrell Dow Pharmaceuticals, Inc.*" (p. 196) and other cases. The *Frye* standard accepts evidence if it is "generally accepted" by the scientific community. The *Daubert* case provided four "nonexclusive criteria... testability (or falsifiability), error rate, peer review and publication, and general acceptance" (p. 197). Of these two, the *Daubert* criteria are most often used. Morse states that: "If neuroscientific evidence is specifically relevant in an individual case, as *Daubert* requires, and it is based on competent science, it will be admitted" (p. 198).

To summarize Morse's position: "Discoveries that increase our understanding and control of human behavior may raise the stakes, but they don't change the game" (p. 198). In general, Morse presents a compelling argument against changing the legal system. There is some question as to what role an increased emphasis on the importance of emotion might mean for the law, since it places so much of the burden of responsibility on being rational. Morse does briefly mention this research, but does not think it will have much impact on the law. Perhaps he is correct, but it would seem that such research would at least cause one to reconsider what it means to be rational, and look more closely at how nonconscious thought processes can influence decision-making. It may be better to think of what the law calls rationality as conscious deliberation. Still, since this work argues that such conscious deliberation is both possible and influential on decision-making, the point remains that no significant changes to the law would be necessary.

Dave Opderbeck (2010) looks at neuroscience and the law from a Christian perspective. He thinks that the law must take the findings of neuroscience into account when making decisions, but that this neuroscientific information should not be emphasized to an extent that is greater than what it merits. It ought to be recognized that there are multiple levels of reality (for more on this, see the discussion of Opderbeck in chapter six):

> The law must account for neurobiology, but it must also do more. For Christian theology, the human "self" is more than the brain, and we may speak as a matter of faith or theological confession of a "soul" that persists beyond the life of the brain, not as a ghost filling up some gap in causation, but as part of a different layer of reality than the physical. The human "will" also is more than biology, but is constituted at a deeper layer of reality than the physical, and is ultimately rooted in the source of all reality, the perichoretic life of the Triune God (p. 28).

As Opderbeck argues, neither neuroscience nor the law can be understood apart from an understanding of God if one is to take a Christian perspective about the law. Therefore, while neuroscience ought to be considered, along with other tools "such as utilitarian welfare economics, behavioral economics, game theory, and public choice theory" (p. 28), these tools must be considered as part of an interdisciplinary approach that ultimately is informed by the "ultimate ends and purposes of the human person, which are rooted in God" (p. 28).

Joshua Greene and Jonathan Cohen (2004) agree with Morse that existing legal principles can handle the new discoveries of neuroscience. However, they also believe that these new discoveries will change the "intuitive sense of justice" (p. 1775) that people have, and which guides our legal decision-making. Cognitive neuroscience will,

> by identifying the specific mechanisms responsible for behaviour, will vividly illustrate what until now could only be appreciated through esoteric theorizing: that there is something fishy about our ordinary conceptions of human action and responsibility, and that, as a result, the legal principles we have devised to reflect these conceptions may be flawed (p. 1775).

This is because the law, though ostensibly taking a compatibilist position on free will, actually has quite an indeterminist (or libertarian) one:

> At present, the gap between what the law officially cares about and what people really care about is only revealed occasionally when vivid scientific information about the causes of criminal behaviour leads people to doubt certain individuals' capacity for moral and legal responsibility, despite the fact that this information is irrelevant according to the law's stated principles. We argue that new neuroscience will continue to highlight and widen this gap. That is, new neuroscience will undermine people's common sense, libertarian conception of free will and the retributivist thinking that depends on it, both of which have heretofore been shielded by the inaccessibility of sophisticated thinking about the mind and its neural basis (p. 1776).

Neuroscience will lead to a rejection of commonsense notions of free will. This will change the law from having a retributivist notion of punishment to a consequentialist notion. Punishment will then be "aimed at promoting future welfare rather than meting out just deserts" (p. 1776). If the common notion of free will is an illusion, then perhaps there can be no such thing as a "just desert."

One advantage to a consequentialist understanding of punishment is that it works for all three definitions of free will mentioned in this paper. It does not matter is one takes an indeterminist, a determinist, or a compatibilist approach. Punishment under consequentialism is not concerned with one's reasons for doing anything. All that is important is what the results of the punishment will be. If a punishment produces a positive result for society as a whole, it is probably a good punishment.

This is not the case with a retributivist form of punishment. For retributivism, only an indeterminist or compatibilist idea of free will is acceptable. Greene and Cohen immediately reject an indeterminist answer, leaving only compatibilism as a possibility. They point out that this is the position on free will that most legal scholars tend to accept. This is why Stephen Morse argues that there need be no changes to the law because of neuroscience. Neuroscientific findings do not threaten a compatibilist notion of free will.

However, because Greene and Cohen believe that, at present, the law is a combination of compatibilist and indeterminist thought, neuroscientific findings should affect the indeterminist part of the equation:

> In our opinion, the 'fundamental psycholegal error' is not so much an error as a reflection of the gap between what the law officially cares about and what people really care about. In modern criminal law, there has been a long tense marriage of convenience between compatibilist legal principles and libertarian moral intuitions. New neuroscience, we argue, will probably render this marriage unworkable (p. 1778).

Even the principle of having rationality that Morse mentions as being necessary for punishment may not be that important. Greene and Cohen argue that what people really want is to know if it really was the person who committed the crime, and not his genes or his upbringing. This represents a dualist understanding of a person that fails to account for the fact that a person cannot be considered separate from his genes and environment.

This concern also shows a gap between what people intuitively want and what the law delivers. The law punishes as long as someone was acting rationally when he or she committed a crime. People intuitively believe that criminals are deserving of more sympathy when their actions are the result of external forces that are controlling them. This intuition results in a belief in a libertarian free will that requires our actions to be caused by "some kind of magical mental causation" (p. 1780). This gap in belief is what creates the "psycholegal error." Greene and Cohen do not see this as being as easily dismissed as Morse does.

As Greene and Cohen see it, our intuitive notion of free will is very much threatened by the determinism that neuroscience seems to support. This is because it allows us to see more clearly how our actions are determined:

> Thus, neuroscience holds the promise of turning the black box of the mind into a transparent bottleneck. There are many causes that impinge on behaviour, but all of them—from the genes you inherited, to the pain in your lower back, to the advice your grandmother gave you when you were six—must exert their influence through the brain. Thus, your brain serves as a bottleneck for all the forces spread throughout the universe of your past that affect who you are and what you do. Moreover, this bottleneck contains the events that are, intuitively, most critical for moral and legal responsibility, and we may soon be able to observe them closely (p. 1781).

As we better understand these influences on the brain, questions like whether someone could have done otherwise or whether someone acted by their own volition will be unimportant, "the idea of distinguishing the truly, deeply guilty from those who are merely victims of neuronal circumstances will, we submit, seem pointless" (p. 1781).

The only reason the problem of free will exists is because of the illusion that exists in our minds, as Daniel Wegner argues. Once that illusion is dispelled (though the authors admit this might not be entirely possible), the problem of free will should seem less important. As it stands now, philosophers who are libertarians offer "panicky metaphysics," determinists "embrace the conclusions of modern science," and compatibilists try to talk themselves into a fragile compromise (p. 1783).

This shows the superiority of a consequentialist approach to punishment. A consequentialist approach does not require a belief in free will. It is only interested in whether or not a punishment will have beneficial effects for society. The results of taking such an approach may even end up acknowledging something that looks a lot like free will:

> In the name of producing better consequences, we will want to make several distinctions among various actions and agents. To begin, we will want to distinguish the various classes of people who cannot be deterred by the law from those who can. That is, we will recognize many of the 'diminished capacity' excuses that the law currently recognizes such as infancy and insanity. We will also recognize familiar justifications such those associated with crimes committed under duress (e.g. threat of death). If we like, then, we can say that the actions of rational people operating free from duress, etc. are free actions, and that such people are exercising their free will (p. 1783).

What is missing is the idea of retribution that currently exists in the law. This is an aspect of the law that can only survive if free will exists in the libertarian sense.

Biologist Anthony Cashmore also argues that the law will change because neuroscience shows that there is no free will. Like Wegner and others, he believes that the conscious will is an illusion. It exists, to provide a reason to feel responsible for our actions: "Consciousness confers the illusion of responsibility. No wonder the belief in freewill is so prevalent in society—the very survival of those 'selfish free-will genes' is predicated on their capacity to con one into believing in free will!" (p. 4502).

This belief in free will has become unnecessary and sometimes harmful as we have come to understand more about how the brain works. For example, the "insanity defense" is allowed by the courts, but it is seldom accepted, even in the case of someone like serial killer Jeffrey Dahmer. It creates unnecessary debates over whether certain crimes are diseases or addictions. Instead, we should move to a more utilitarian basis for our legal system (much like Greene and Cohen argue for):

> The primary difference would be the elimination of the illogical concept that individuals are in control of their behavior in a manner that is something other than a reflection of their genetic makeup and their environmental history. Furthermore, psychiatrists and other experts on human behavior should be eliminated from the initial judicial proceedings—the role of the jury would be to simply determine whether or not the defendant was guilty of committing the crime; the mental state of the defendant would play no part in this decision. However, if a defendant were found guilty, then a court-appointed panel of experts would play a role in advising on matters of punishment and treatment (p. 4503).

Again, the aspect of retribution currently in the law would be eliminated.

The views presented by these two articles are very similar, and the suggestions they make are predicated on the belief that free will does not exist. In both cases, they are primarily arguing against the libertarian idea of free will, but they also do not seem to find much value in a compatibilist understanding. This puts their ideas into conflict with those advanced here. This work has argued why a libertarian understanding of free will is unnecessary, but embraces a compatibilist one. In fact, having reasons for acting does not diminish but increases our responsibility. Some of these reasons are generated by the brain, but the brain is not ultimately responsible for its actions, it is the person who possesses that brain. To think otherwise is to commit the mereological fallacy. Furthermore, it has hopefully been shown that our conscious thoughts do influence the unconscious aspects of our brains as much as they are influenced by them. This should make our actions less determined by uncontrollable forces and more by our selves. Obviously I believe such an understanding is more accurate than what is argued by Cashmore and Greene and Cohen.

If such a belief is indeed supported by the evidence this should reduce the need to make the radical changes to the law these articles suggest. This does not mean that they do not contain some valuable suggestions. In Greene and Cohen's article, they recognize the need to make distinctions between the capacities of individual people, and that establishing a criteria for what it means for someone to be considered rational and operating without duress can result in a distinction that makes it look as though we do have free will. Neuroscience can help us in creating such distinctions, as it can help us to understand how the brain influences our behavior and how its defects can result in aberrations. It is probably best to think of our freedom as a spectrum, with a rational mind unimpeded by significant defects or external circumstances at one end of it. We can then determine how various internal and external factors can act to diminish that freedom. This is similar to Patricia Churchland's idea of a "parameter space" of in-control behavior mentioned in chapter five. The more freedom is diminished, the less one can be held responsible for his or her actions. This is not much different than what the law currently aspires to do.

Neuroscience can simply help us make these distinctions more accurately.

Adina Roskies (2010) seems to agree with this position. She writes:

> It is highly unlikely that moral or even legal responsibility will be threatened in its entirety. Indeed, recent experimental results suggest that people are more apt to give up other central beliefs than they are to jettison practices of blame and holding people responsible for their actions (Roskies & Nichols, 2008). However, it is likely that certain types of actions that we now treat as voluntary will be recognized as compelled, and that parts of the law will become more instrumental and less retributive in nature (Greene & Cohen, 2004).

In other words, Greene and Cohen may be right that there could be some shift towards not having retributive punishment in some cases where we do so now, but the overall, so long as we maintain the existence of free will, this aspect of retributivism will not disappear from our legal system.

Even if Cashmore and Greene and Cohen are correct about how neuroscience shows that the law needs to change, it is very likely that such a change would be difficult to make. As Oliver Goodenough (2006) points out, the idea of free will is one very strongly rooted in our psyches and provides a useful fiction: "However counterfactual the free will proposition may be in a deterministic world... it is a strategic fiction that underlies the productivity of a punishment rule... Our free will intuitions may be false in the world of deterministic science and yet nonetheless effective in the world of strategic interaction" (p. 263). This seems similar to Michael Gazzaniga's position.

Regardless of what one believes the role of neuroscience to be in determining the law, its influence is growing. One way this is occurring is with the possibility of brain-based lie detector tests. This is accomplished by functional magnetic resonance imaging (fMRI) that measures brain activity. There are currently at least two companies, No Lie MRI, Inc. and Cephos Corp., that offer such services commercially (Wagner, 2010, p. 13). Anthony Wagner argues that of all the studies that have been done on fMRI based lie detection, none of them have been able to present conclusive evidence of this method's validity at the level of the individual. Most studies have been conducted using group level analysis: "These showed brain regions where a differential response between deception and truth conditions were identified using statistical procedures that displayed consistent effects across subjects" (p. 13).

In group studies, some subjects were asked to lie and others to tell the truth. The difference in areas of the brain that are active when lying or telling the truth was then measured. The areas activated in these studies were not always the same:

> No brain region was associated with deception versus truth telling in all studies. There are undoubtedly multiple reasons for this across-study variability, including differences in: (a) the types of lies examined (*e.g.*, Ganis *et al.*, 2003); (b) the particular stimuli used and/or tasks performed (*e.g.*,modified versions of the guilty knowledge test using playing cards (see below for details) as compared to tasks probing autobiographical memories for events in the subject's recent past); (c) data acquisition procedures (*e.g.*, magnet field strength, acquired functional data resolution, number of trials per condition and number of subjects both of which impact a study's statistical power); and (d) the statistical procedures applied to the data including use of different statistical thresholds, and differences in whether the lie and truth conditions were directly compared or whether they were only indirectly compared with reference to the neutral baseline (p. 15).

However, across the studies done, there has been a correlation of some areas of activity in the brain at a rate that is above that of chance. These areas, such as the ventrolateral prefrontal cortex, are known to be "support mechanisms that are engaged during a variety of cognitive tasks, including working memory and inhibitory control" (p. 15). The studies done tended not to be able to explain why a particular area of the brain was active when it was.

In the studies done on individuals, there have been three methods used. The first uses two different tests, either the Guilty Knowledge Test (GKT) or Concealed Information Test (CIT). These tests ask subjects to either lie or tell the truth about possessing a particular item. The second method is "a 'mock theft' paradigm where subjects give truthful versus deceptive answers about the location of money in a room or about which of two objects the subject took at the outset of the experiment." The third method is "a 'mock sabotage' study involving the instructed destruction of property" (p. 16).

The results produced by these tests seem to indicate a fairly high reliability rate (anywhere between 71-90 percent). However, with each test there are methodological issues that call their validity into question. What is actually being measured may be something different than whether or not a subject is lying. Because of this, Wagner concludes that there is "no data that provides unambiguous evidence regarding the sensitivity and specificity of fMRI-based neuroscience methods in the detection of lies at the individual-subject or the individual-event levels" (p. 22). In addition, there are several issues that these studies do not address, such as how easy these tests might be to trick, whether the level of stakes for being caught matter, and whether the results are the same across the population. For these reasons, he points out that more studies are needed before such methods could possibly be used as evidence.

Jennifer Chandler (2010) agrees with Wagner. She also argues that for these tests to be admissible as evidence they must reach a "legal threshold of reliability" (p. 25), which they currently do not. If these neuroscientific tests eventually do reach this level of reliability, they still may be adopted with some resistance, because, like DNA evidence and polygraph evidence in the past, they lead to the "dehumanization" of the legal system. Still, in the end, if these methods prove reliable they must be adopted because "technologies that are widely accepted as reliable cannot be permitted to remain outside the justice system to deliver their own verdicts incompatible with those of the courts" (p. 25).

In the case of U.S. v. Semrau (2010), which was heard by the United States District Court for the Western District of Tennessee, Eastern Division, it was decided that an fMRI lie detector test conducted by the Cephos Corporation was inadmissible as evidence. In this case, Dr. Lorne Semrau, a psychologist, had defrauded the United States government of 3 million dollars as a result of billing errors to Medicare and Medicaid while providing mental health services. Dr. Semrau claimed this was because of the confusing coding system for the Medicaid and Medicare billing processes. To prove his innocence, Dr. Semrau hired Dr. Steven Laken of Cephos Corp. to conduct an fMRI lie detector test. The results of this test showed that Dr. Semrau was not lying when he claimed to have made the billing mistakes as a result of confusion and not deliberation. Dr. Laken claimed that "based on his prior studies, 'a finding such as this is 100% accurate in determining truthfulness from a truthful person'" (p. 18).

Using the Daubert criteria, the court decided that this evidence was inadmissible, because, though Dr. Laken could be considered an expert in his field, the evidence did not meet all the criteria that the

Daubert case established. Among the reasons the court cited were that this form of evidence does not have a sufficiently verifiable error rate outside a laboratory setting, standards for real life application have not been established yet, and that it is not yet widely accepted by the scientific community.

In India, a brain scan method using electroencephalographic (EEG) sensors has already been used in the case of Aditi Sharma to convict her of the murder of her fiancée, whose food she poisoned with arsenic at a McDonald's. This scan was admitted to show that Sharma had "experiential knowledge" about murdering her fiancée (Farrell, 2010, p. 1). This means that she had first hand knowledge of the murder, and had not simply heard about it from someone else. In the test conducted, which is known as the "Brain Electrical Oscillations Signature test" or BEOS: "Researchers compare brain activity for stimuli, which are known to be familiar or unfamiliar to the subject. New stimuli are then presented and compared to these baselines to determine whether they are familiar or unfamiliar to the subject" (p. 3).

Other countries, such as Singapore and Israel have also shown an interest in using such tests in their legal system. The United States has reportedly been interested in using fMRI and EEG scans when questioning suspected terrorists: "At least one anonymous source within the American intelligence community has indicated that brain scans are already being employed in this fashion" (p. 4).

The test that was used in the case in India does not seem to have been widely accepted by the scientific community, so in the United States it would probably be inadmissible under the Daubert criteria.

Michael Gazzaniga, for example, thinks that this technique is "shaky at best" (p. 5). In the Indian court system, the test is only admissible if it is conducted by the consent of the defendant.

Farrell argues that the collection of such evidence could eventually prove to be a threat to an individual's rights. If in the future, these, or similar, tests are proven to be reliable, there will be a temptation to conduct them on defendants without their consent. This would be a mistake. The evidence of our brains is different than other evidence that might be collected from our bodies, such as blood samples, "a person’s thoughts, including memory, should be viewed as a part of the person’s will" (p. 6). Thoughts should not be seen as separate from a person, and therefore, to require someone to submit to a brain scan would be forcing them to give evidence against themselves. Because of this, brain scans done without a defendant's consent should be ruled inadmissible in a court.

Neuroscience has also influenced the law in other areas. In the case of Roper v. Simmons (2005), neuroscientific evidence on the difference between the brains of adolescents and adults was thought to be influential on the U.S. Supreme Court's decision to declare the execution of minors unconstitutional (Farrell, p. 3). This ruling was recently confirmed the U.S. Supreme Court case of Graham v. Florida:

> No recent data provide reason to reconsider the Court’s observations in *Roper* about the nature of juveniles. As petitioner’s *amici* point out, developments in psychology and brain science continue to show fundamental differences between juvenile and adult minds. For example,parts of the brain involved in behavior control continue to mature through late adolescence (2010, p. 18).

Because an adolescent brain has not reached its full development, adolescents should not bear the same responsibility for their actions that adults do. This would seem to be an area where neuroscience can provide great benefit to the justice system.

Conclusion

On the basis of the evidence examined here, it would seem that the greatest impact neuroscience will make on the law, at least in short term, is in the collection of evidence. Brain scans may well improve to the point where they can reliably determine whether a defendant is telling the truth. They can also continue to assist in determining whether sufficient brain development has occurred to hold someone completely responsible for their actions, as the U.S. Supreme Court did in Roper v. Simmons and Graham v. Florida. As of yet, neuroscience does not seem to pose a threat to free will as it is understood in the legal system. Even Greene and Cohen admit that the law's understanding of free will is unlikely to change in the short term.

If neuroscientific methods of collecting evidence do continue to improve, then the question of one's free will may come under question as a matter of expediency. Brian Farrell rightly points out that one's consent should be required for a brain scan to be admissible as evidence, because it is "a part of a person's will." However, if the idea of a will being free is discarded or greatly reduced, then perhaps there will be more justification for conducting neuroscientific tests against a defendant's consent. Even though I do not agree with it, one can make an argument for there being a lack of free will using neuroscientific evidence. This is what Anthony Cashmore has done in the article cited above. Such arguments may become more popular when they can seem to provide a potential benefit to society. It is hard to argue against collecting evidence that might convict guilty people if it can be collected very reliably.

Although there may be some appeal to such a notion, for the most part it is not a comforting thought. The erosion of freedom is almost always dangerous. Neurodynamic theory shows, and common sense and an appreciation of history confirm, that too little freedom causes the stagnation of a society. Allowing the admission of brain scans without a defendant's consent would represent a substantial erosion of our freedom. Adina Roskies's research shows that people are very unwilling to give up central beliefs in things like the freedom of the will, because this belief allows us to hold others responsible for their actions. However, in this instance such a belief may come into conflict with the greater effectiveness that neuroscientific technology might be able to provide. A reliable neuroscientific lie detector test would help to make sure that the right people were held responsible, even if it meant giving up some freedom, so that all defendants could be scanned. If they are not guilty, they have nothing to hide, after all. Such an argument will hopefully not be persuasive. It is the position here that the costs would outweigh the benefits.

CONCLUSION

A work that concerns itself with free will ought to, in all good conscience, give a clear answer to the question of whether we have it or not. Unfortunately, this question that does not permit a simple yes or no answer. It is more accurate to conceive of the freedom of the will as being on a spectrum. Some may have free will, while other may not. Who one places on the end of the spectrum of freedom, and how extensive that spectrum is, depends on what one believes. For a Christian like St. Augustine, those who have free will are those who have received it by the grace of God. For a Buddhist like B. Allen Wallace, those who have the most free will are those who have most purified their minds and cultivated their insights. For an atheist like Daniel Dennett, so long as we believe that we can be held responsible for our actions, we have enough free will. It is a matter of belief, but the beliefs differ. If there is a belief in the freedom of the will, then it is possible to possess.

Belief precedes understanding. Out of my belief that I have freedom has come this explanation of how I understand it. My interpretation of the philosophers and neuroscientists spoken of here is a result of that belief. This act of belief and the subsequent act of writing this work have allowed me to create the explanation that exists here and my understanding of what freedom is has emerged from the act of this belief and creation. However similar my belief might be to others, it is uniquely my own, and cannot (and probably would not if it could) be duplicated in its entirety in anyone else.

However, though this interpretation is in a certain respect unique and therefore subjective, it cannot exist without the objective facts of the world, as imperfectly as we understand them. To see how this has worked, it may be helpful to revisit just what it is that has been said here:

In chapter one, we discussed the problem of free will, examining why it was an important question and how such a complex question could possibly be answered. Three broad categories for free will were defined. One could be an indeterminist, who believes that the will acts as an uncaused cause; one could be a determinist, who believes that all actions have an antecedent cause, and that this means we have no free will; or one could be a compatibilist, who believes that all actions have a cause, and yet there is freedom in our actions. Of these three categories, most philosophies of free will tend to fall somewhere within the compatibilist one.

Two sets of criteria for determining if we have free will were also established. The first set, drawn from the work of German philosopher Henrik Walter, stated that the will could be considered free if three conditions are met: we need to be able to act otherwise; we need to be acting for understandable reasons; and we need to be the originator of our actions. Under this definition, we do not have free will. What we consider to be free will in this work would look closer to what Walter terms "natural autonomy.

The second set of criteria came from John Searle. Under his criteria, essentially we are free if our conscious deliberation can influence the behavior of our neurons, instead of just being influenced by them. Under his criteria, we would indeed have free will as it is defined here.

We made these determinations primarily on the basis of a neurodynamic conception of how the brain works. Understanding our freedom in this way preserves the responsibility we need to be held accountable for our actions. Furthermore, seeing the importance of freedom for the brain shows the importance of freedom on a larger scale.

In chapter two, we looked at the philosophical ideas on free will held by Aristotle, St. Augustine, and St. Thomas Aquinas. The theme we saw running through their thought is that the way we understand the world is a result of the interactions we make with it. Aristotle showed us that habits of virtue develop from taking virtuous actions in the world. St. Augustine showed us that understanding is the result of the need to justify our beliefs and must be informed by those beliefs. St. Thomas Aquinas showed us that meaning emerges as a result of the assimilative process that occurs between the self and the world, and that freedom emerges as a result of a dynamic interaction between the will and the intellect.

Because our interactions with the world are what create meaning, it is by a dynamic process that meaning occurs. Because of this interaction, neither what is within us nor what is in our environment should be thought of as causally determinative. In this dynamic interaction, we can see a kind of freedom, as all factors may be valuable, but none exercise a tyranny over how we understand things. It was also pointed out that this way of understanding the world is similar to a neurodynamic approach.

In chapter three, we looked at the philosophical ideas on free will held by Rene Descartes, David Hume, Jonathan Edwards, William James, Maurice Merleau-Ponty, and Hans Jonas. In this chapter, we saw that the ability to otherwise is probably not an important aspect of free will. Rene Descartes argued that, at best, it is more of a defect than it is something we should want. It would seem that when we are confused we have the greatest range of possibilities, because our paths are not clear. At our best, we see the path that will produce the best results, and to do otherwise would be foolish. Jonathan Edwards also showed why the ability to do otherwise was not necessary when he explained that actions are more worthy of praise when we do them for good reasons. If someone does a good action for a reason he does not understand, it is not as valuable as someone who did it because he wanted to.

David Hume showed us the importance of emotions for the decision-making process, an insight that is confirmed by many different neuroscientific studies.

William James showed the importance of focused attention for the will. We are the most free when we are focused on a difficult idea and contemplate the implications of it. This idea of focused attention being central to the will would later be important to the work of Jeffrey Schwartz concerning neuroplasticity and Leslie Paul Thiele for the education of our practical judgment.

Here too we saw the theme of interaction with the world leading to the creation of meaning in the work of Maurice Merleau-Ponty and Hans Jonas. Merleau-Ponty argued that because there is this interaction, there is never absolute freedom and never absolute determinism. We have argued that what exists as the result should be thought of as a kind of freedom. Hans Jonas shows us that our dynamic interaction with the world can allow us to create new meaning by our imagination.

In chapter four, we looked at a neurodynamic theory of free will. A neurodynamic theory shows that the question of free will versus determinism is the result of a poor understanding of how causality works. Causality is a circular, dynamic process in which what influences something is also influenced by it in turn. This is certainly the case in our brains, as neurons interact with each other, and group into neural populations, where individual neurons influence the population they belong to and are influenced by it, and neural populations influence and are influenced by global brain states. The dynamics of our brain support a "biological capacity to choose" which is an "essential and unalienable property of human life." In such a process, our conscious deliberation would seem to be able to influence the working of the neurons that consciousness emerges from, and this is sufficient to at least meet the criteria for free will of John Searle.

In chapter five, we looked at other neuroscientific alternatives to a neurodynamic theory of free will. The work and ideas of Henrik Walter, Gerald Edelman, Patricia Churchland, Daniel Dennett, Michael Gazzaniga, Jeffrey Schwartz, and Daniel Wegner were considered here. Though there are many valuable ideas contained in these works, and many of them present a viewpoint that is compatible with a neurodynamic understand of free will, a neurodynamic view was still preferred. There were two reasons for this. The first was that the foundations of a neurodynamic approach could be found in the philosophers considered in earlier chapters, particular St. Thomas Aquinas. The second was that this theory resembled the way other complex systems in the world work. In many ways, complex systems behave in similar ways. Given the complexity of the brain, it makes sense to understand the brain, at least in part, by a consideration of how such systems work.

From here we moved on to consider some of the implications of a neuroscientific understanding of free will. In chapter six, we looked at implications neuroscience has for religion. Often neuroscientists reject the idea of free will because it is associated with the soul.

Since the soul is usually understood as being an entirely spiritual entity, more scientifically compatible explanations for the soul were considered. The emergent dualism of William Hasker and the constitutive materialism of Kevin Corcoran were examined. In this chapter we also looked at claims that science has helped to show that religion is unnecessary and even dangerous. The arguments of the new atheists, particularly Daniel Dennett were discussed, and contrasted with the natural theology of Alistair McGrath, who showed how Christianity and science can be consonant with each other when understood in the right way.

In chapter seven, we looked at the implications of using neuroscience to understand politics. It was shown how a neurodynamic understanding of politics emphasizes the need for freedom, but that this freedom needs to be somewhat balanced with constraint. Too much freedom will result in anarchy and too much constraint will result in stagnation. The work of Leonid Perlovsky speculated that it was possible to find a way to measure the levels of freedom and constraint in a society to determine its health, although quantifying such measures seems very difficult. Drew Weston and George Lakoff pointed out that neuroscientific research shows the importance of emotion in decision-making, and that politicians should concentrate on making emotional appeals to the electorate rather than rational ones. They also showed how easily our nonconscious thought processes can be manipulated. Leslie Paul Thiele's work showed how neuroscientific insights can be valuable for the cultivation of the practical judgment necessary to make good political decisions. It also discussed how our unconscious thought processes can be better educated, so they are not as easily manipulated. Central to this process of education was the use of narrative. William Connolly's work suggested that neuroscientific research supported arguments for pluralism in society, so that freedom could be better respected. The chapter concluded with a discussion of whether freedom was the most important goal a society should have.

In chapter eight, we looked at how neuroscience has affected the law. Issues of neuroscience and free will may become a central issue for legal systems as our knowledge of how the brains works grows. Whether or not one thinks significant changes to the law will occur depends on how much one believes neuroscientific knowledge has diminished the freedom of our wills. We looked at the work of Steven Morse, who argued that the law should not change much by neuroscientific discoveries, and who had a compatibilist view of free will. To contrast this, we then considered the work of Joshua Greene and Jonathan Cohen, who argued that neuroscientific knowledge should lead the law towards a more consequentialist form of punishment, and who had a more determinist view of free will. We then looked at how neuroscience has already begun to affect the law, examining how neuroscientific methods of evidence collection have been used in recent court cases.

What this work should show is the possibility of the freedom of the will, and also the need for it. This is not a freedom that should be absolute, because without constraint we will have anarchy, but it is very important for a healthy society. One thing neuroscientific research has shown is how to better manipulate people's decisions. This is a way of reducing our freedom. Fortunately, it has also shown how our unconscious thought processes can be effectively educated to increase our freedom. In particular, Jeffrey Schwartz and Leslie Paul

Thiele's research in this area would seem to be helpful. Schwartz emphasizes mindfulness meditation for cultivating a better mind, and Thiele emphasizes the study of narrative to cultivate better judgment. Both techniques would seem to have promise when done correctly.

Like William James, I hope that this effort can convince others to believe that they are free. Not only does this seem to be the truth, it is also a good idea. One reason for this is shown in the work of Vohs and Schooler (2008) and Baumeister et al. (2009). These articles explain that there is a difference between those who believe they have freedom and those who do not. In a study done by Vohs and Schooler, they found that weakening someone's belief in free will tended to make them more likely to cheat on a computer test. Vohs and Schooler point out that these results seem to confirm a general trend that has been observed in the rise of people cheating as they believe they are less in control of their actions:

> A recent meta-analysis that took into account cohort effects (Twenge, Zhange, & Im, 2004) revealed substantial changes in Locus of Control scores from the 1960s to the 1990s. The Locus of Control scale (Rotter, 1966) assesses lay beliefs about whether internal (personal) or external (situational) factors are responsible for one's outcomes in life (Rotter, 1966). People's belief that they do not control their own outcomes (external locus of control) jumped more than three quarters of a standard deviation over the decades Twenge et al. studied.

This can also be seen in the rise of students cheating on tests in school. Although there are likely other factors involved, it does not seem unreasonable to conclude that we are more likely to behave badly if we do not think we are in control of our actions. Vohs and Schooler do point out that the results of this study may not apply to more important situations: "Our experiments measured only modest forms of ethical behavior, and whether or not free-will beliefs have the same effect on more significant moral and ethical infractions is unknown" (p. 53).

The study by Baumeister et al. shows that not only are people more likely to cheat if they do not believe in free will, but such a belief also increases their aggression and decreases their willingness to cooperate with others.

Although it may seem so as initially stated here, it should not be necessarily be concluded that a belief in determinism is what led these research subjects to their poor behavior. Though those who had been exposed to pro-determinist messages did exhibit worse behavior overall, the difference between the groups was not overwhelming. Other factors could quite possibly have been involved as well. Even with these caveats, it does seem better to believe in free will, and perhaps these studies can help to demonstrate why this is so. This may be why Daniel Wegner admits that though conscious will is an illusion, it is a useful one.

In the end, I suppose I cannot make a definitive claim that we have free will. I hope that we do, and I believe so. If I am correct, my belief at least makes me more free. Still, what all this should show is the need for humility. There is no clear answer, only what I have found to be one that is better than the alternatives. It is not uncommon in books that concern bioethical issues to recommend humility, but, much as there are many other books written about free will, the fact that others have done this is no reason not to do the same here. It is an important thing to remember and an easy thing to forget.

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